Structural Augmented Reasoning for Language Models

by

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Abstract

Language models utilize external information to make predictions and answer questions when their inputs lack the required knowledge. Explicit knowledge structures can expose relationships that help select and use this information. However, augmentation with an existing knowledge graph depends on its coverage, and supervised model adaptation requires labeled data and fine-tuning. This thesis investigates structural augmentation from knowledge-enriched representation learning to task-related graph retrieval and reasoning.

We first propose KGLink to enrich column representations with knowledge-graph descriptions and learn their correspondence to annotation labels. To remove the dependence on a predefined knowledge graph and task-specific fine-tuning, we propose GORAG to construct and update a task graph for dynamic few-shot classification. Connected retrieval supplies candidate labels to a large language model. We then extend retrieval beyond candidate labels to facts that support an answer. AGRAG selects connected textual facts and supporting passages for question answering and generation. MGRAG constructs query-specific multimodal graphs, matches query relations to semantic paths, and uses graph structure to guide the reuse and recomputation of cached model states.

Experiments across these tasks show improvements in prediction and answer quality, with efficiency gains in graph indexing and context processing. Ablation studies examine the contributions of the structural components. A related study of hyper-relational knowledge graphs analyzes information preservation under graph transformations.

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This is to certify that I have examined the above PhD thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.

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CHAPTER 1

INTRODUCTION

1.1 Motivation

Language models utilize information in tables, text, and images to make predictions and respond to queries. Table pre-training and visual instruction tuning have extended their applications beyond plain text [1, 2]. However, the required information is often incomplete or distributed across sources [3, 4]. A table may list people without identifying their occupations. A classification task may introduce labels absent from its initial training examples. A question may require facts from several passages or a relationship expressed jointly by text and images. Hence, selecting relevant evidence and preserving the relationships among facts are crucial real-world problems [5].

The usefulness of external information depends on how it relates to the current task [3, 6]. A person's occupation can help identify a table column, while a keyword's association with a label can help classify a text. For question answering, several facts may need to be combined to answer the query [7]. Explicit knowledge structures represent these relationships and make them searchable when selecting information for a language model [8]. This motivates our study of structural augmentation, from enriching model representations with existing knowledge graphs to constructing task-related graphs for retrieval and reasoning.

We first investigate how an existing knowledge graph can supply information missing from a model's input. In column type annotation, a column of people accompanied by birth and death dates may lack the information needed to identify their common occupation. Linking the names to a knowledge graph can supply that information, but the graph may assign several valid types to each person. These graph types may also differ from the labels required by the annotation task. To address these issues, we propose KGLink to select relevant graph descriptions and integrate them with a pre-trained table encoder. Through supervised fine-tuning, the encoder learns their correspondence to the target labels [9].

KGLink's use of external knowledge depends on the coverage of the existing graph and the labeled data available for fine-tuning. When a suitable graph is unavailable or new labels arrive with few examples [10], the system needs another way to obtain and use knowledge. Few-shot

prompting shows that a language model can adapt its predictions through input examples [11], while meta-learning studies adaptation from limited supervision [12]. These approaches motivate us to explore using a large language model's semantic capabilities with a graph constructed from task data. To support this setting, we propose GORAG for dynamic few-shot text classification. GORAG constructs weighted keyword–label relations, retrieves candidate labels by connecting query keywords, and updates the graph as new labels and texts arrive [13]. The system incorporates new information through index updates without requiring a predefined knowledge graph or model fine-tuning.

Building a graph from task data extends structural augmentation to settings without a predefined knowledge graph. However, for question answering and generation, retrieving candidate labels is no longer sufficient. The model needs facts that jointly support an answer [5], and a summary can require several complementary facts [14]. Individually relevant passages can repeat facts or leave their connections implicit, while a short path between query anchors can omit required details. To address these issues, we propose AGRAG to construct a graph from the text corpus and select a connected evidence subgraph using query-dependent node influence and semantic edge costs. Subgraph expansion and hybrid passage retrieval supply additional facts and context for answer generation [15].

On the other hand, multimodal question answering requires joint reasoning over evidence from text and images [4]. Different questions about the same image can require different visual details [16, 17]. Instruction-aware visual extraction further makes the selected features depend on the requested task [18]. A relation stated directly in a question may also correspond to several relations in the evidence. To address these issues, we propose MGRAG to build query-specific evidence graphs and match query relations to semantic paths [19]. Query-aware construction selects relevant facts, while path matching checks their correspondence to the requested relationships. The selected structure also guides the reuse and selective recomputation of the LLM's key-value (KV) cache when the evidence is processed for answering. Attention-based cache retention provides related approaches to limiting context-processing resources [20, 21].

In this thesis, we study how explicit knowledge structures can help language models obtain and use the evidence required by a task. We begin with knowledge-enriched representation learning, then reduce the dependence on existing knowledge graphs and task-specific fine-tuning through graph-based retrieval. We further investigate connected textual evidence and query-aware multimodal matching as two extensions to more complex answer requirements.

1.2 Research Scope and Structural Augmentation

1.2.1 Evidence, structure, and model roles

Structural augmentation uses explicit relationships among entities, types, keywords, labels, passages, or query components. We study knowledge-graph edges, keyword–label associations, passage links, and paths that express the relationships requested by a query.

Graphs record entities and their connections, while embeddings provide similarity scores or contextual features. We combine them by serializing graph descriptions for a table encoder, weighting connections for label retrieval, scoring nodes and edges for subgraph selection, and comparing path semantics for multimodal matching.

The source of the structure changes with the task. KGLink retrieves descriptions from an existing knowledge graph and fine-tunes a pre-trained encoder using annotated columns. GORAG builds its task graph from labeled examples and updates it using subsequent predictions. AGRAG and MGRAG construct evidence graphs from source documents. GORAG, AGRAG, and MGRAG use large language models for classification, extraction, filtering, or answer generation without task-specific model fine-tuning.

1.2.2 Adapting structure to task requirements

The move from representation learning to retrieval changes how the system adapts to new information. KGLink learns how graph descriptions correspond to annotation labels through model training. GORAG incorporates new labels and vocabulary into its retrieval index. AGRAG and MGRAG organize source content into evidence for individual questions. Adaptation therefore concerns both the relationships stored in a graph and the evidence selected for the current input.

We evaluate these choices through annotation accuracy and F1, classification accuracy across label rounds, task-specific answer measures, and context-processing costs.

1.2.3 Structural information preservation

Our study of hyper-relational knowledge graphs examines which information is preserved when facts are transformed into graph representations. It distinguishes qualifier capture from graph reception and develops FormerGNN to combine them. The background chapter presents this study to explain how representation choices affect the relationships available to downstream models.

1.3 Research Questions

1.3.1 RQ1: External knowledge for table understanding

The first research question asks how graph-derived information and a pre-trained language model can be combined when a table lacks informative context and the graph's type vocabulary differs from the annotation task. The input consists of a table, an external knowledge graph, and labeled columns from the target task. The output is a semantic type for each target column.

Two difficulties motivate the question. Knowledge-graph linking can return multiple candidate descriptions at different granularities. The target task may require a different abstraction from a graph's available types. Meanwhile, a table encoder can only directly read the content in its input sequence. Surrounding dates or numeric values may provide little evidence about an entity's occupation or role. The method must expose external information while learning how it relates to the task's labels.

KGLink answers the question through cell linking, cross-column neighborhood filtering, row selection, candidate-type construction, and graph-derived feature sequences. These components supply evidence to the table encoder. Contextual masked-type representation training relates that evidence to the target annotation vocabulary. Chapter 3 explains the method and reports the SemTab and VizNet experiments, ablations, no-linkage cases, and efficiency evidence.

1.3.2 RQ2: Task-driven graph retrieval for dynamic classification

The second question asks how to construct and maintain task-related knowledge for dynamic few-shot classification without an existing external knowledge graph or model fine-tuning. The input includes the examples and labels available at the current round, the current graph index, and an unseen text. The output is a label prediction and an index update for subsequent processing.

As labels arrive, the index must retain earlier classes and incorporate new examples. Query predictions can also supply associations for later retrieval. The order of prediction and insertion determines which information is available to each query.

GORAG answers this question through weighted keyword–label indexing, terminal-based connected retrieval, and online updates after classification. The graph selects an input-specific candidate set, and the language model predicts a label using the original text and retrieved label descriptions. Chapter 4 describes the round and shot settings, the retrieval construction, and the CAD, WOS, and IFS-Rel results. The chapter also examines the cost of indexing and retrieval and the effect of the proposed components.

1.3.3 RQ3: Connected evidence for textual generation

The third question extends task-related graph retrieval from candidate labels to the facts needed for an answer. It asks how to select connected textual evidence with sufficient information to support question answering and generation. The input is a text corpus and a query whose answer can involve several information items. The output includes an evidence subgraph, passage context, and the generated answer.

A graph-ranking score identifies relevant nodes but does not by itself specify an evidence structure. Conversely, a small structure connecting query anchors may omit additional facts needed for a summary. The system must also account for information that remains in the original text after graph extraction. These requirements motivate a combination of graph selection and passage retrieval.

AGRAG answers the question by constructing a graph with statistically selected entities, extracted relations, passage nodes, and synonym links. Query-mapped facts condition the influence scores; semantic similarity supplies edge costs. The method constructs an initial terminal-connecting structure and expands it according to the reported cost-score rule. A hybrid dense and BM25 channel supplies additional passages. Chapter 5 evaluates these choices across textual QA tasks and classification.

1.3.4 RQ4: Semantic matching and computation over multimodal evidence

The fourth question extends task-related graph retrieval to evidence distributed across text and images. It asks how query-aware construction and semantic path matching can identify the required relationships when the query and evidence express them through different topologies. It also examines how the retrieved structure can guide context recomputation. The input includes a multimodal corpus, a query, and the model states available for reuse. The output is a matched evidence subgraph and an answer generated under the selected recomputation policy.

A query can contain unknown entities, represented by semantically meaningful placeholders. Evidence may express the requested relation through intermediate entities. The relevant details can also depend on the question posed about an image. These conditions make both the construction of the graph and the matching operation query-dependent. Once selected evidence enters the generator, tokens associated with connected events supply information about which computations to refresh.

MGRAG answers the question through caption-based coarse ranking, incremental query-

aware graph construction, and path-based semantic subgraph matching. It then combines query attention, graph-sequence attention, and graph topological features to score tokens for selective KV recomputation. Chapter 6 reports QA results on MMQA and six MRAMG subsets, comparisons of matching methods, component ablations, and cache quality–cost measurements.

1.4 Contributions

We make four contributions that address the research questions. Table 1.1 links each contribution to its method and evaluation.

1. **Knowledge-enriched column representations.** KGLink selects graph-derived evidence using table relationships and integrates that evidence with a pre-trained table encoder. Its contextual representation task connects the graph information to the annotation labels.
2. **Online structured label retrieval.** GORAG constructs and maintains weighted associations between keywords and labels from task data, removing the requirement for a predefined external knowledge graph. Connected retrieval generates input-specific candidate labels, and online indexing incorporates new information without fine-tuning the language model.
3. **Influence-aware connected evidence selection.** AGRAG combines statistical entity selection, query-conditioned influence, semantic edge costs, an expanded evidence subgraph, and hybrid passage retrieval.
4. **Query-aware multimodal matching and recomputation.** MGRAG incrementally constructs evidence graphs for individual queries, matches query relations to semantic paths, and uses graph information in selective KV recomputation.

Table 1.1: Research questions, structural operations, and evaluations.

RQ	Structural operation	Method location	Published evaluation
RQ1	KG-derived candidate types and features	Chapter 3, graph processing and model integration	SemTab and VizNet annotation, ablations, efficiency
RQ2	Weighted online keyword–label retrieval	Chapter 4, indexing and connected retrieval	CAD, WOS, IFS-Rel; rounds, shots, and costs
RQ3	Influence and cost-guided evidence subgraph	Sections 5.2–5.5	Section 5.7; QA and classification
RQ4	Query-aware construction, semantic paths, and KV scores	Chapter 6, matching and recomputation	MMQA and six MRAMG subsets, matching and cache studies

Table 1.2: Publications incorporated into the thesis. The hyper-relational study provides a separate background section; the four other studies form the core research chapters.

Publication	Venue	Thesis location
KGLink [9]	ICDE 2024	Chapter 3
Understanding the Embedding Models on Hyper-relational Knowledge Graph [22]	CIKM 2025	Chapter 2, structural information section
GORAG [13]	WWW 2026	Chapter 4
AGRAG [15]	ICDE 2026	Chapter 5
MGRAG [19]	PVLDB 2026	Chapter 6

Experiments evaluate representation quality, adaptation to evolving labels, answer quality, and context-processing cost. On VizNet, KGLink obtains a weighted F1 of 96.07 compared with 95.06 for Doduo. In GORAG's fourth WOS round, accuracy with ten support examples per label is 0.4929 using Qwen2.5-7B; the one-shot result in the same round is 0.4420. AGRAG obtains Medical contextual-summarization coverage of 0.690 and accuracy of 0.604 with Qwen2.5-14B-Instruct. Its hybrid-retrieval ablation reduces coverage in that task to 0.420, showing the contribution of source passages to information coverage. MGRAG reaches MMQA accuracy of 0.7442 with Qwen2.5-VL-7B-Instruct. The associated serving comparison reports 4.3 hours for MGRAG and 8.8 hours for vLLM with HippoRAG2.

The author led the research presented in this thesis under the supervision of Prof. Lei Chen. The work benefited from discussions and collaboration with Dr. Haoyang Li, Dr. Hao Xin, and Dr. Shimin Di. Each research chapter identifies its coauthored publication source.

1.5 Publications and Thesis Organization

The thesis draws on five publications. KGLink was published at ICDE 2024. *Understanding the Embedding Models on Hyper-relational Knowledge Graph* was published at CIKM 2025. GORAG was published at WWW 2026, AGRAG at ICDE 2026, and MGRAG in Volume 19 of the Proceedings of the VLDB Endowment. Table 1.2 records their roles in the thesis. Each research chapter begins with its own publication statement.

Chapter 2 establishes the shared concepts and reviews knowledge-enhanced understanding, graph-based retrieval, and multimodal context processing. It includes the hyper-relational study to examine structural information preservation. Chapter 3 studies representation learning with an existing knowledge graph. Chapter 4 develops task-related graph retrieval without a predefined knowledge graph or model fine-tuning. Chapters 5 and 6 extend this direction to connected textual evidence and query-aware multimodal matching, respectively.

Chapter 7 compares the points at which structure enters these pipelines. It examines static and query-dependent evidence, the relationship between quality and computational cost, and the conditions affecting interpretation. Chapter 8 returns to the research questions and identifies directions for future work.

CHAPTER 2

BACKGROUND AND RELATED WORK

This chapter reviews knowledge structures for language-model understanding and reasoning. We first introduce how explicit relationships organize external information and how this information enters a language model. We then review knowledge-enhanced table annotation, few-shot adaptation, and graph-based retrieval. Multimodal retrieval and context processing extend this direction to text and images. Section 2.7 presents our study of information preservation in hyper-relational knowledge graphs.

2.1 Knowledge Structures for Language Models

A knowledge graph represents entities and their relationships explicitly [23]. A conventional relational fact is a triple (s, r, o), where s is a subject entity, r a relation, and o an object entity. These triples form a directed graph with entities as nodes and relations as labeled edges. For example, an occupation relation can connect a person to a profession, supplying information absent from a table that contains only names. Resources such as Wikidata make these statements accessible beyond the document in which an entity is mentioned [24].

The relationships needed by a task can also be constructed from its own data. HippoRAG extracts relational triples from passages to build a graph for retrieval [5]. A classification graph can instead connect keywords to labels according to their associations in task examples, as in GORAG [13]. The meaning of an edge therefore depends on how it is constructed. An extracted factual relation describes the source content, while a keyword-label edge records an association used for classification. AGRAG and MGRAG further link extracted facts to source chunks, allowing retrieval to return both structured relationships and the original text or images [15, 19].

Accessing an existing graph requires identifying which nodes correspond to the input. Entity linking maps a mention in a table cell or passage to an indexed entity. Since a name can refer to several entities, its surrounding context helps disambiguate the candidates. BLINK retrieves candidate entities using dense representations of mention contexts and entity descriptions, then reranks them with a cross-encoder [25]. KGLink uses connections between table columns to filter linked candidates [9]. When a graph is built from source documents, entity resolution also

determines whether mentions in different passages share a node. AGRAG adds synonym edges between similar entity strings to connect expressions indexed as separate nodes [15].

Once the input is mapped to the graph, neighborhoods, paths, and subgraphs provide different ways to select information. A neighborhood supplies facts around a linked entity. A path can connect several facts through intermediate entities, while a subgraph can collect multiple related paths. Think-on-Graph uses a language model to explore entities and relations iteratively and select reasoning paths from an existing KG [26]. For tasks requiring several supporting facts, the retrieval decision must account for their connections as well as their individual relevance. Source links provide access to details that remain in the original passages after relation extraction [15].

Selected knowledge can enter a language model through learned representations. Transformer attention computes contextual token representations, which BERT learns through bidirectional pre-training [27, 28]. KnowBERT retrieves entity embeddings through an integrated entity linker and combines them with contextual word representations through word-to-entity attention [29]. KEPLER jointly optimizes knowledge-embedding and language-modeling objectives using encoded entity descriptions [30]. These approaches incorporate knowledge during representation learning. Graph structure can also enter through a learned graph encoder and projection module, as in Graph Neural Prompting [31].

Knowledge can also be expressed as additional input text. K-BERT injects KG triples into sentences and controls their interaction through soft positions and an attention visibility matrix [32]. In a prompting pipeline, selected graph facts become context for a generative language model, as studied by KnowGPT [8]. The graph used for selection can therefore differ from the input received by the model. A graph may index entities and relations, while the model reads their descriptions, serialized triples, or associated source passages. This conversion determines which relationships and source details are available for prediction.

How knowledge enters the model also affects task adaptation. Supervised fine-tuning updates model parameters using labeled examples. In-context learning supplies task instructions or examples in the input without gradient updates for that task [11]. Retrieval selects external evidence for the current input [33]. An index can incorporate new documents, labels, or associations, changing the available context while the prediction model remains fixed. GORAG uses supplied labels to initialize its graph and subsequent predictions to associate newly observed keywords with classes [13]. Its updates change the retrieval source used by later queries.

For table annotation, graph coverage and differences between graph descriptions and target labels affect how retrieved knowledge can be used [9]. The next section reviews this setting.

2.2 Knowledge-enhanced Table Understanding

Column-type annotation assigns semantic labels to table columns. Similar values can require different labels under different annotation vocabularies. A column of people may be labeled Name or assigned an occupation category. Surrounding columns and external facts can help determine the intended type.

WebTables demonstrated that relational tables extracted from the Web form a large collection of small databases and that corpus-wide schema statistics support table search and schema applications [34]. This setting motivates methods that recover semantics from cell values, surrounding columns, and other tables when a target table has limited metadata.

Sherlock trains a neural type detector from statistical properties, character distributions, word embeddings, and paragraph vectors [35]. Sato adds topic features and structured prediction to model which column types occur together [36]. The two approaches differ in whether table context contributes to the prediction.

Doduo serializes multiple columns for a pre-trained language model and learns column-type and column-property annotations from their contextual representations [37]. Self-attention lets a column representation depend on values in other columns, subject to the input-length limit.

Table pre-training can encode additional structural information. TURL learns contextual representations of relational tables through self-supervised pre-training [1]. TAPAS augments a language model with table-aware inputs and learns to answer questions through cell selection and aggregation [38]. TableFormer introduces learned attention biases for table structure and table–text alignment while addressing sensitivity to row and column order [39]. These designs show how serialization, structural biases, and pre-training tasks affect the table information available to a predictor.

Knowledge-based methods obtain evidence through a different route. HNN combines learned table features with knowledge-base lookup and inter-column semantics [40]. A knowledge resource can identify a relation between mentioned objects that the table's strings do not explain. RECA uses related tables as an additional contextual source [3]. These approaches extend the information available within the target table through external semantic facts or evidence from other tables. Their retrieval units and preparation costs differ accordingly.

ColNet combines knowledge-base-derived supervision with convolutional representations of cell and column semantics [41]. TCN models contextual information within and across Web tables through a table convolutional network [42]. Together with RECA, these approaches illustrate different uses of external context, from knowledge-base concepts to relationships among

tables. KGLink instead selects graph-derived descriptions for an encoder that learns the target annotation vocabulary.

KGLink retrieves candidate entities for cell mentions and filters them through connections between columns [9]. Candidate-type text and a separate KG feature sequence enrich the column representation. A contextual type-representation objective learns the correspondence between graph descriptions and annotation labels. Chapter 3 presents the method and experiments.

These methods obtain context from different sources. Statistical features summarize column contents, contextual encoders model values jointly, RECA retrieves related tables, and knowledge-based methods retrieve external facts. Their coverage determines which ambiguities they can resolve.

2.3 Few-shot Adaptation and Evolving Label Sets

Few-shot learning studies prediction from limited labeled examples, often using experience from related tasks to support adaptation [43]. The support set provides examples of the current classes, while the query set tests the resulting decisions. A method must specify which information is learned beforehand and which information becomes available for a new task.

Metric-based methods organize support information in a learned representation space. Matching Networks use an attention-based prediction over support examples, while Prototypical Networks compare query representations with class prototypes [44, 45]. Relation Networks learn a comparison function between examples [46]. Model-agnostic meta-learning instead learns an initialization that supports adaptation through a small number of gradient updates [12]. The learned metric, comparison function, and initialization provide different mechanisms for using a small support set.

An expanding label set adds the need to retain earlier categories. Dynamic Few-Shot Visual Learning studies recognizing novel categories while preserving base categories, and Attention Attractor Networks regularize the addition of new classifier weights in incremental few-shot learning [47, 48]. These visual-learning settings establish related adaptation problems; incremental few-shot text classification introduces successive rounds of text labels and limited examples [10].

Large language models introduce another adaptation interface. GPT-3 demonstrates prediction from task descriptions and examples supplied in the prompt, without gradient updates for the demonstrated task [11]. GORAG uses a language model for this prediction role, but maintains keyword–label associations in an external graph. Its online updates change which label

evidence is retrieved for subsequent queries.

2.4 Retrieval-augmented Generation

Retrieval-augmented generation conditions a generative model on information selected from an external collection [33]. A retrieval stage maps the query to indexed evidence, and a generation stage uses the evidence to produce an answer. The external collection supplies evidence at prediction time. Updating the collection or its retrieval rule changes the generator's context even when its parameters stay fixed.

Retrieval can also participate in model training. REALM trains a language model with a latent knowledge retriever, RETRO conditions generation on chunks retrieved from a large text collection, and Atlas studies few-shot knowledge-intensive tasks with a pre-trained retrieval-augmented model [49–51]. These systems make the retrieval source part of the model's information supply, while differing in their training objectives and task adaptation.

A passage-based RAG pipeline divides a corpus into chunks, indexes them, and concatenates the retrieved chunks with the query. Small chunks can separate a fact from its explanation. Large chunks preserve more context but consume more input tokens. Overlap retains information across chunk boundaries at the cost of repeating some text.

Sparse retrieval scores lexical evidence. BM25 combines within-document term frequency with corpus frequency and document-length normalization [52]. It distinguishes a query term that occurs widely from one that is concentrated in fewer indexed items. Dense retrieval instead represents queries and passages as vectors and scores their correspondence in a learned space. Dense Passage Retrieval uses a dual-encoder framework with separate question and passage encoders [6]. The two representations provide different signals, so hybrid retrieval can select context using both lexical overlap and embedding similarity.

Retrieval need not compress each passage into a single vector. ColBERT independently encodes query and passage tokens, then uses a late-interaction operation to score fine-grained matches [53]. HyDE addresses retrieval without relevance labels by generating a hypothetical document and encoding it as a search representation [54]. The generated document guides retrieval; the retrieved source documents supply the evidence used downstream. These methods change the representation used to find context without requiring a graph index.

For a query q and indexed evidence item e, write the retrieval score abstractly as $s(q, e)$. A conventional fixed-budget result contains the highest-scoring k items. Evidence is scored at the chosen item granularity, and the retrieval cutoff controls how many items are returned. A

graph-based system can alter either assumption. It can score relationships as well as items, or determine the context by a structural condition such as connecting query anchors. The selected nodes and relationships are then serialized as context for the language model.

Retrieval evaluation measures whether the selected context contains supporting evidence. Generation evaluation measures the resulting answer through correctness, reference overlap, or model-based assessment. An answer improvement can arise from selecting better evidence, organizing the context, or using it more effectively.

Question-answering benchmarks expose different evidence requirements. Natural Questions contains naturally occurring search queries with long and short answer annotations on Wikipedia pages, while TriviaQA pairs trivia questions with independently gathered evidence documents [55, 56]. MuSiQue constructs multihop questions by composing connected single-hop questions so that one reasoning step depends on information from another [57]. These collections help distinguish locating supporting text from reading or generating an answer from it.

Complex questions can require retrieval to change as reasoning proceeds. IRCoT interleaves retrieval with generated intermediate reasoning steps, using those steps to search for further evidence [7]. This iterative approach obtains additional context through successive retrieval decisions. Graph-based retrieval instead makes extracted relationships available within the index, where several supporting facts can be selected through connected paths or propagation.

An evolving collection adds a maintenance problem. Newly available documents, examples, or labels must enter the index before retrieval can use them. Updating stored evidence can involve new embeddings, new graph nodes, changed edge weights, or derived summaries. The information supplied by an update also matters. A labeled support example carries a provided class, while a query processed online may carry only a model prediction. GORAG makes this distinction part of its dynamic classification protocol [13]. Its index expands as observations arrive, and retrieval uses the graph state available at the current prediction.

Retrieving more passages can increase both answer coverage and model-processing cost. The retrieval budget therefore affects the information available to the generator even when the corpus and model remain fixed.

A larger context window does not ensure that all supplied evidence is used equally well. Lost in the Middle shows that answer performance can depend strongly on where relevant information occurs in a long input [58]. Evidence selection and organization therefore remain relevant even when the model can accept additional passages.

The indexed object can differ from the retrieved context. A document can contribute several

graph facts, a traversal can return candidate labels, and a prompt can combine selected relationships with source passages. Graph retrieval makes these connections available during evidence selection.

2.5 Graph-based Evidence Retrieval

Graph indexes connect entities, passages, labels, and other evidence items through extracted facts, shared mentions, or semantic similarity. Retrieval selects the nodes and connections relevant to the current query.

GraphRAG builds an entity graph and community summaries for query-focused summarization [14]. Community summaries provide a retrieval unit above individual passages and support questions about broader corpus themes.

HippoRAG combines language-model extraction, a knowledge graph, and personalized PageRank [5]. Query entities seed propagation through the graph, producing a ranking of related evidence. This ranking does not require all selected nodes to belong to one terminal-connecting structure.

LightRAG combines graph and vector representations, low- and high-level retrieval, and incremental index updates [59]. Its retrieval modes select specific entity information and broader relational or conceptual information.

KnowGPT studies knowledge-graph-based prompting by extracting useful knowledge and constructing context-aware prompts [8]. It makes graph knowledge available through the prompt supplied to a language model. This connects the retrieval decision to how the selected information is presented, complementing methods that rank nodes, return source passages, or summarize graph communities.

Table 2.1 compares the evidence organization of these approaches with passage retrieval. It identifies the evidence object made available to each retrieval computation.

Personalized propagation ranks evidence by influence from query anchors. Terminal-connecting retrieval instead selects paths that include a specified set of nodes, admitting intermediate nodes when needed. The former locates relevant graph regions; the latter constructs connections among required evidence items.

The retrieved graph must be converted into model input. Entity retrieval requires finding associated source passages. Path retrieval requires serializing entities and relations. Community retrieval can return derived summaries. These choices determine which original details and relationships remain available to generation.

Table 2.1: Representative retrieval organizations and the evidence structures they expose. Each row refers to the cited method's design.

Method	Indexed representation	Query-related retrieval operation
DPR [6]	Dense passage representations	Question–passage similarity scoring
GraphRAG [14]	Entity graph and community summaries	Use of graph-derived summaries for query-focused summarization
HippoRAG [5]	Knowledge graph linked to passages	Personalized PageRank from query-related entities
LightRAG [59]	Graph and vector representations	Dual-level retrieval of specific and broader information

Our research moves from retrieving candidate classes to retrieving facts that support an answer. GORAG constructs a task graph and retrieves labels by connecting keywords extracted from the query [13]. AGRAG selects connected textual evidence while considering node influence and edge cost [15]. MGRAG constructs query-related multimodal graphs and matches query relations to semantic paths [19]. A set of highly ranked nodes need not contain paths connecting the requested facts, while intermediate nodes can supply useful relationships even when their direct query similarity is lower.

Passage nodes link extracted relations to their original wording. Combining graph serialization with source chunks gives the generator both connected facts and surrounding context. AGRAG uses this combination for text, and MGRAG retains source chunks containing the original images.

2.5.1 Reasoning structures and source evidence

Explicit structure also appears in generated reasoning. Chain-of-thought prompting supplies intermediate reasoning demonstrations, and Zero-shot-CoT uses a reasoning instruction without task-specific demonstrations [60, 61]. Plan-and-Solve prompting first develops a plan and then carries out its subtasks [62]. These approaches organize how a model produces intermediate steps.

Tree of Thoughts searches over alternative intermediate thoughts, while Graph of Thoughts allows dependencies and combinations among generated information units [63, 64]. Their nodes represent model-generated reasoning states. In an evidence graph, nodes and edges instead describe information extracted or indexed from task data. Combining a reasoning structure with retrieved facts therefore requires tracking the source of each information item.

The NLGraph study evaluates graph problems expressed in natural language and finds that language-model performance depends on problem complexity and graph presentation [65]. This

motivates specifying which structural operations are performed by graph algorithms and which decisions are left to a language model. In our retrieval pipelines, graph construction and selection define the evidence passed to the final predictor.

2.6 Multimodal Retrieval and Context Serving

Multimodal question answering requires evidence represented in more than one modality. MultiModalQA includes questions over text, tables, and images and studies combinations of evidence across them [4]. The evidence needed for a question can occur in different source objects. A relationship stated in text may identify the image whose visual content contains the answer. Retrieval must therefore consider both the relevance of individual objects and the relationship connecting them to the question.

The VQA task asks for an answer to a natural-language question about an image, making visual relevance depend on the question [17]. OK-VQA adds questions that require knowledge beyond the image content [66]. MultiModalQA extends the evidence requirement to reasoning across source modalities. The distinction matters for retrieval because a visually informative image can still lack the particular detail or relationship requested by the question.

Vision–language pre-training supplies several ways to align the modalities. ViLBERT uses interacting visual and textual streams, while BLIP combines understanding and generation objectives with caption bootstrapping [67, 68]. BLIP-2 connects frozen image and language models through a trainable querying component [69]. These architectures determine how visual content becomes accessible to language-based prediction.

Flamingo supports few-shot prediction with interleaved visual and textual inputs [70]. LLaVA studies visual instruction tuning, and InstructBLIP introduces instruction-aware visual feature extraction [2, 18]. Such models can condition their processing on a task instruction, but a retrieval pipeline must additionally identify which source objects contain the supporting evidence. MGRAG uses query-aware graph construction to organize relationships among those objects.

Images can be ranked through captions or visual embeddings. Captions expose selected visual content in a text embedding space, while the original images preserve details for multimodal generation. MGRAG ranks images through captions and supplies selected original chunks to the answering model [19].

MRAMG-Bench studies retrieval-augmented generation in which answers can combine text and images [71]. Its multimodal output requirement adds another distinction between finding

supporting evidence and presenting the response. The MGRAG evaluation uses six MRAMG subsets together with MMQA. Chapter 6 specifies the retrieval and answer-assessment protocol used for these datasets.

Once evidence is retrieved, the system must process it as model context. In an autoregressive Transformer, the prefill stage computes representations for the supplied input, and decoding generates answer tokens. Key and value tensors from processed input positions can be cached for later attention computation [27, 72]. Reusing these tensors concerns the computation associated with selected evidence. It is a separate decision from choosing which evidence belongs in the context.

Attention computation, cache retention, and cache recomputation address different costs. FlashAttention reduces memory traffic through an IO-aware implementation of exact attention [73]. H2O retains a combination of recent and heavily attended tokens, while Scissorhands exploits persistent token importance to compress the KV cache [20, 21]. These retention policies decide which states remain available; selective recomputation decides which available states to refresh when the context changes.

CacheGen addresses the storage and transfer of context caches through compression and streaming [74]. The relevant resource includes the cost of fetching cached tensors as well as computing context representations. CacheBlend studies combining precomputed caches for several chunks by selectively recomputing part of their key and value states [72]. Its motivation includes interactions between chunks that were not present when their caches were independently computed. These methods place context serving within the evidence pipeline, where retrieval order and the assembled input can affect the computation required for generation.

MGRAG supplements attention-based recomputation scores with features of the retrieved graph [19]. The graph therefore guides both context selection and the computation reused when that context is processed.

2.7 Structural Information in Hyper-relational Knowledge Graphs

This section presents our work *Understanding the Embedding Models on Hyper-relational Knowledge Graph*, published at CIKM 2025 [22]. It examines how graph representations and embedding architectures preserve information used for link prediction. Transforming source information into a graph determines which relationships a downstream model can access. The

study analyzes this representation problem through fact decomposition, qualifier capture, and graph reception.

2.7.1 Facts, Qualifiers, and Decomposition

A hyper-relational fact extends a main triple with a set of qualifier pairs,

$$F = (s, r, o, \mathcal{Q}), \qquad \mathcal{Q} = \{(qr_i, qe_i)\}_{i=1}^{n}. \tag{2.1}$$

Here F is the fact, (s, r, o) is its subject–relation–object main triple, and $\mathcal{Q}$ contains n qualifier pairs. In pair i, qr_i is a qualifier relation and qe_i its associated entity. Qualifiers attach information to a particular statement. For example, degree and major can distinguish two education statements about the same person. Their association with the intended main triple is part of the information represented by the fact. StarE retains the semantic roles of qualifiers and triples in message passing and introduces WD50K for hyper-relational link prediction [75].

Related work represents facts with more than two participating entities in several ways. NaLP models the relatedness among role–value pairs, GETD extends tensor decomposition to n-ary relational knowledge bases, and RAM learns role-aware representations [76–78]. These representations expose the roles occupied by entities within a fact.

NeuInfer represents a primary triple together with auxiliary attribute–value descriptions and studies inference from complete or partial facts [79]. HAHE combines global hypergraph structure with local sequential representations of hyper-relational facts [80]. These designs motivate examining both information within a fact and information received from the wider graph, the two aspects studied below.

The CIKM study asks how much of an HKGE model's performance comes from its base KGE architecture and how much comes from specialized qualifier processing. It evaluates a data-level alternative to modifying the model. Three decomposition methods convert a hyper-relational fact into ordinary triples while retaining different information. Prune keeps only the main triple. Direct adds triples connecting the subject to qualifier entities through their qualifier relations. Hyper adds further relations whose names combine the main relation with each qualifier relation and connect the qualifier entity to the main object.

Figure 2.1 reproduces the original comparison, including database reification. The study distinguishes main-triple information, intra-qualifier information, and inter-qualifier information. A decomposition can retain qualifier entities and relation labels while changing how their association with the main fact is expressed. Database reification introduces auxiliary entities and

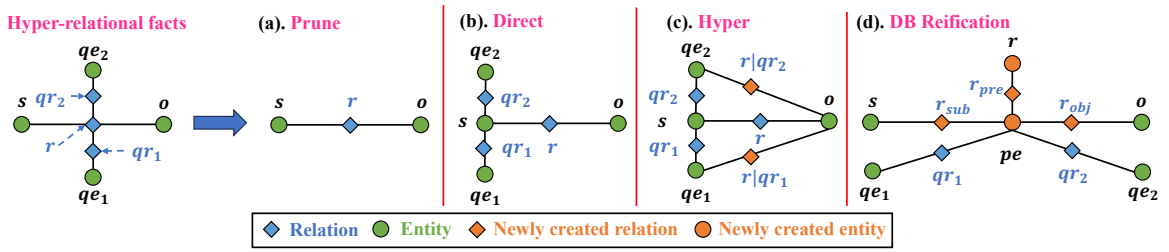


Figure 2.1: Original comparison of hyper-relational decomposition methods. Prune retains the main triple; Direct adds subject–qualifier links; Hyper adds combined relations; database reification introduces auxiliary objects. Reproduced from Fig. 1 of the CIKM study [22].

relations, also changing the original main-triple topology. These differences determine which neighborhoods and paths an ordinary KGE model receives after conversion.

Qualifier associations matter when a subject participates in several facts with the same relation. For example, a degree qualifies a particular university attendance statement. Separate subject–degree and subject–university triples retain the entities but change how that association is represented. Hyper adds connections intended to preserve these associations. A qualifier integrator instead receives the main fact and its qualifier pairs together.

We compare conventional KGE models under these transformations with specialized HKGE models. Some GNN-based configurations obtain comparable scores on Cleaned JF17k and WD50k. Adding qualifier-derived edges changes the graph paths available to an encoder, so information preservation depends on both the transformed data and the model that processes it.

2.7.2 Qualifier Capture and Graph Reception

The analysis separates qualifier capture from graph reception. Qualifier capture concerns how the model represents qualifier pairs and their association with the main statement. Graph reception concerns how it receives information from the wider graph, including dependencies beyond one fact. Models operating on a fact in isolation have a different receptive field from graph encoders that propagate information across several hops. These properties are visible in the operations used to construct representations.

CompGCN jointly models entity and relation representations in a multi-relational graph [81]. NBFNet aggregates path information for link prediction [82]. Their graph computations offer different ways to receive neighborhood and path information after decomposition. The CIKM analysis also identifies an information-compression issue when qualifier information is aggregated into fixed-size main-triple representations. Graph reception supplies wider context, while qualifier integration preserves information within each fact. The study evaluates designs that retain both sources for prediction.

Table 2.2: Projection of the CIKM study's Table 6 onto Cleaned JF17k and WD50k. All model rows and the original metric precision are retained; the original bold and underline markings indicate best and second-best values.

	Cleaned JF17k			WD50k		
Model	MRR	H@1	H@10	MRR	H@1	H@10
StarE	0.381	0.284	0.577	0.346	0.268	0.494
QUAD	0.402	0.305	0.602	0.349	0.275	0.487
HAHE	0.405	0.310	0.585	0.360	0.288	0.496
HyperFormer	<u>0.425</u>	<u>0.328</u>	0.613	<u>0.372</u>	<u>0.293</u>	0.522
NBFNet + Prune	0.415	0.313	**0.630**	0.360	0.271	**0.530**
CompGCN + Hyper	0.389	0.294	0.590	0.350	0.277	0.488
FormerGNN	**0.432**	**0.335**	<u>0.624</u>	**0.377**	**0.299**	<u>0.527</u>

FormerGNN combines a Transformer-based qualifier integrator operating on the original hyper-relational facts with a graph encoder operating on the pruned main-triple graph. The graph encoder generates a topology representation for the subject. The model concatenates this representation with representations of the main triple and qualifier pairs and passes them to a Transformer decoder. This organization supplies qualifier and graph information jointly to the prediction stage. In the reported configuration, HyperFormer provides the qualifier integrator and NBFNet provides the graph encoder [82, 83].

2.7.3 Representative Published Evidence

Table 2.2 reproduces the Cleaned JF17k and WD50k columns of the study's Table 6. MRR measures mean reciprocal rank, and H@1 and H@10 measure whether the target entity appears within the corresponding rank cutoff. These are link-prediction measures, distinct from the annotation and answer metrics of the later chapters.

FormerGNN obtains MRR 0.432 on Cleaned JF17k and 0.377 on WD50k, compared with HyperFormer's 0.425 and 0.372. NBFNet + Prune has the highest H@10 in both selected datasets, at 0.630 and 0.530. The experiment uses a single NVIDIA A100-40GB GPU. Parameters follow the original repositories or are selected by grid search when dataset-specific settings are unavailable. For comparisons involving CompGCN, the study replaces its decoder with a Transformer to align it with the corresponding HKGE configurations.

The complete comparison also includes Wikipeople and FBAUTO, with different dataset characteristics and metric rankings. The study shows how changing the organization of graph facts changes the information available to a prediction model.

2.8 Research Questions Emerging from the Background

Knowledge-enhanced annotation motivates our first question about supplying missing context and learning its correspondence to labels. Dependence on existing graphs and supervised training then motivates graph construction from task data and adaptation through retrieval. Extending retrieval beyond classification introduces questions about connected evidence and answer coverage for text, and query-aware construction and semantic matching for multimodal sources. The following chapters investigate these questions.

CHAPTER 3

KNOWLEDGE-ENRICHED TABLE UNDERSTANDING WITH KGLINK

This chapter is based on the published paper *KGLink: A Column Type Annotation Method that Combines Knowledge Graph and Pre-Trained Language Model* [9]. We begin our study of structural augmentation by investigating how an existing knowledge graph can supply information missing from a table. We propose KGLink to select candidate types and graph-derived feature sequences, then learn their correspondence to annotation labels through a contextual representation objective.

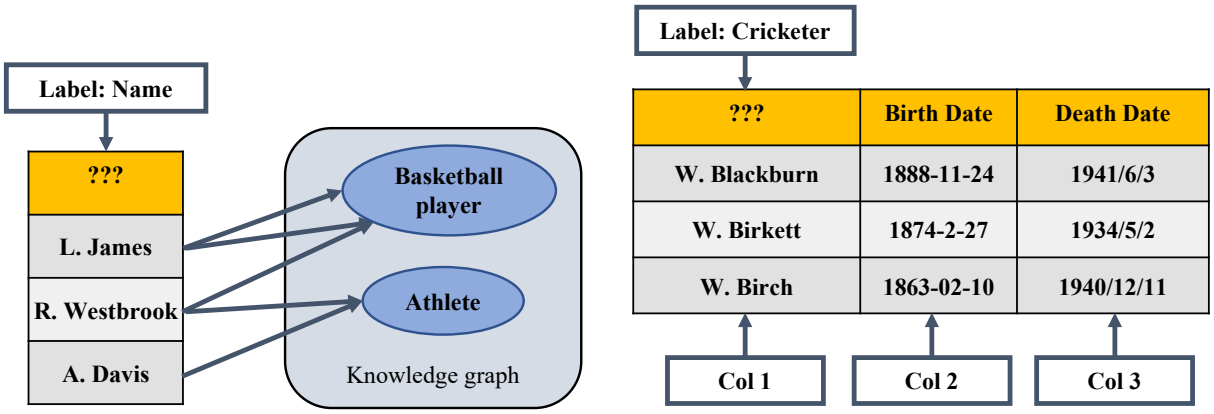
3.1 Motivation and Task

A table may contain too little context to determine a column type. For example, names of cricket players accompanied by birth and death dates do not reveal their occupation. Linking those names to a knowledge graph can supply occupation and sport relations that are absent from the table.

External knowledge introduces a second difficulty. The type expected by an annotation dataset need not coincide with a type attached to an entity in the graph. A column of basketball players can plausibly be described as *Basketball player*, *Athlete*, or *Name*. An annotation benchmark may use only the last of these labels. A factually relevant graph type alone may therefore leave the required output label ambiguous. The model must learn which description the task expects. KGLink refers to this mismatch as the type granularity gap. The gap includes cases where the desired dataset label does not occupy a convenient ancestor position in the retrieved graph's type hierarchy.

To address these two difficulties, KGLink combines graph processing with supervised representation learning. Graph processing selects relevant external descriptions, while a supervised classifier learns the dataset labels. An auxiliary task trains a contextual representation of the target type so that the classifier can use graph descriptions even when their wording differs from the annotation vocabulary.

We evaluate single-label column annotation on SemTab and modified VizNet. The encoder



(a) Type granularity (b) Missing context

Figure 3.1: Two annotation difficulties in the original KGLink examples. A graph description such as Athlete can differ from the target label Name, while dates beside a person's name provide little occupation information. Original subfigures reproduced from KGLink [9], Fig. 2.

predicts a dataset label from the cells, table context, and selected graph descriptions.

3.2 Problem Formulation and Framework

Let T be a table with row set R and column set C. A cell mention m_c^r is the content of row r and column c. The task assigns a label l_c from the dataset's type set D to each target column. All columns in the same table can contribute context to that decision. The external knowledge graph supplies entity labels and their one-hop relations. During training, column labels supervise both classification and the auxiliary representation task. During evaluation, those labels are absent from the model input.

KGLink has a graph-processing stage followed by a language-model stage. The graph stage retrieves candidate entities for cell mentions, filters these entities using connections between columns, ranks rows by their linkage quality, and derives candidate types from the remaining entities' neighborhoods. It also constructs a feature sequence for each target column. The language-model stage serializes the processed table, encodes the table and feature sequences, combines the resulting representations, and predicts the column labels. A second training input contains the true type labels and provides contextual supervision for representations generated from masked type positions.

Figure 3.2 shows the two stages and their training objectives.

Candidate types summarize descriptions supported by linked cells across a column. The feature sequence retains an entity and its neighboring relations from a selected cell. Both enter the learned classifier, which predicts a label from the dataset vocabulary.

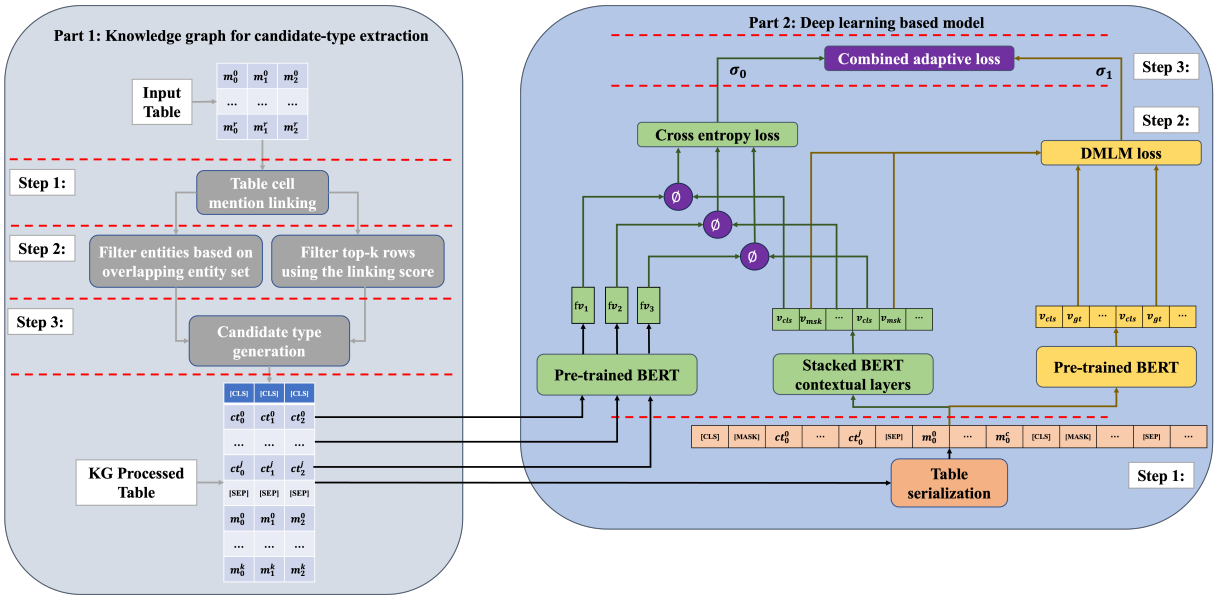


Figure 3.2: KGLink's graph-processing and language-model components. Candidate-type text, column feature vectors, and a contextual type-representation objective connect external knowledge to annotation. Original image reproduced from KGLink [9], Fig. 3.

Graph descriptions and cell values share the encoder input budget. We therefore select rows before serialization and split wide tables by columns under the input limits.

3.3 KG-derived Candidate Types

3.3.1 Cell Mention Linking

The first step retrieves a candidate entity set $E_{m_c^r}$ for each suitable cell mention m_c^r in row r and column c. KGLink uses BM25 to score lexical matches between the mention's words and indexed entity labels [52]. The implementation indexes Wikidata with Elasticsearch. The score incorporates query-term frequency, inverse document frequency, and label-length normalization. Multiple entities can match the same mention, so the retrieval output is a candidate set rather than a single disambiguated entity.

Short strings are often ambiguous. An album title can match an ordinary word, and a name can refer to several people. We use relationships between cells in the same row to filter these lexical candidates.

Named-entity detection identifies numeric and date mentions, which receive a linking score of zero. Failed lookups also receive zero. The original cell values remain in the encoder input, allowing classification when KG evidence is missing.

We retrieve at most ten entities per cell mention, then apply neighborhood filtering and row

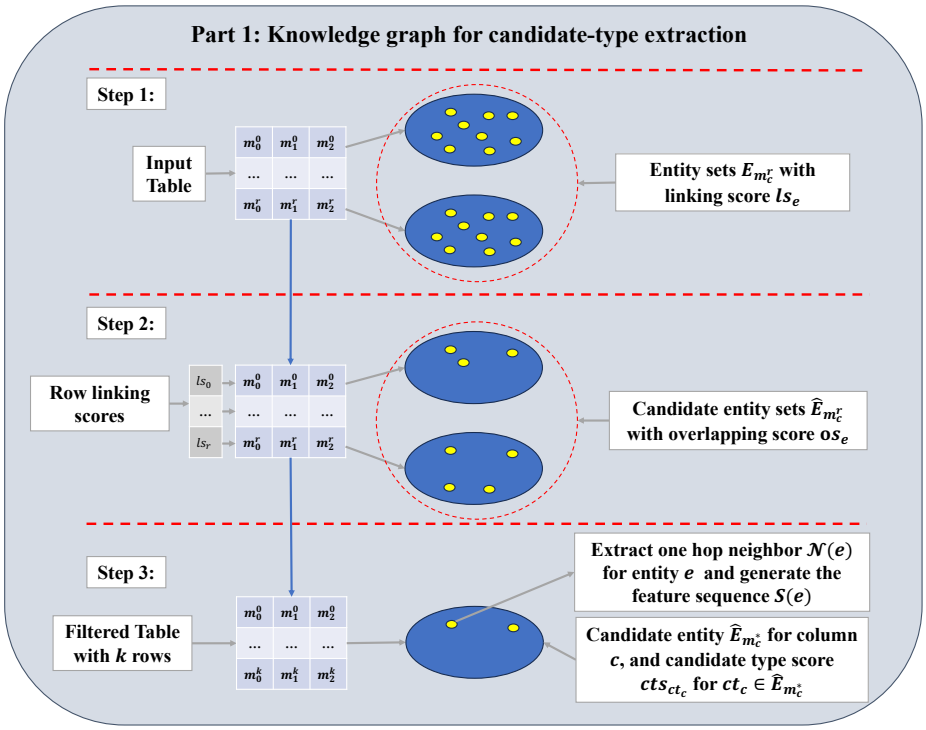


Figure 3.3: The original KG-processing pipeline. Cell linking is followed by entity filtering, row selection, and candidate-type generation. Reproduced from KGLink [9], Fig. 4.

ranking before constructing the encoder inputs.

3.3.2 Cross-column Neighborhood Filtering

Cells in the same row describe related objects. An album title beside a musician's name supplies a useful relation that either mention lacks in isolation. KGLink uses this relation by checking whether a candidate entity for one cell occurs in the one-hop neighborhood of candidates for another cell in the same row. Let $\mathcal{N}(E)$ denote the one-hop neighbors of a candidate entity set. The filtered set for a cell is

$$\widehat{E}_{m_{c_1}^r} = \bigcup_{c_2 \in C \setminus \{c_1\}} \left(E_{m_{c_1}^r} \cap \mathcal{N}(E_{m_{c_2}^r}) \right). \tag{3.1}$$

Here $\widehat{E}_{m_{c_1}^r}$ is the filtered candidate set for the mention in row r and target column c_1; $E_{m_{c_2}^r}$ contains the candidates for another column c_2, and $\mathcal{N}$ returns their one-hop neighbors. The union allows support from any other column. The intersection retains entities retrieved for the target mention that also have the required neighborhood support.

Figure 3.4 uses *Rust* and *Peter Steele*. A graph connection between an album candidate and the musician candidate supports their interpretation as the entities intended by the row. The role of the edge is to constrain which external facts are relevant to the table. KGLink subse-

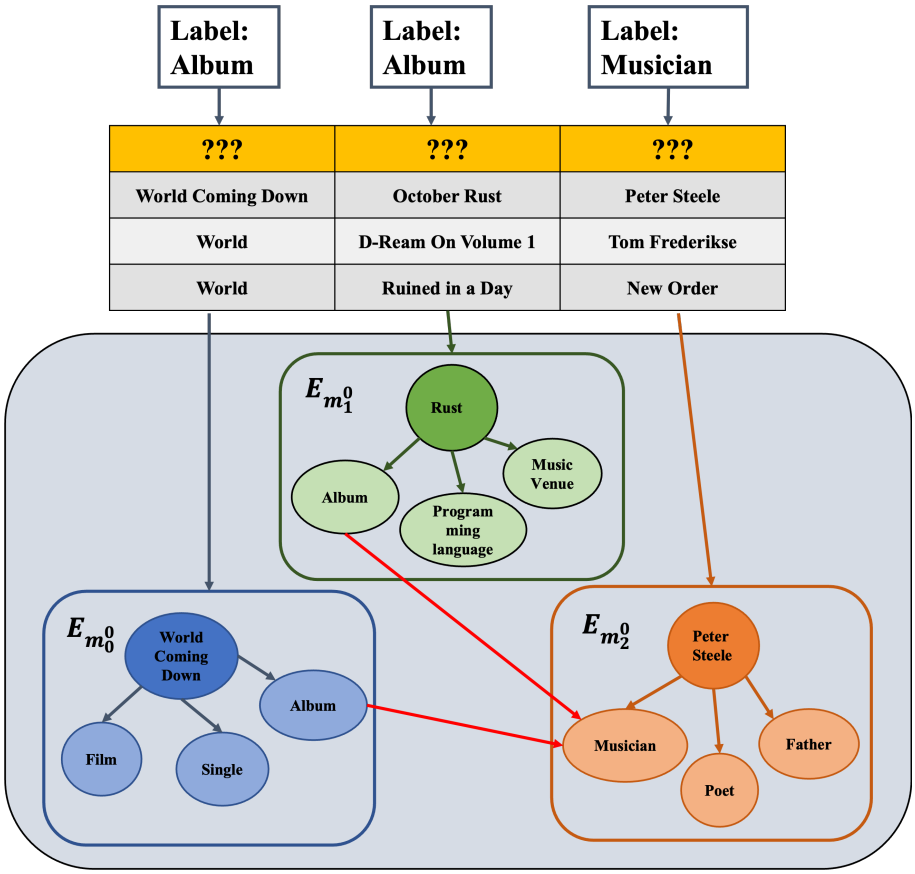


Figure 3.4: Cross-column evidence in the original album-and-musician example. Graph links connect candidate interpretations of cell mentions in the same row. Original image reproduced from KGLink [9], Fig. 5.

quently associates each retained entity with an overlapping score that counts its neighborhood support from other columns. Entities with more such support contribute more to candidate-type selection.

Filtering depends on both the initial entity candidates and the relations present in the knowledge graph. A missing relation can discard a useful candidate, while an incorrect candidate can survive through a matching contextual link. The encoder also receives the original table values and the separate feature sequence.

Filtering uses relationships between cells in the same row. Candidate-type generation then aggregates recurring descriptions across rows in a column.

3.3.3 Row Linking Quality and Input Selection

KGLink derives a cell score from the best-scoring retained entity. For a nonempty filtered candidate set, the score is

$$ls_{m_c^r} = \max_{e \in \hat{E}_{m_c^r}} ls_e, \tag{3.2}$$

where $ls_{m_c^r}$ is the score of the mention in row r and column c, $\widehat{E}_{m_c^r}$ is its filtered entity set, and ls_e is the BM25 linking score of candidate entity e. Unlinked cells use zero as described in the evaluation protocol. A row's score sums the scores of its cells,

$$ls_r = \sum_{c \in C} ls_{m_c^r}. \tag{3.3}$$

Here ls_r is the linking score of row r, C is the column set, and $ls_{m_c^r}$ is the score of its cell in column c. Rows are sorted by this quantity, and only the top k rows are retained for prediction.

Linkage ranking favors rows with graph-supported candidates, while taking the first k rows, with the same row budget k, follows the source order. The row-filter experiment compares these rules under the same input budget.

The highest-ranked row also supplies the selected cell used for feature-sequence generation. Row ranking therefore changes both the table values and the KG entity passed to the encoder.

3.3.4 Candidate Types and Feature Sequences

Candidate-type generation searches beyond an entity's explicit type attribute. A person's graph entry may be typed as *Human*, while *Musician* appears as an occupation neighbor. The latter can be more informative for a table column. KGLink collects filtered entities across the target column and examines their one-hop neighborhoods. It scores a potential type by aggregating the overlapping scores of entities from other rows that connect to that type. Repeated support within the column increases the candidate's score.

A label-based filter excludes candidates assigned the named-entity schema categories `PERSON` or `DATE`. The reported implementation uses spaCy for this step. This filter distinguishes a descriptive candidate type from a particular person's name or a date encountered in the neighborhood. KGLink retains the selected candidate-type labels as text and places them before the cell values in the processed column. The reported implementation generates up to three candidate types per row from the KG entities in each column. Their later contribution depends on how the encoder interprets them in the table context.

If no candidate type is returned, padding occupies the corresponding position. Numeric columns use numeric summary information in this part of the input, while their original values remain available in the table sequence.

The feature sequence provides a second representation of the retrieved knowledge. From the filtered and ranked table, KGLink selects a cell for each target column, chooses its highest-linking entity, and serializes that entity with its one-hop predicates and neighboring entities. If

e is the selected entity, p a predicate, and o a neighboring object, the sequence consists of the entity label followed by predicate–object descriptions. Encoding this text yields a feature vector for the column. A padding sequence is used when the column has no retrieved entity.

3.4 Language Model Integration

3.4.1 Table Serialization and Column Representations

KGLink adapts the multi-column serialization used by Doduo [37]. A special `[CLS]` token begins each column's segment. The segments are concatenated into one input sequence, followed by a `[SEP]` token. The contextual representation at each column's `[CLS]` position provides that column's vector. Because the sequence contains multiple columns, self-attention can use information outside the predicted column.

For table T with column set C, write the tokenized contents of column i as $\mathrm{tok}_i^1, \ldots, \mathrm{tok}_i^{m_i}$, where m_i is its token count. The serialized sequence $S(T)$ has the form

$$\begin{aligned} S(T) = [\mathrm{CLS}]\ \mathrm{tok}_1^1 \cdots \mathrm{tok}_1^{m_1} \quad [\mathrm{CLS}]\ \mathrm{tok}_2^1 \cdots \\ [\mathrm{CLS}]\ \mathrm{tok}_{|C|}^1 \cdots \mathrm{tok}_{|C|}^{m_{|C|}} \quad [\mathrm{SEP}]. \end{aligned} \tag{3.4}$$

Each column's `[CLS]` position produces a contextual vector using the serialized columns. Candidate-type text precedes the corresponding cell values, allowing both to interact with neighboring columns through self-attention. KGLink separately encodes the feature sequence into a column-specific vector and combines it with the contextual column vector through the composition function ϕ_{comp}, shown as ϕ in Figure 3.2. The resulting representation is mapped into the dataset's classification space.

The principal experiments fine-tune BERT [28]. The model has a sequence limit of 512 tokens; the reported implementation limits each column to 64 tokens and each encoded table to eight columns. Tables with more columns are divided after candidate-type generation and are encoded separately. These limits define the context available to the classifier. In particular, retaining 25 table rows does not imply that every token in those rows survives serialization, since tokenization and the column limit also constrain the final sequence.

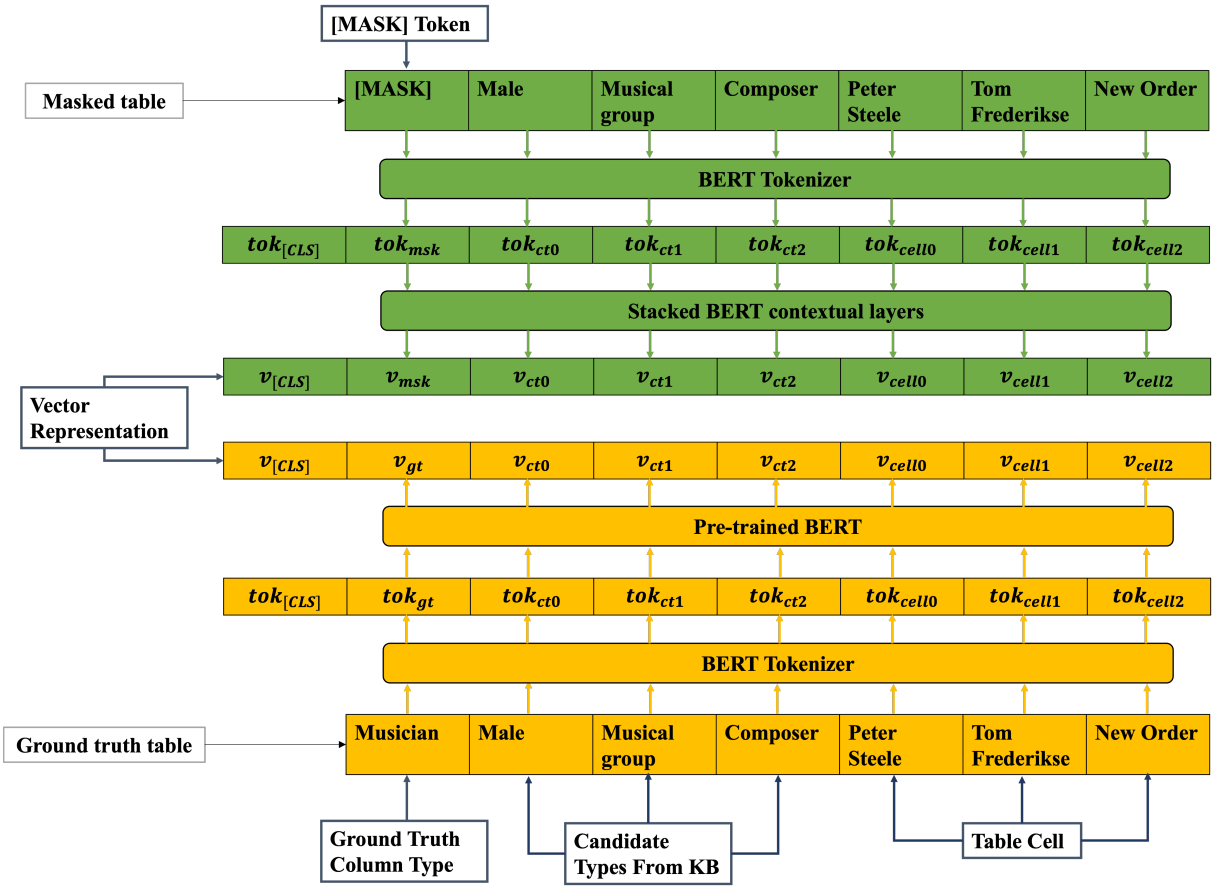


Figure 3.5: Contextual type-representation training with masked and ground-truth table inputs. The true-type table is constructed during training. Original image reproduced from KGLink [9], Fig. 6.

3.4.2 Contextual Type Representation Learning

The auxiliary task trains a type representation at a masked position. During training, KGLink constructs two versions of the processed table. The ground-truth version places the annotated type label before the candidate types in each target column. The masked version replaces that label with `[MASK]`. Both versions retain the column values and candidate-type context. Encoding the two sequences produces contextual representations for the true type and its masked counterpart.

Figure 3.5 illustrates the two inputs and their contextual outputs.

The paired inputs teach the encoder how the label relates to the table context. A label such as *Musician* is encoded together with the column values and graph-derived descriptions, rather than as an isolated word. The masked input must produce its type representation from the evidence that will be available at prediction time. Distilled masked language modeling provides the auxiliary objective [84]. The model projects contextual representations into vocabulary space with a temperature parameter, then uses the ground-truth and masked outputs in the reported

training loss.

With contextual vector H and temperature $\mathscr{T}$, vocabulary projection gives

$$Y = \mathrm{Softmax}\Big(W_o \frac{H}{\mathscr{T}}\Big), \tag{3.5}$$

where W_o projects the contextual vector H into vocabulary space, $\mathscr{T}$ is the temperature, and Y is the resulting probability vector. Denote the masked and ground-truth outputs by Y_{msk} and Y_{gt}. The main paper writes the auxiliary objective as

$$\mathscr{L}_{\mathrm{DMLM}}(Y_{\mathrm{msk}}, Y_{\mathrm{gt}}) = -\sum_{j=1}^{V} y^j_{\mathrm{msk}} \log y^j_{\mathrm{gt}}, \tag{3.6}$$

where V is the vocabulary size and y^j_{msk} and y^j_{gt} are the jth components of the masked and ground-truth output vectors, respectively. The objective uses the contextualized representations produced from the two table inputs.

Training uses both true-type and masked inputs; evaluation uses the masked input. The model thus learns from annotated types during training and predicts from table values and retrieved descriptions at inference, including for numeric columns and columns without KG candidates.

The shared encoder is trained jointly for dataset-label classification and vocabulary-space type representation. The classification head produces the final column label.

3.4.3 Classification and Adaptive Task Weighting

Let z_c be the classification logits obtained from the combined column and feature representations. For a target label l_c, the classification loss is cross-entropy,

$$\mathscr{L}_{\mathrm{CE}}(z_c, l_c) = -\log \frac{\exp\Big(z_{c,l_c}\Big)}{\sum_{j \in D} \exp\big(z_{c,j}\big)}. \tag{3.7}$$

Here $z_{c,j}$ is the logit for label j in the dataset label set D, and z_{c,l_c} is the logit for the target label l_c of column c. KGLink jointly optimizes this loss and the auxiliary representation objective using uncertainty weighting [85]. With trainable task parameters σ_0 and σ_1, the combined objective is

$$\mathscr{L}_{\mathrm{total}} = \frac{\mathscr{L}_{\mathrm{CE}}}{2\sigma_0^2} + \frac{\mathscr{L}_{\mathrm{DMLM}}}{2\sigma_1^2} + \log\big(\sigma_0 \sigma_1\big). \tag{3.8}$$

Here $\mathscr{L}_{\text{total}}$ combines classification loss $\mathscr{L}_{\text{CE}}$ and representation loss $\mathscr{L}_{\text{DMLM}}$, with trainable parameters σ_0 and σ_1, respectively. This indexing follows Figure 3.2; the source equation assigns the two indices in the opposite order. The temperature used in the representation objective is set to two in the reported experiments.

The trainable task weights balance the representation and classification losses during optimization.

3.5 Computational Structure

The graph-processing cost depends on table dimensions, candidate-set sizes, and neighborhood sizes. Linking traverses table cells. Cross-column filtering compares candidate information between columns within a row. Candidate-type scoring visits neighborhoods and aggregates support across rows. The implementation's limits on retrieved entities and serialized columns constrain these operations in the evaluated setting. The external Wikidata index is prepared with Elasticsearch and supplies lexical retrieval scores.

Let n_R and n_C be the maximum numbers of rows and columns per table, n_T the number of tables, n_E the maximum number of retrieved entities per mention, and n_N their maximum one-hop neighborhood size. In these count symbols, the original analysis reports the preprocessing expression

$$O\big(n_R n_C n_E (n_N + n_C n_E n_T)\big)\,. \tag{3.9}$$

Neighborhood aggregation and cross-column candidate filtering contribute to preprocessing. Increasing the candidate-set limit adds graph work even when the encoder input remains capped.

Encoding cost depends on the serialized input length and the encoder's hidden dimension. KGLink encodes table inputs and column feature sequences; training also includes the ground-truth and masked constructions used by the representation task. The classification head maps column vectors into the label space, while the auxiliary projection operates over the language model's vocabulary. Thus, increasing the label set affects classification, whereas increasing the vocabulary-space projection concerns the representation objective.

Reducing the row count need not shorten a sequence already truncated to its token limit. Graph preprocessing and encoder computation can therefore respond differently to the same input-selection change.

3.6 Experimental Evaluation

3.6.1 Datasets and Protocol

The evaluation uses SemTab 2019 and a modified VizNet subset [36, 86]. SemTab includes tables derived from Wikipedia, DBpedia, and the T2Dv2 Gold Standard sources. The experiment uses rounds 1, 3, and 4, with 3,048 tables, 7,587 columns, and 275 column types. Round 2 has columns with multiple ground-truth labels and is excluded from this single-label comparison. The reported averages are 69 rows and 4.5 columns per table.

The modified VizNet corpus targets column annotation rather than the full range of visualization tasks in the original repository [87]. The experiment uses its multi-column portion so that methods can exploit relations between columns. Its annotation scheme is coarser than SemTab's and includes numeric types. The evaluated portion contains 32,265 tables, 73,034 annotated columns, and 77 types.

The data are split into training, validation, and testing portions in the ratio 7:1:2 while maintaining each class's original sample proportion. Each experiment is averaged over three runs. Accuracy measures exact label prediction, and weighted F1 weights class-level performance by sample frequency.

Training uses AdamW with optimizer parameter $\epsilon = 10^{-6}$ and an initial learning rate of 3×10^{-5}. The learning rate decays linearly without warm-up. The reported maximum training lengths are 50 epochs on SemTab and 20 on VizNet, with early stopping. Batch size is 16 for both datasets; dropout is 0.1 for SemTab and 0.2 for VizNet. The implementation uses PyTorch and the Transformers library on an NVIDIA Tesla V100 GPU.

The baseline set includes MTab, TaBERT, Doduo, HNN, Sudowoodo, and RECA [3, 37, 40, 88–90]. TaBERT and Doduo use KGLink's experimental settings. HNN and RECA follow their original settings. Sudowoodo receives the full labeled training set in this comparison. VizNet labels are translated to Wikidata entities for MTab, and MTab's weighted F1 is unreported because of the difference in problem definition.

3.6.2 Main Annotation Results

Table 3.1 reports the main comparison. KGLink obtains accuracy 87.12 and weighted F1 85.78 on SemTab, and accuracy 96.28 and weighted F1 96.07 on VizNet. MTab has the highest SemTab accuracy, 89.10. KGLink has the highest reported weighted F1 on SemTab and the highest values of both reported metrics on VizNet.

Table 3.1: Column-type annotation results reported for KGLink. Scores and missing entries reproduce the original main-result table.

	SemTab		VizNet	
Model	Accuracy	Weighted F1	Accuracy	Weighted F1
MTab	**89.10**	-	38.21	-
TaBERT	72.69	71.21	94.68	94.07
Doduo	84.06	82.43	95.40	95.06
HNN	66.54	65.12	66.89	68.82
Sudowoodo	79.34	79.24	91.57	91.08
RECA	86.12	84.91	93.25	93.18
KGLink	87.12	**85.78**	**96.28**	**96.07**

Table 3.2: Reported KGLink variant scores. The abbreviations *w/o msk*, *w/o ct*, and *w/o fv* denote variants without masked representation generation, KG-derived information, and the feature vector, respectively. The table reproduces the numeric columns and variant names of the original ablation table.

	SemTab		VizNet	
Variant	Accuracy	Weighted F1	Accuracy	Weighted F1
KGLink *w/o msk*	86.14	84.54	95.95	95.67
KGLink *w/o ct*	86.27	84.56	95.83	95.48
KGLink *w/o fv*	87.02	85.68	95.98	95.70
KGLink DeBERTa	**87.24**	**85.81**	**96.98**	**96.37**
KGLink	87.12	85.78	96.28	96.07

Doduo provides a useful comparison for the language-model pathway because KGLink adapts its serialization and uses BERT in the main configuration. Doduo obtains 84.06 accuracy on SemTab and 95.40 on VizNet. KGLink improves on this baseline with graph-augmented inputs and joint training.

The dataset dependence is also visible in MTab's results. Its accuracy is 89.10 on SemTab and 38.21 on VizNet under the label translation used in the experiment. These measurements motivate separating graph knowledge from the final annotation vocabulary.

3.6.3 Representation Learning and KG Evidence

Table 3.2 compares KGLink variants. The *w/o msk* variant removes the masked representation-generation task. Its scores are lower than the BERT-based full model on both datasets. The *w/o ct* variant removes the KG-derived information, including candidate types and the feature vector, while keeping the multitask language-model pathway. Its accuracy is 86.27 on SemTab and 95.83 on VizNet. Removing KG inputs reduces accuracy on both datasets.

Table 3.3: Reported accuracy on the VizNet test subset whose tables have no extracted KG information.

Model	Numeric accuracy	Non-numeric accuracy
KGLink	**97.04**	**90.92**
HNN	44.05	18.37
TaBERT	96.57	90.27
Doduo	96.28	89.50
RECA	96.89	61.54
Sudowoodo	96.21	67.72

The *w/o fv* variant removes the KG-derived feature vector, following the original method description. It obtains 87.02 accuracy and 85.68 weighted F1 on SemTab, and 95.98 accuracy and 95.70 weighted F1 on VizNet. Candidate types provide the remaining KG input in this configuration. The full model has higher scores on both datasets.

Replacing BERT with DeBERTa produces the fourth variant [91]. It obtains 87.24/85.81 on SemTab and 96.98/96.37 on VizNet. The architecture uses the same graph-processing design with a different encoder in this experiment.

3.6.4 Annotation without Retrieved Knowledge

The original study constructs a test subset from VizNet in which the entire table has no KG linkage. This condition is stricter than requiring only the target column to lack a link. With multi-column encoding, a target column can otherwise benefit from graph information retrieved for its neighbors. The subset contains 315 tables and 612 columns across 26 labels, comprising 556 numeric columns and 56 non-numeric columns.

Table 3.3 compares annotation accuracy without retrieved KG evidence. KGLink obtains 97.04 numeric accuracy and 90.92 non-numeric accuracy. TaBERT and Doduo also achieve high values on this subset. The table encoder supports annotation when external graph evidence is absent.

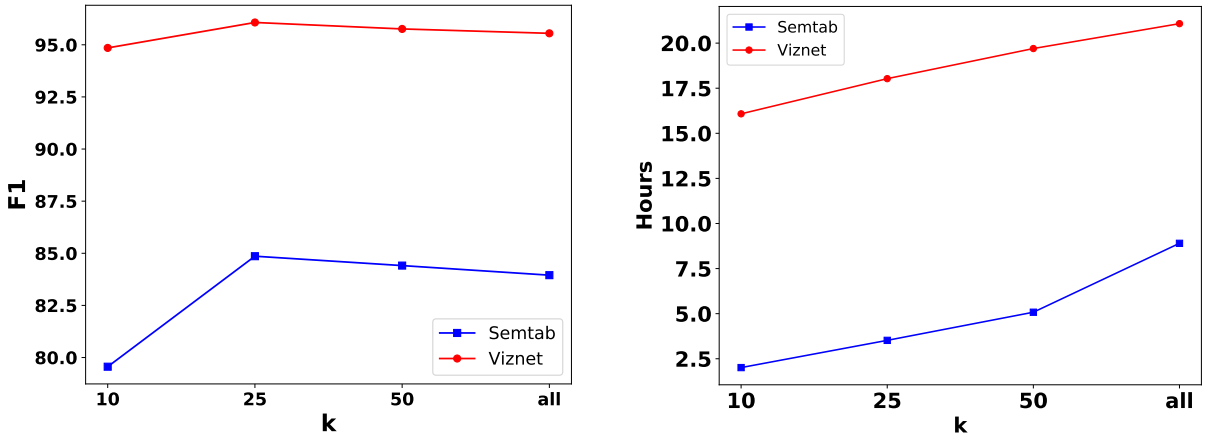
3.6.5 Type Granularity in the Reported Cases

Among SemTab classes with more than ten test samples, Athlete, Protein, and Film have the three largest reported accuracy increases when the representation-generation task is added.

Protein columns can retrieve KG descriptions such as Gene or Enzymes, while Film columns can retrieve Television series, Scholarly article, or Book. These descriptions are related to the cell values but differ from the annotation labels. The contextual representation task learns how

Table 3.4: Reported comparison between linkage-ranked row selection and selection in original table order. The row budget k is the number of retained rows.

	SemTab		VizNet	
Selection	Accuracy	Weighted F1	Accuracy	Weighted F1
Top-k row filter	**87.12**	**85.78**	**96.28**	**96.07**
Original top-k rows	85.93	84.39	96.14	95.97



(a) Weighted F1

(b) Time cost

Figure 3.6: Reported effect of the retained row budget k. The original two subfigures compare annotation quality and time cost on SemTab and VizNet. Reproduced from KGLink [9], Fig. 10.

to use them jointly with the column contents.

The Athlete example shows the same granularity gap. KGLink retains the graph description as input and learns the annotation category from supervised examples.

3.6.6 Row Selection and Input Budgets

Table 3.4 shows that linkage-ranked rows improve annotation over taking the first rows on both datasets, with a larger change on SemTab. The filter also selects the cell used for the feature sequence, so the comparison measures both effects of row ranking.

The parameter study considers row budgets $k = 10, 25, 50$, and a condition labeled *all*. The source describes *all* as retaining at most 64 rows under its encoding configuration. The reported default row budget is $k = 25$, selected from this study. The original curves compare weighted F1 and time cost as the retained row budget changes. Figure 3.6 shows the measured relation between input selection, weighted F1, and time cost.

The training-data study uses proportions $p = 0.2, 0.4, 0.6, 0.8, 1.0$ while keeping the test set unchanged. It compares KGLink with the version without the representation-generation task. The experiment measures data efficiency by varying available supervision against a fixed eval-

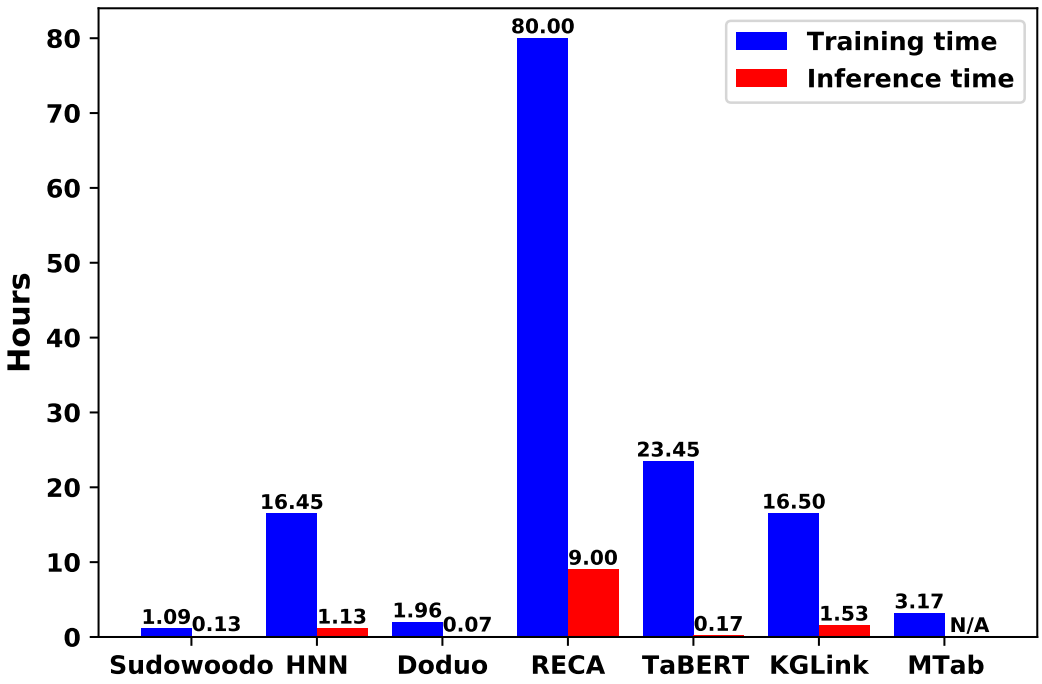


Figure 3.7: Training and inference time reported on VizNet. Original image reproduced from KGLink [9], Fig. 7.

uation set.

3.6.7 Runtime in the Evaluated Configuration

Figure 3.7 reproduces the original VizNet runtime chart. KGLink's reported training and inference times are 16.50 and 1.53 hours. The corresponding RECA values are 80.00 and 9.00 hours. The figure also shows methods with shorter measured times, including Doduo at 1.96 hours for training and 0.07 hours for inference. These results show the cost of KGLink's graph-augmented annotation. External retrieval and the auxiliary training objective add computation beyond the table encoder alone.

The chart labels MTab's inference entry as N/A. Training and inference times describe different preparation and prediction procedures. The comparison uses VizNet and the original experimental hardware.

3.7 Discussion and Chapter Summary

We proposed KGLink to supply missing table context and learn the correspondence between graph descriptions and annotation labels. Cross-column filtering selects entity candidates, recurring neighborhoods supply candidate types, and the contextual representation objective trains

the encoder to use this information.

Removing either the masked representation task or KG information lowers the full BERT configuration's annotation scores. Linkage-based row selection also improves the results. On tables without retrieved KG evidence, the trained encoder continues to annotate from the table values.

KGLink's external evidence is limited to the entities and relations available through linking and neighborhood retrieval. Its supervised predictor learns the training label set, and the experiments use single-label annotation with bounded table inputs.

To support tasks without a suitable knowledge graph or sufficient fine-tuning data, we next investigate constructing a graph from task examples and retrieving candidate labels for a large language model. Chapter 4 develops this approach for dynamic few-shot classification.

CHAPTER 4

ADAPTIVE GRAPH RETRIEVAL FOR DYNAMIC CLASSIFICATION WITH GORAG

This chapter is based on *GORAG: Graph-based Online Retrieval Augmented Generation for Dynamic Few-shot Social Media Text Classification*, published in the Proceedings of the ACM Web Conference 2026 [13]. We propose GORAG to construct and maintain a weighted graph of keyword–label associations from task data. Connected retrieval selects candidate labels for a large language model, and online insertion adds unseen terms after classification.

4.1 Dynamic Few-shot Classification

KGLink uses an existing knowledge graph to enrich representations through supervised fine-tuning. For applications without a suitable graph or sufficient training data, we instead build a graph from the task's own examples and use a large language model for prediction. Retrieved labels and descriptions supply context, allowing new information to be incorporated without parameter fine-tuning. We study this approach in dynamic few-shot classification, where the target categories evolve as new examples arrive.

Dynamic classification must distinguish new categories from those introduced earlier. For example, an abuse classifier may initially recognize threatening language and later add identity insults. With few examples per class, the system must recognize the new category while retaining access to earlier labels.

Few-shot classification provides a small labeled support set for each class. In the dynamic setting, new classes arrive in successive rounds [10]. The index must incorporate their examples while preserving evidence for the cumulative label set.

Adding more examples to the prompt can introduce irrelevant context as the label set grows. Passage retrieval alone also leaves the generator to choose among all labels. We instead retrieve a candidate label set for each query.

To select labels jointly from the query keywords, we represent keywords and labels in a weighted graph. A keyword can connect to several labels, with an edge cost for each association. Connecting several query keywords selects labels jointly from their neighborhoods.

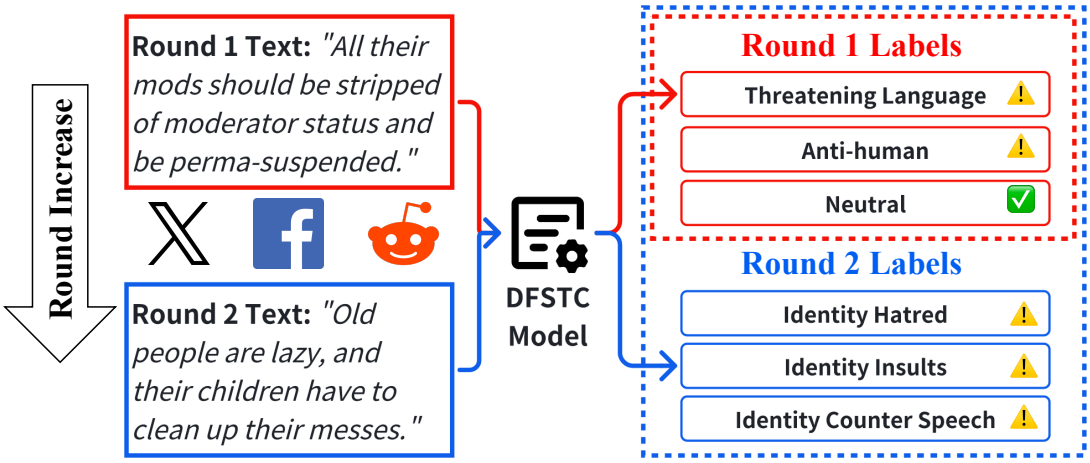


Figure 4.1: Two rounds of the dynamic classification task. New abuse categories extend the available labels. Reproduced from Figure 1 of GORAG [13].

GORAG also updates its index after classifying an unlabeled query. Keywords that were absent from the index can then become available to later queries. The predicted label associates the newly observed keywords with a class for subsequent retrieval. Support examples initialize the graph using their supplied labels.

GORAG combines weighted indexing, connected candidate-label retrieval, and online insertion. The following sections define these operations and evaluate their classification accuracy and resource costs.

4.1.1 Rounds and information availability

Let $\mathcal{Y}_{\text{new}}^{r}$ denote the labels introduced at round r. The current label universe is

$$\mathcal{Y}^{r} = \bigcup_{i=1}^{r} \mathcal{Y}_{\text{new}}^{i}. \tag{4.1}$$

Here $\mathcal{Y}^{r}$ is the union of the label sets $\mathcal{Y}_{\text{new}}^{i}$ introduced in rounds $i = 1, \ldots, r$. The few-shot support data provide k annotated texts per class, with k equal to 1, 5, or 10 in the main experiments. A query text t must be assigned a label from $\mathcal{Y}^{r}$. The superscript r identifies the current classification round.

Support texts carry supplied labels. Query texts become visible as the system processes them, and their predictions can be used for subsequent index updates. Corpus statistics used in keyword scoring refer to training and query texts seen so far. The online association nevertheless inherits prediction errors when the classifier chooses an incorrect label.

Semantic label names supply another source of information. Where names are available,

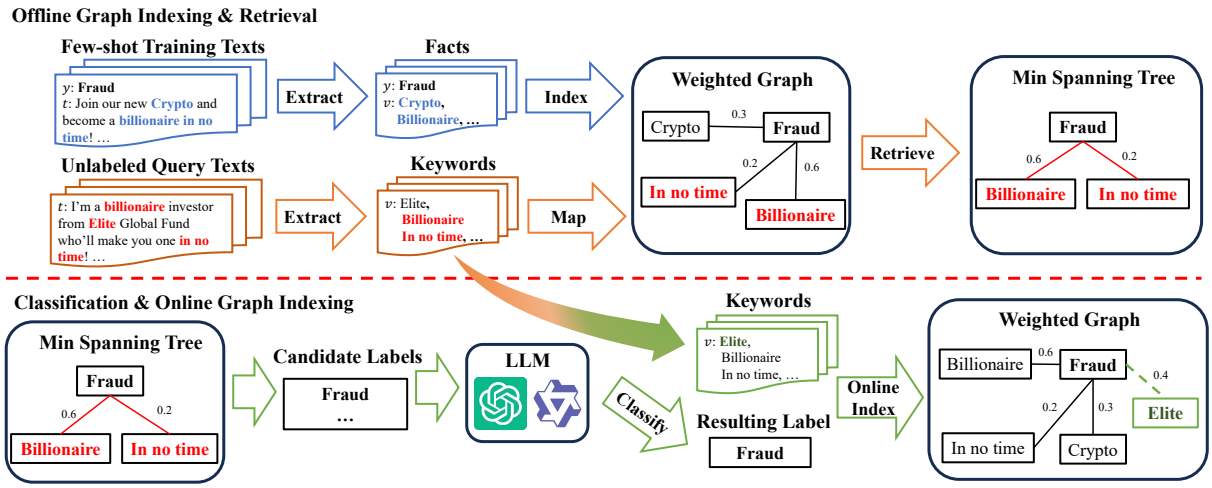


Figure 4.2: GORAG's offline indexing, query-dependent retrieval, classification, and online update. New query keywords are associated with the predicted class after classification. Reproduced from Figure 2 of GORAG [13].

GORAG uses a language model to generate representative descriptions or keywords for labels. These descriptions enter the classification context, and label-derived keywords can support graph initialization. The zero-shot experiment later in this chapter uses this source without any labeled examples.

4.2 Weighted Graph Indexing

GORAG maintains an undirected weighted graph $\mathcal{G}^r = (\mathcal{V}^r, \mathcal{E}^r, \mathcal{W}^r)$. Here $\mathcal{V}^r$ and $\mathcal{E}^r$ are the node and edge sets, and $\mathcal{W}^r$ contains their edge costs. Its node set contains keyword nodes and label nodes. A keyword node represents an extracted word or short phrase. A label node represents a target class. Label nodes make candidate classes directly accessible through their keyword associations. Support documents contribute keyword–label associations to this representation.

A keyword–label edge is introduced when a keyword occurs in a text associated with that label. Repeated occurrences across texts contribute to the edge weight. A keyword shared by several classes can therefore have several label neighbors, with a different cost for each association. Query retrieval will use these costs when choosing connections between mapped keywords.

The graph stores keyword–label associations, while the classifier receives the original query, retrieved labels, and their descriptions. The full query retains syntax, word order, and discourse omitted by keyword extraction. The keyword-only ablation removes this full-text input.

4.2.1 Keyword extraction and correlation scores

GORAG scores terms using normalized TF–IDF [92]. Its extraction vocabulary includes words and phrases of fewer than three words. Let $\mathcal{T}_{\leq r}$ denote the corpus seen so far, and let $\mathrm{df}(v)$ be the number of its texts containing term v. The reported correlation score is

$$CS(v,t) = \frac{\mathrm{count}(v,t)}{|t| \log\big(|\mathcal{T}_{\leq r}| + 1\big)} \log \frac{|\mathcal{T}_{\leq r}| + 1}{\mathrm{df}(v) + 1}. \tag{4.2}$$

Here $CS(v,t)$ is the correlation score of term v in text t, $\mathrm{count}(v,t)$ counts its occurrences, $|t|$ is the text length, $\mathrm{df}(v)$ counts the seen texts containing the term, and $\mathcal{T}_{\leq r}$ is the corpus observed by round r. The expression combines frequency within the text with frequency across the observed corpus. Its corpus term changes when additional texts are observed. Terms with $CS(v,t) > \tau$ are retained, with $\tau = 0.5$ in the original study. The method uses the score both to select keywords and to calculate association costs.

Short phrases preserve lexical combinations that unigram extraction would split. Only extracted terms can become retrieval terminals, although the classifier still receives the full query text.

4.2.2 Keyword–label edge costs

For keyword v and label y, let $\mathcal{T}^{r,v,y}$ be the seen texts that contain v and are associated with y. GORAG assigns the edge cost

$$w^r_{v,y} = \frac{\sum_{t_j \in \mathcal{T}^{r,v,y}} (1 - CS(v,t_j))}{|\mathcal{T}^{r,v,y}|}. \tag{4.3}$$

Here $w^r_{v,y}$ is the cost of the edge between keyword v and label y, $\mathcal{T}^{r,v,y}$ is their associated set of seen texts, and $CS(v,t_j)$ is the correlation score in text t_j. Subtracting the correlation score from one gives stronger keyword associations lower connection costs. Averaging makes the edge depend on all contributing observations rather than on one occurrence. Offline observations use the supplied class of a support text; online observations use a predicted class when new query keywords are inserted.

The same keyword can have different edge costs for different labels. The unit-weight ablation replaces all costs with one while preserving adjacency, testing whether these association-specific costs help retrieval.

Each observed text contributes a correlation value to its keyword–label pair. Averaging these

values summarizes repeated associations without concatenating every support text into the classification prompt.

4.2.3 Merging label rounds

At each round, we merge the new graph with the accumulated index by taking the unions of their node and edge sets. Earlier labels and their associations remain available for subsequent retrieval.

GORAG also introduces edges between newly introduced and earlier label nodes. These bridges support connections between parts of the graph indexed in different rounds. Let $\mathscr{M}_y^r$ denote the keyword neighbors of label y. For a new label y_n and an earlier label y_o, GORAG reports

$$w^r_{y_n,y_o} = \frac{1}{2}\left(\frac{\sum_{v\in\mathscr{M}^r_{y_n}} w^r_{v,y_n}}{2|\mathscr{M}^r_{y_n}|} + \frac{\sum_{v\in\mathscr{M}^r_{y_o}} w^r_{v,y_o}}{2|\mathscr{M}^r_{y_o}|}\right). \tag{4.4}$$

Here $w^r_{y_n,y_o}$ is the bridge cost, $\mathscr{M}^r_{y_n}$ and $\mathscr{M}^r_{y_o}$ are the keyword-neighbor sets of the new and earlier labels, and $w^r_{v,y}$ is a keyword–label edge cost. The formula uses costs incident to both labels. These factors determine the relative cost of traversing a label bridge rather than a keyword association.

Label bridges connect graph regions introduced in different rounds. Unlike keyword edges, which record observed term–class associations, they are created during graph merging.

4.3 Adaptive Candidate Label Retrieval

For a query t, GORAG extracts keywords using the indexing procedure and partitions them according to whether they already occur in the current graph node set $\mathscr{V}^r$. Write these sets as $\mathscr{V}^t_{\text{exist}}$ and $\mathscr{V}^t_{\text{notexist}}$. The existing keywords become the terminals for current retrieval. The unseen keywords are held for online insertion after classification. This partition restricts current retrieval to keywords with indexed label associations.

Retrieval seeks a low-cost connected structure that contains the mapped terminals. Label nodes encountered in this structure form the candidate set $\widehat{\mathscr{Y}}^r_t$. Connecting several terminals can select intermediate labels and keywords. The graph therefore determines the number of candidate labels through the returned structure, rather than requiring the same number of labels for all queries. A short query and a query containing several mutually distant indexed terms can produce different retrieval structures.

This terminal-connecting objective is related to the Steiner tree problem. In that problem, the required terminals are a subset of graph vertices, and the solution may use additional vertices to connect them. An ordinary minimum spanning tree instead spans all vertices of its input graph. These objects play distinct roles in GORAG's procedure. GORAG uses these intermediate structures to construct a heuristic connection among the query terminals.

4.3.1 Tree and auxiliary-graph operations

GORAG uses a greedy terminal-connecting procedure based on Mehlhorn's algorithm [93]. The mapped keywords serve as terminals. An auxiliary graph H represents connections between terminals, with edge weights given by path costs. A minimum spanning tree of H selects a collection of terminal connections. Expanding these connections into their underlying paths produces the graph structure used for candidate selection. The language model then receives the labels encountered in this structure.

Once the expanded structure S_t^r is obtained, the candidate set is

$$\widehat{\mathcal{Y}}_t^r = \mathcal{V}(S_t^r) \cap \mathcal{Y}^r. \tag{4.5}$$

Here S_t^r is the retrieved structure, $\mathcal{V}(S_t^r)$ its node set, $\mathcal{Y}^r$ the current label set, and $\widehat{\mathcal{Y}}_t^r$ the resulting candidates. Representative keywords and label descriptions are added to the classification prompt for the labels selected by this intersection.

4.4 Classification and Online Indexing

GORAG constructs a classification input by concatenating the query text, extracted query keywords, retrieved candidate labels, and representative keywords associated with those labels. With $\mathcal{K}^t$ denoting query keywords and $\mathcal{K}_{\widehat{\mathcal{Y}}_t^r}$ the representative label information, the input is

$$c_t = \text{Concat}\left(t, \mathcal{K}^t, \widehat{\mathcal{Y}}_t^r, \mathcal{K}_{\widehat{\mathcal{Y}}_t^r}\right). \tag{4.6}$$

Here c_t concatenates query t, its keywords $\mathcal{K}^t$, candidate labels $\widehat{\mathcal{Y}}_t^r$, and their representative information $\mathcal{K}_{\widehat{\mathcal{Y}}_t^r}$. A language model applies the classification instruction p_{classify} to this input and produces the predicted label y_t^*. The graph narrows and describes the alternatives; the language model performs the final text classification.

After the prediction, each keyword in $\mathcal{V}_{\text{notexist}}^t$ is inserted into the graph and connected to the

predicted label y_t^*. The update can be written as

$$\mathcal{V}^r \leftarrow \mathcal{V}^r \cup \mathcal{V}^t_{\text{notexist}}, \tag{4.7}$$

$$\mathcal{E}^r \leftarrow \mathcal{E}^r \cup \{(v, y_t^*) : v \in \mathcal{V}^t_{\text{notexist}}\}. \tag{4.8}$$

Here $\mathcal{V}^r$ and $\mathcal{E}^r$ are the current node and edge sets, $\mathcal{V}^t_{\text{notexist}}$ contains the query keywords absent from the index, and y_t^* is the predicted label. New associations receive weights through the reported weighting mechanism. Later queries can then map terms that were previously absent from the index. The update changes the retrieval source without fine-tuning the language model's parameters.

The prediction precedes insertion, so a new keyword's association can affect later queries but not the prediction that created it. Incorrect predicted labels can propagate through those associations. The offline ablation suppresses online expansion, measuring its contribution relative to an index that receives no such query-dependent additions.

4.4.1 Computational organization

The graph separates work performed at a label update from work performed for each query. Offline indexing processes the supplied texts, creates or updates keyword associations, generates label information when available, and merges the round's graph. The cross-round label bridges introduce interactions between the newly indexed and earlier label regions. In the source's abstract operation count, let $\mathcal{V}^r_{\text{new}}$ be the nodes in the new round graph, $\mathcal{D}^r$ its labeled training texts, and $\mathcal{Y}^r$ and $\mathcal{Y}^{r-1}$ the current and previous label sets. Construction is expressed as

$$O\left(|\mathcal{Y}^r||\mathcal{Y}^{r-1}| + |\mathcal{V}^r_{\text{new}}||\mathcal{D}^r|\right). \tag{4.9}$$

The two terms count label merging and keyword–text processing. Language-model token costs additionally depend on prompt and output lengths and are measured in the token study.

Each query maps its keywords to the index, retrieves a connecting structure, and constructs a classification prompt. New terms are inserted after prediction, so the index can grow within a round. The prompt contains only the labels selected for the current query.

Table 4.1: Dataset statistics for the four-round experiments, transposed from the original table. IFS-Rel has 40 test samples per label. Round label counts refer to newly introduced labels.

Statistic	CAD	WOS	IFS-Rel
Average text tokens	154	200	105
Round 1 test texts	240	2,417	640
Round 1 new labels	4	32	16
Round 2 test texts	60	2,842	640
Round 2 new labels	3	53	16
Round 3 test texts	286	2,251	640
Round 3 new labels	3	30	16
Round 4 test texts	231	1,886	640
Round 4 new labels	3	18	16
Total test texts	817	9,396	2,560

4.5 Experimental Design

We evaluate GORAG on CAD, WOS, and IFS-Rel. CAD contains social media posts annotated for contextual abuse [94]. WOS contains academic texts associated with research-area labels [95]. IFS-Rel supplies relation-classification examples derived from news text in an incremental few-shot benchmark [10]. These datasets exercise different label vocabularies and text lengths within the dynamic classification setting.

The four-round split introduces 13 labels across CAD's rounds, 133 across WOS, and 64 across IFS-Rel. The test collections contain 817, 9,396, and 2,560 examples, respectively. IFS-Rel uses 40 test examples per label. Its balanced test allocation differs from the changing class frequencies in the other collections.

The label counts in Table 4.1 describe labels introduced in each round. The classifier operates over their cumulative union. Thus the second WOS round adds 53 labels to the 32 labels already introduced, rather than replacing the earlier set by a new 53-label task. This distinction matters when relating the dataset statistics to the size of the classification instruction and to the graph state retained from earlier rounds.

The main few-shot study uses 1, 5, and 10 labeled examples per class. CAD and WOS RAG experiments use Qwen2.5-7B, while IFS-Rel uses GPT-4o. The source reports an Intel Xeon Gold 5220R CPU at 2.20GHz and a single NVIDIA A100-SXM4-40GB GPU for the experiments. IFS-Rel inference uses the GPT-4o service.

Classification accuracy requires an exact match between the model output and the ground-truth label name or number. An output that cannot be matched to the required class is not counted as correct.

Table 4.2: WOS classification accuracy with one support example per label. Qwen2.5-7B is the RAG backbone. Values and emphasis reproduce the corresponding columns of GORAG's main table.

Method	Round 1	Round 2	Round 3	Round 4
Entailment	0.3695	0.3994	0.4510	0.4030
NaiveRAG	0.3885	0.2267	0.1821	0.1653
LongLLMLingua	0.3806	0.2155	0.1770	0.1468
GraphRAG	0.3852	0.2213	0.1816	0.1641
LightRAG	0.3930	0.2202	0.1743	0.1625
HippoRAG2	0.4133	0.2552	0.2205	0.1906
GORAG	**0.4866**	**0.4790**	**0.4736**	**0.4420**

4.5.1 Compared methods and controls

The study compares six baselines. Entailment constructs text–label entailment pairs and fine-tunes a classifier [10]. NaiveRAG retrieves text chunks from the indexed support information. Retrieved context is compressed by LongLLMLingua before classification [96]. GraphRAG, LightRAG, and HippoRAG2 supply graph-based retrieval alternatives [14, 59, 97]. The graph retrieval systems are adapted to return classification evidence.

Entailment uses RoBERTa-large for five training epochs, a batch size of 16, a learning rate of 10^{-6}, and Adam. LongLLMLingua is evaluated with compression rates of 0.75, 0.8, and 0.85, with 0.8 selected in the reported comparison. GraphRAG and LightRAG use local search. The source uses the nano-graphrag implementation for GraphRAG and NaiveRAG and the other baselines' official implementations.

4.6 Classification Results

The tables beginning with Tables 4.2 and 4.5 preserve the WOS and IFS-Rel portions of the main accuracy table. Each dataset is split into one-, five-, and ten-shot tables, with rounds as columns. Each value is a fractional accuracy, and the labels available to the model expand with the round.

On WOS with one support example per label, GORAG reports accuracies of 0.4866, 0.4790, 0.4736, and 0.4420 across the four rounds. Entailment reports 0.3695, 0.3994, 0.4510, and 0.4030 in the same columns. The corresponding GraphRAG values are 0.3852, 0.2213, 0.1816, and 0.1641. GORAG maintains higher accuracy than these baselines as the label set grows.

Additional support examples improve GORAG's WOS accuracy in the main table. At the fourth round, its one-, five-, and ten-shot values are 0.4420, 0.4717, and 0.4929. At the first round they are 0.4866, 0.4990, and 0.5134.

Table 4.3: WOS classification accuracy with 5 support examples per label. Qwen2.5-7B is the RAG backbone. Values and emphasis reproduce the corresponding columns of GORAG's main table.

Method	Round 1	Round 2	Round 3	Round 4
Entailment	0.3823	0.4471	0.4857	0.4387
NaiveRAG	0.3904	0.2154	0.1475	0.1556
LongLLMLingua	0.3823	0.2202	0.1567	0.1382
GraphRAG	0.3897	0.2197	0.1770	0.1634
LightRAG	0.3806	0.2216	0.1767	0.1626
HippoRAG2	0.4245	0.2656	0.2210	0.1860
GORAG	**0.4990**	**0.5109**	**0.4996**	**0.4717**

Table 4.4: WOS classification accuracy with 10 support examples per label. Qwen2.5-7B is the RAG backbone. Values and emphasis reproduce the corresponding columns of GORAG's main table.

Method	Round 1	Round 2	Round 3	Round 4
Entailment	0.4187	0.4222	0.4787	0.4442
NaiveRAG	0.3897	0.2187	0.1799	0.1649
LongLLMLingua	0.3901	0.2198	0.1608	0.1493
GraphRAG	0.3906	0.2219	0.1786	0.1625
LightRAG	0.3815	0.2145	0.1799	0.1632
HippoRAG2	0.4208	0.2666	0.2242	0.1915
GORAG	**0.5134**	**0.5284**	**0.5234**	**0.4929**

Table 4.5: IFS-Rel classification accuracy with one support example per label. GPT-4o is the RAG backbone. Values and emphasis reproduce the corresponding columns of GORAG's main table.

Method	Round 1	Round 2	Round 3	Round 4
Entailment	0.3391	0.4008	0.3061	0.2971
NaiveRAG	0.7343	0.5394	0.4719	0.5233
LongLLMLingua	0.8750	0.5734	0.4703	0.5819
GraphRAG	0.6406	0.4406	0.3828	0.4297
LightRAG	0.7578	0.4906	0.3802	0.4531
HippoRAG2	0.7313	0.4875	0.4297	0.4984
GORAG	**0.9688**	**0.5938**	**0.4844**	**0.6165**

Table 4.6: IFS-Rel classification accuracy with 5 support examples per label. GPT-4o is the RAG backbone. Values and emphasis reproduce the corresponding columns of GORAG's main table.

Method	Round 1	Round 2	Round 3	Round 4
Entailment	0.6046	0.5516	0.4703	0.4395
NaiveRAG	0.7344	0.5406	0.4719	0.4938
LongLLMLingua	0.8906	0.5922	0.4741	0.4903
GraphRAG	0.7813	0.4688	0.3734	0.4422
LightRAG	0.8281	0.5155	0.4010	0.4804
HippoRAG2	0.7375	0.5016	0.4328	0.5008
GORAG	**0.9688**	**0.6172**	**0.4948**	**0.4988**

Table 4.7: IFS-Rel classification accuracy with 10 support examples per label. GPT-4o is the RAG backbone. Values and emphasis reproduce the corresponding columns of GORAG's main table.

Method	Round 1	Round 2	Round 3	Round 4
Entailment	0.5859	0.5828	0.5193	0.4895
NaiveRAG	0.7406	0.5469	0.4688	0.5000
LongLLMLingua	0.9219	0.5806	0.5225	0.5038
GraphRAG	0.8125	0.4719	0.3141	0.3922
LightRAG	0.8594	0.5156	0.4427	0.4610
HippoRAG2	0.7406	0.4938	0.4438	0.4953
GORAG	**0.9688**	**0.6250**	**0.5344**	**0.5469**

Table 4.8: Ablation experiments on WOS for GORAG's variants.

Dataset	**Model**	**Round**			
		R1	R2	R3	R4
WOS	GORAG *offline&unit*	0.2979	0.2287	0.2645	0.2249
	GORAG *offline*	0.3063	0.2302	0.2455	0.2156
	GORAG *unit*	0.4706	0.4394	0.4407	0.3899
	GORAG *keyword*	0.4746	0.4606	0.4455	0.4030
	GORAG	**0.4862**	**0.4649**	**0.4814**	**0.4210**

IFS-Rel uses GPT-4o. In its first round, GORAG reports 0.9688 for all three support sizes, while the fourth-round scores are 0.6165, 0.4988, and 0.5469. The five-shot fourth-round value is below the one-shot value, so increasing support size does not produce a monotonic improvement in every setting. LongLLMLingua reports 0.5819, 0.4903, and 0.5038 in these fourth-round columns. HippoRAG2 reports 0.4984, 0.5008, and 0.4953; its five-shot result exceeds GORAG's five-shot result in that column.

CAD uses Qwen2.5-7B. Its GORAG first-round accuracy is 0.6125 in all three support settings. The fourth-round one- and five-shot accuracies are 0.3133 and 0.3146. Accuracy decreases in later rounds as the cumulative label set grows.

WOS improves with additional support in the displayed rounds, whereas IFS-Rel is non-monotonic across support sizes and CAD loses accuracy as labels accumulate.

4.7 Ablations and Resource Use

The WOS ablation study uses a one-shot Qwen2.5-7B setting. The offline variant disables online indexing. The unit variant assigns cost one to every graph edge. The offline-and-unit variant removes both features. The keyword variant uses the extracted keywords in place of the complete query text for classification.

In the ablation table, the full method reports 0.4862, 0.4649, 0.4814, and 0.4210 across

Table 4.9: Token cost on IFS-Rel dataset with 1-shot setting.

Model	AVG Token per Query	AVG Cost per Query (USD)
GPT-4o	634.6	$0.005
LongLLMLingua	2771.3	$0.021
NaiveRAG	4537.4	$0.034
LightRAG	7866.3	$0.059
GraphRAG	9296.3	$0.070
GORAG	**557.5**	**$0.004**

rounds. The offline variant reports 0.3063, 0.2302, 0.2455, and 0.2156. The unit variant reports 0.4706, 0.4394, 0.4407, and 0.3899. Disabling online insertion causes a larger accuracy drop than replacing weighted edges with unit costs.

The offline-and-unit variant reports 0.2979, 0.2287, 0.2645, and 0.2249. Its third- and fourth-round values exceed those of the offline variant. The effects of removing these components are therefore not strictly additive across rounds. The keyword variant reports 0.4746, 0.4606, 0.4455, and 0.4030, below the full method in each displayed column. The full query provides information useful to the classifier in these experiments even though keywords determine the retrieval terminals.

The publication gives different full-method values in the main and ablation WOS tables without explaining their provenance. Component comparisons here use the full-method row from the ablation table.

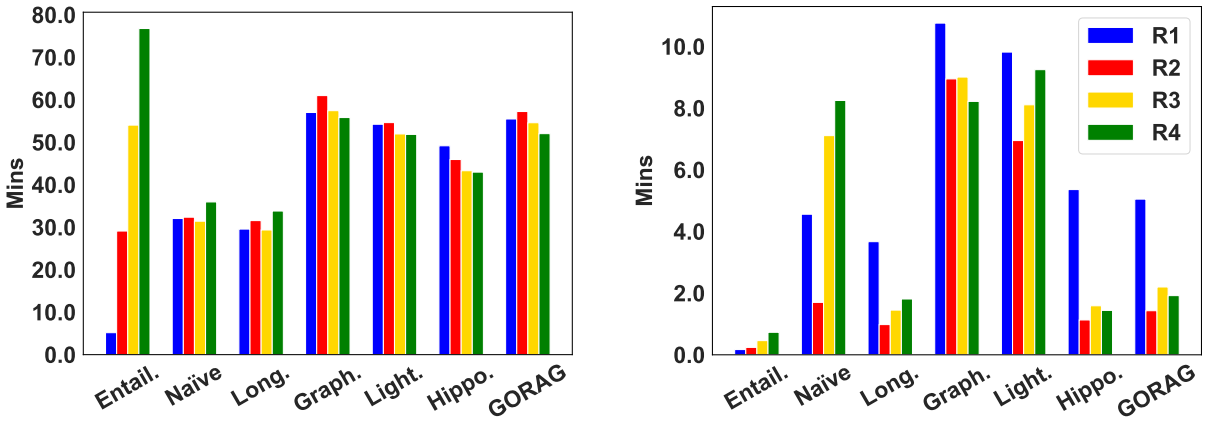
4.7.1 Prompt tokens and graph growth

The IFS-Rel one-shot token study reports average tokens per query and the corresponding US-dollar costs under the original study's pricing conditions. GORAG uses 557.5 tokens per query and reports a cost of $0.004. The direct GPT-4o comparison uses 634.6 tokens and reports $0.005. LongLLMLingua uses 2,771.3 tokens, NaiveRAG 4,537.4, LightRAG 7,866.3, and GraphRAG 9,296.3. Their reported costs are $0.021, $0.034, $0.059, and $0.070, respectively.

GORAG's candidate-label context uses fewer tokens than the retrieval baselines in this one-shot experiment.

Graph growth is measured separately during the first IFS-Rel round with GPT-4o. The reported index contains 1,644 nodes and 1,771 edges at the 25-percent checkpoint, and 5,299 nodes and 5,503 edges at the final checkpoint. The corresponding reported memory values are 13.3 KB and 46.4 KB. These memory values cover the graph index alone.

The original runtime study compares total time on WOS and CAD under the one-shot setting, using Qwen2.5-7B. Online insertion adds work after each prediction, while statistical keyword



(a) WOS. (b) CAD.

Figure 4.3: Total time in minutes for the one-shot four-round experiments with Qwen2.5-7B. The original R1–R4 legend in panel (b) applies to both panels. Reproduced from Figure 3 of GORAG [13].

Table 4.10: 0-shot experiments for GORAG and RAG baselines on IFS-Rel dataset under 0-shot setting.

Model	**R1**	**R2**	**R3**	**R4**
GPT-4o [98]	0.9219	0.5109	0.4063	0.5936
LongLLMLingua [96]	0.9281	0.5297	0.4234	0.5806
LightRAG [59]	0.6718	0.3047	0.2135	0.0977
GraphRAG [14]	0.5625	0.1875	0.1458	0.0938
GORAG	**0.9328**	**0.5406**	**0.4375**	**0.6063**

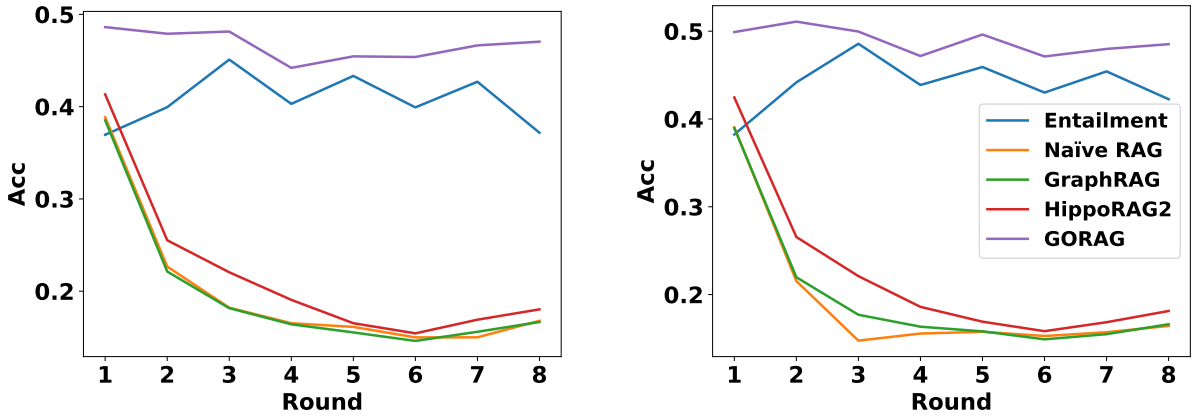
indexing and smaller model inputs can change other parts of the pipeline.

4.8 Zero-shot and Round Generalization

The zero-shot IFS-Rel experiment supplies label names and language-model-generated descriptions without annotated support texts. GORAG extracts keywords from these descriptions to initialize its graph. The experiment consequently tests whether description-derived associations can support candidate selection and subsequent classification.

GORAG reports zero-shot accuracies of 0.9328, 0.5406, 0.4375, and 0.6063. The direct GPT-4o comparison reports 0.9219, 0.5109, 0.4063, and 0.5936. LongLLMLingua reports 0.9281, 0.5297, 0.4234, and 0.5806. The graph-based comparisons in the same table are lower under this initialization protocol. These values support the use of label descriptions as an initial retrieval source for this experiment.

The round-generalization study uses WOS split into four, six, and eight rounds under one- and five-shot settings. The displayed curve combines rounds one to four from the four-round



(a) One-shot.

(b) Five-shot.

Figure 4.4: WOS classification across the reported round splits. Points 1–4 use the four-round split, 5–6 the six-round split, and 7–8 the eight-round split. The legend in panel (b) applies to both panels. Reproduced from Figure 4 of GORAG [13].

split, rounds five and six from the six-round split, and rounds seven and eight from the eight-round split. These points represent results from different partitions of the label set. They do not form the trajectory of one index updated through eight consecutive rounds.

4.8.1 A retrieved classification case

Figure 4.5 reproduces a contextual-abuse example from the original study. The query contains the terms highlighted in red. These terms map to keyword nodes, and the selected connecting structure passes through an insult label. The classifier outputs the abuse category shown in the figure. This example exposes the relationship between the extracted query evidence, the intermediate label node, and the final classification.

The insult label connects more than one highlighted query term. Other keyword neighbors remain outside the selected structure, showing how connected retrieval narrows the available label evidence.

4.9 Discussion and Chapter Summary

We proposed GORAG for dynamic few-shot classification. Weighted keyword–label connections determine the candidate set for each query, and the language model classifies the original text using those labels and their descriptions.

Disabling online insertion lowers accuracy, showing the contribution of expanding the index beyond the support sample. Because these additions carry predicted labels, the method couples

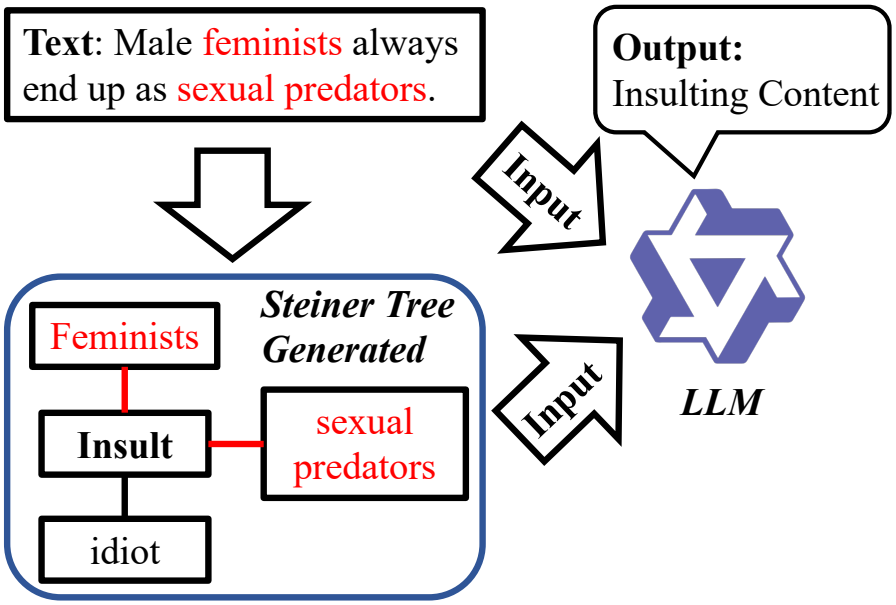


Figure 4.5: A contextual-abuse example and its selected graph connections. Reproduced from Figure 5 of GORAG [13].

current classification with the evidence available to later queries.

Keyword extraction restricts the query terms available as retrieval terminals. Sparse support data and limited label descriptions constrain initial graph coverage.

GORAG constructs and updates task-related knowledge without requiring a predefined external graph or model fine-tuning. This supports classification as labels and vocabulary evolve. For question answering and generation, the output requires facts and explanations beyond a candidate label set. The following chapters investigate two extensions of task-related graph retrieval. Chapter 5 studies connected textual evidence and answer coverage, while Chapter 6 studies query-aware construction and semantic matching across text and images.

CHAPTER 5

CONNECTED EVIDENCE SELECTION FOR LANGUAGE MODELS WITH AGRAG

This chapter is based on *AGRAG: Advanced Graph-based Retrieval-Augmented Generation for LLMs*, published at ICDE 2026 [15]. We propose AGRAG to construct a graph from a text corpus and retrieve connected facts with supporting passages. Query-conditioned graph influence, semantic edge costs, and neighborhood expansion collect evidence beyond the initially mapped facts.

5.1 Research Problem and Evidence Requirements

GORAG uses relationships extracted from task data to retrieve candidate classes. For question answering and generation, the retrieved structure must provide facts that jointly support the answer. A question can require information from several parts of a corpus, and a summary can need multiple complementary facts. Independently retrieved passages may repeat one fact or leave the relationships between different facts implicit. A graph records these relationships, but retrieval must still select a subgraph that contains enough information for the answer.

To address these evidence requirements, we propose AGRAG to select connected facts using query relevance and graph influence. Statistical extraction determines the entity vocabulary, semantic costs rank connections, and neighborhood expansion adds facts around the query anchors. A hybrid passage channel supplies context omitted by the graph representation.

Consider a corpus containing statements about an illness, the organ that it affects, and the routes through which it spreads. A query may match facts mentioning the illness and the organ. Connecting those facts can identify a small initial structure, but answering the question may require a neighboring fact that specifies another anatomical component. Figure 5.2 illustrates why evidence retrieval may need to expand beyond an initial terminal-connecting structure.

The input is a text corpus $\mathcal{T}$ and a natural-language query q. The corpus is divided into chunks $\mathcal{T}_c$. The indexed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has node set $\mathcal{V}$ containing entities and passages and edge set $\mathcal{E}$ containing several kinds of relations. Query processing identifies an initial set of relevant triple facts, assigns node influence scores and edge costs, and produces a subgraph for

generation. A hybrid retriever supplies additional passages from the same corpus. The output is an answer conditioned on the query, a textual serialization of the retrieved graph, and the hybrid passages.

Entity–relation edges represent facts, while passage nodes link those facts to source text. We supply both the selected graph and retrieved passages to the generator so that it can use relational connections alongside their surrounding explanations.

The chapter evaluates the resulting pipeline on tasks with different answer requirements. Fact retrieval emphasizes individual facts. Complex reasoning combines facts. Contextual summarization requires coverage of several evidence items. Creative generation evaluates responses grounded in retrieved material while allowing more flexible output. Text classification supplies a further setting in which graph retrieval identifies candidate labels.

5.2 Framework and Graph Construction

5.2.1 The three-stage pipeline

AGRAG separates data preparation, graph retrieval, and hybrid retrieval. Data preparation builds the graph and the passage indexes. Graph retrieval conditions the graph on a query through semantic mapping and influence scores, then selects a connected evidence subgraph. Hybrid retrieval combines dense passage similarity with BM25. Figure 5.1 reproduces the original architecture.

Corpus indexing performs chunking, entity and relation extraction, passage linking, and dense and sparse indexing. Query processing maps the question to facts, computes influence scores and edge costs, expands the evidence subgraph, and retrieves hybrid passages.

The statistical component replaces the entity extraction step used in the compared graph pipelines. Language models remain involved in relation extraction, filtering raw mapped triples, and generating answers. The entity-extraction ablation changes the nodes supplied to subsequent relation extraction and retrieval.

5.2.2 Chunking and statistical entity extraction

The corpus is split into overlapping text chunks,

$$\mathscr{T}_c = \text{Split}(\mathscr{T}, l_t, l_o), \tag{5.1}$$

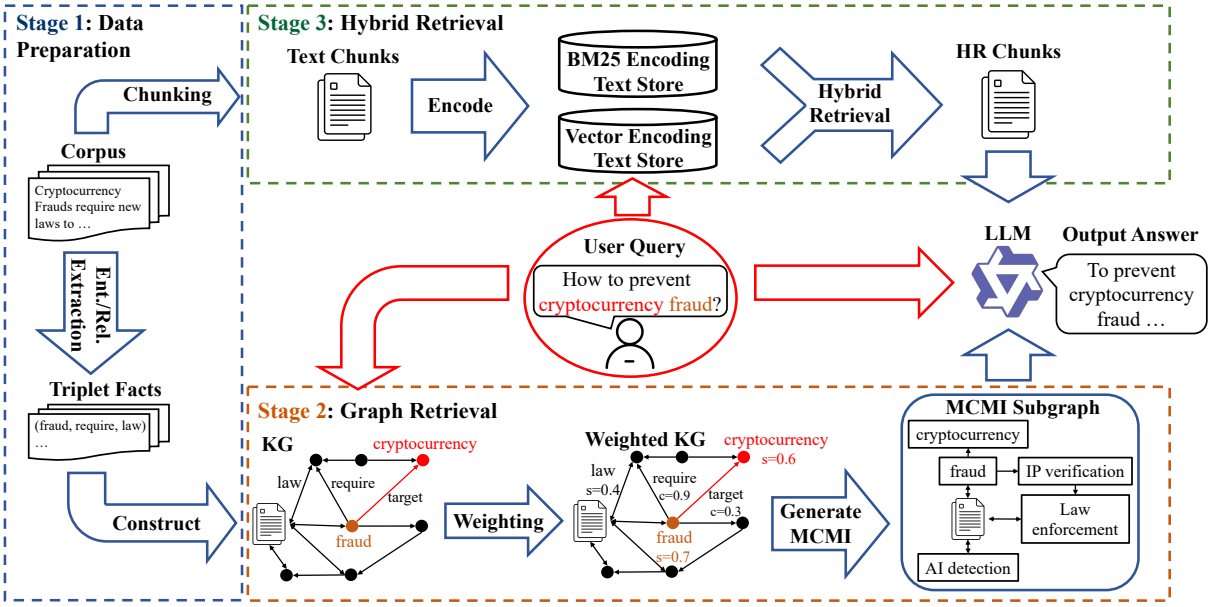


Figure 5.1: The AGRAG pipeline. The original figure separates corpus preparation, query-conditioned graph retrieval, and hybrid passage retrieval. Reproduced from Figure 1 of the source paper [15].

where $\mathcal{T}$ is the source corpus, $\mathcal{T}_c$ is its collection of chunks, l_t is the chunk length, and l_o is the overlap. The reported main setting uses 256 tokens per chunk and an overlap of 32 tokens. Overlap allows adjacent chunks to share text at their boundaries; the graph subsequently connects chunks through the entities they contain.

AGRAG considers words and phrases up to a maximum n-gram length b. For a candidate term v in chunk t, the extraction rule modifies TF–IDF [92] to define the score

$$\mathrm{ER}(v,t) = \frac{\mathrm{count}(v,t)}{|t| \log\big(|\mathcal{T}_c| + 1\big)} \log \frac{|\mathcal{T}_c| + 1}{|\{t_j \in \mathcal{T}_c : v \in t_j\}| + 1}. \tag{5.2}$$

Here $\mathrm{ER}(v,t)$ scores candidate term v in chunk t, $\mathrm{count}(v,t)$ counts its occurrences, $|t|$ is the chunk length, and $\mathcal{T}_c$ is the chunk collection. The set in the denominator contains the chunks t_j that include v. The first factor depends on within-chunk frequency and length. The second distinguishes terms according to their distribution across chunks. The extracted set is

$$\mathcal{V}_t = \{v : \mathrm{ER}(v,t) > \tau\}, \tag{5.3}$$

where $\mathcal{V}_t$ is the extracted entity set for chunk t, v ranges over candidate terms up to n-gram length b, and τ is the selection threshold. The main paper reports $b = 3$ and $\tau = 0.5$.

Statistical extraction selects strings present in the corpus. Term frequency does not directly measure factual importance, so a generic expression can score differently from a rare but relevant

phrase. We examine the n-gram boundary and threshold in the sensitivity study.

Shared entities connect passages in the graph. These links are useful when the extracted strings identify compatible entities across their occurrences; ambiguous mentions can instead create misleading connections.

5.2.3 Relations, passage nodes, and synonyms

For each chunk, the language model receives the selected entity set and the chunk text, and extracts relations,

$$\mathscr{E}_t = \mathrm{LLM}(\mathscr{V}_t, t, p_{\mathrm{re}}). \tag{5.4}$$

Here $\mathscr{E}_t$ is the set of relations extracted from chunk t, $\mathscr{V}_t$ is its entity set, and prompt p_{re} asks the language model for relations between those entities. The resulting entity–relation–entity triples provide the graph's factual edges. Statistical extraction selects entity candidates from observed terms. The model then extracts relations conditioned on those candidates and their context.

Chunks are also inserted as passage nodes. A *Contains* edge connects a passage to an entity extracted from it. These links preserve the association between the symbolic representation and the source chunk. They also allow paths to cross passages through shared entities. Passage nodes retain context shared by several extracted facts.

The graph includes synonym edges between entity nodes whose embedding similarity exceeds a threshold φ. Such an edge expresses an embedding-based correspondence between strings. It permits retrieval to connect different expressions that would otherwise form separate nodes. Extracted relations come from the language model's reading of a chunk, whereas synonym edges come from a similarity test.

AGRAG additionally introduces a pseudo node linked to passage nodes. The reported cost of each pseudo edge is 10. This connection gives the graph a route between otherwise disconnected portions while penalizing its use relative to semantically relevant factual edges. Pseudo edges provide connectivity between passages without asserting a factual relation.

5.3 Query Mapping and Influence-aware Retrieval

5.3.1 Mapping a query to triple facts

At query time, AGRAG first identifies facts that are semantically close to the query. A text encoder $\mathscr{M}$ maps the query and a textual rendering of each triple to vectors. With $\mathbf{q} = \mathscr{M}(q)$

and $\mathbf{f} = \mathscr{M}(f)$, the mapping score is

$$\mathrm{MS}(\mathbf{q}, \mathbf{f}) = \cos(\mathbf{q}, \mathbf{f}). \tag{5.5}$$

Here MS is cosine similarity, $\mathbf{q}$ is the encoded query q, and $\mathbf{f}$ is the encoded triple fact f. The highest-scoring k_a facts form the raw mapped set $\mathscr{F}_{\text{raw}}$. The main setting uses $k_a = 5$. This stage locates candidate facts in the indexed graph before graph propagation begins.

A model-based filtering step then evaluates the raw facts with respect to the query,

$$\mathscr{F} = \mathrm{LLM}(\mathscr{F}_{\text{raw}}, q, p_{\text{tf}}). \tag{5.6}$$

Here $\mathscr{F}_{\text{raw}}$ is the set of candidate facts, q is the query, and p_{tf} is the filtering prompt. The filtered set $\mathscr{F}$ supplies query anchors. Dense mapping produces the candidate list, and the model checks its relevance to the question. The filtering ablation removes this second step.

Facts around the anchors can contribute to an answer even when their direct query similarity is lower. We propagate influence from the mapped facts to score this surrounding evidence and use semantic edge costs when selecting its connections.

5.3.2 Node influence and edge costs

AGRAG uses a personalized PageRank-style propagation [99] to assign query-dependent influence scores,

$$\mathbf{s} = \lim_{i \to \infty} \mathbf{s}^{(i)}, \qquad \mathbf{s}^{(i)} = (1 - d)\mathbf{p} + d\mathbf{P}\mathbf{s}^{(i-1)}. \tag{5.7}$$

Here $\mathbf{s}^{(i)}$ is the influence vector at iteration i, $\mathbf{s}$ its limiting value, $\mathbf{p}$ the personalization vector determined by the filtered mapped facts, $\mathbf{P}$ the graph propagation matrix, and d the damping factor. The reported setting uses $d = 0.5$ and stops the iteration when the change is below 10^{-7}. The score of a node v is $s_v = \mathbf{s}[v]$. Passage-node scores are multiplied by 0.05 to balance their contribution relative to entity nodes.

The personalization vector assigns one to nodes participating in the mapped facts and zero to the other nodes. The graph operator is specified by

$$\mathbf{P}[u][v] = \begin{cases} 1/|\mathscr{N}(u)|, & (u, v) \in \mathscr{E}, \\ 1/|\mathscr{V}|, & \text{otherwise.} \end{cases} \tag{5.8}$$

Here $\mathbf{P}[u][v]$ is the entry for nodes u and v, $\mathscr{N}(u)$ is the one-hop neighbor set of u, and $\mathscr{E}$ and

$\mathscr{V}$ are the graph edge and node sets. Nonadjacent node pairs receive the constant term in this operator. The recurrence uses this matrix with the damping value in Equation 5.7, followed by passage-node scaling.

We compute node influence on the indexed graph before expansion. These scores remain assigned to the nodes while the retrieval procedure selects which nodes and edges to include.

Direct similarity scores a triple against the question. Influence also reflects its connections to the query anchors, allowing evidence connected to several mapped facts to receive retrieval weight.

For an edge $e = (u, v)$ associated with fact f_e, AGRAG defines

$$c_e = \frac{1 - \mathrm{MS}(\mathbf{q}, \mathbf{f}_e)}{2}. \tag{5.9}$$

Here c_e is the cost of edge e, and $\mathrm{MS}(\mathbf{q}, \mathbf{f}_e)$ is the cosine similarity between the encoded query and the encoded fact associated with that edge. Higher similarity gives a lower factual edge cost. Subgraph selection combines these edge costs with node influence scores.

Pseudo edges retain their assigned penalty, providing an expensive route when factual connections alone cannot connect the anchors.

5.4 MCMI Formulation and Evidence Expansion

Let $\mathscr{V}_{\text{term}}$ contain the nodes participating in the mapped facts. A feasible subgraph contains these terminals and connects each terminal to a designated root terminal. The main paper defines the Minimum Cost Maximum Influence (MCMI) objective as

$$r(\mathscr{G}') = \sum_{e=(u,v)\in\mathscr{E}'} \frac{c_e}{s_u + s_v}. \tag{5.10}$$

Here $\mathscr{G}' = (\mathscr{V}', \mathscr{E}')$ is a feasible subgraph, c_e is the cost of edge $e = (u, v)$, and s_u, s_v are the endpoint influence scores. The objective $r(\mathscr{G}')$ scales each included edge cost by the influence of its endpoints. We construct an initial terminal-connecting tree, then expand its neighborhood using marginal cost-to-score comparisons.

The initial structure connects the query-mapped information. The source constructs an auxiliary graph on the terminals, uses shortest-path costs between them, selects a tree on that auxiliary structure, and expands its edges back to paths in the evidence graph. This stage supplies connections among anchors instead of returning the anchors as an unordered set. The detailed theorem

in the main paper assigns its 2-approximation statement to this MCST construction stage.

Subsequent expansion examines neighboring candidates. A candidate node v connected by an edge of cost $c_{(\mathrm{MCMI},v)}$ is considered through the ratio

$$\frac{c_{(\mathrm{MCMI},v)}}{s_v}. \tag{5.11}$$

Here $c_{(\mathrm{MCMI},v)}$ is the cost of the candidate connection and s_v is the influence score of the candidate node v. The procedure selects the lowest such ratio and compares it with the current subgraph's reported cost-score quantity. The acceptance condition is

$$\frac{c_{(\mathrm{MCMI},v)}}{s_v} < r(\mathcal{G}_{\mathrm{MCMI}}). \tag{5.12}$$

Here $\mathcal{G}_{\mathrm{MCMI}}$ is the current subgraph, $r(\mathcal{G}_{\mathrm{MCMI}})$ is its cost-score quantity, and $c_{(\mathrm{MCMI},v)}/s_v$ is the candidate connection cost divided by node influence. The candidate and its connecting edge are added when this condition holds. The subgraph quantity is recalculated and the procedure considers the updated neighborhood. Expansion stops when the lowest candidate ratio is greater than or equal to the current subgraph quantity. The comparison uses the node score of the prospective addition, whereas Equation 5.10 uses the two endpoints of each included edge. The candidate ratio therefore ranks possible additions using their assigned node scores, while the current subgraph is evaluated by Equation 5.10.

Expansion adds neighboring facts that can supply answer components absent from the initial tree. The resulting structure can contain cycles, as shown in the MCMI example, and its selected relations are serialized for generation.

5.4.1 The original connected-evidence example

The running example in Figure 5.2 shows why connecting the query anchors may leave required answer facts outside the retrieved structure. Its mapped triples include *(Kidney, Work, Kidney cancer)* and *(Kidney, Spread, Tumors)*, using the relation strings printed in the original illustration. Those triples locate the query's subject region. The expanded structure additionally contains relations involving blood vessels, neighboring tissues, and other entities shown in the graph. The original illustrated answer identifies blood vessels as the requested component.

In this example, the added relation supplies an answer component absent from the mapped facts. Influence scores and edge costs determine whether expansion includes that neighboring evidence.

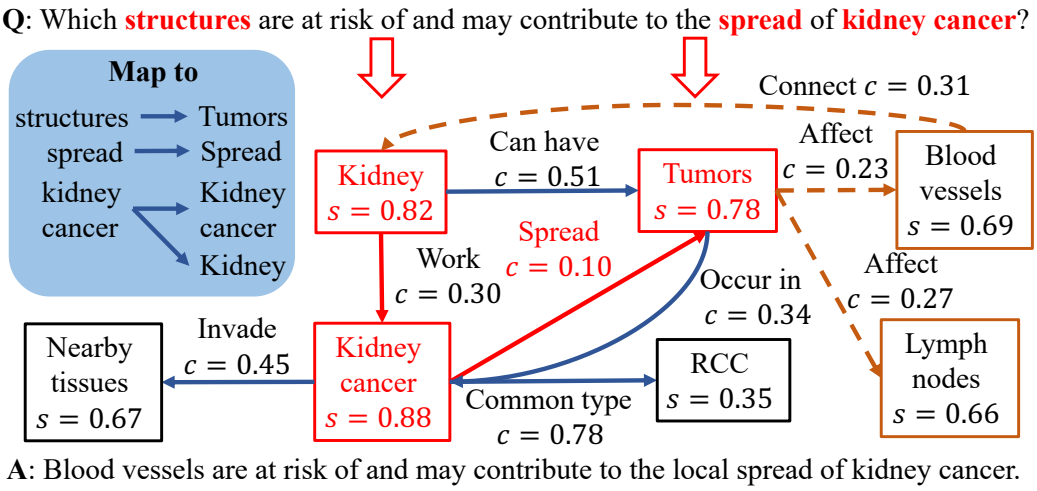


Figure 5.2: The original AGRAG evidence-expansion example. Red triples are query-mapped facts; orange triples show additional neighboring evidence in the illustrated MCMI structure. Reproduced from Figure 2 of the source paper [15].

The theoretical claims have distinct scopes. The main paper reports NP-hardness for the stated subgraph optimization problem and a 2-approximation result for the initial terminal-connecting stage. The expansion procedure adds further evidence after that stage. The ablation replaces the expanded subgraph with the initial Steiner-tree representation.

The graph returned by retrieval is serialized as $g = \mathrm{StrConcat}(\mathscr{V}_{\mathrm{MCMI}}, \mathscr{E}_{\mathrm{MCMI}})$, where $\mathscr{V}_{\mathrm{MCMI}}$ and $\mathscr{E}_{\mathrm{MCMI}}$ are its node and edge sets and StrConcat serializes them as text. This representation contains the selected entities and relations. It makes the selected relationships available within the model's input sequence.

5.5 Hybrid Retrieval and Generation

A graph index captures extracted relationships, whereas many details remain in the original text. AGRAG therefore also retrieves chunks through a hybrid score,

$$\mathrm{HS}(q,t) = \frac{\mathrm{MS}(\mathbf{q},\mathbf{t}) + \mathrm{BM25}(q,t)}{2}, \tag{5.13}$$

where q is the query, t a chunk, $\mathbf{q}$ and $\mathbf{t}$ their dense representations, MS their cosine similarity, and BM25 the lexical score. The top k_r chunks according to this hybrid score form $\mathscr{T}_{\mathrm{HR}}$. The source reports $k_r = 5$ in the main setting.

Dense retrieval scores semantic similarity, while BM25 scores lexical overlap [52]. We average them and combine the selected passages with the graph serialization.

The final answer is generated through

$$A_q = \mathrm{LLM}(q, g, \mathcal{T}_{\mathrm{HR}}, p_{\mathrm{gen}}). \tag{5.14}$$

Here A_q is the generated answer, q the query, g the serialized selected graph, $\mathcal{T}_{\mathrm{HR}}$ the hybrid passages, and p_{gen} the generation prompt. The generator receives the question, selected graph facts, and hybrid passages. Graph-only and hybrid-only ablations test the contributions of the two evidence channels.

For classification, the retrieval interface is adapted to return candidate labels associated with passages. One training passage per label supplies the indexed corpus. The generator selects a label from the retrieved candidate set. This classification index is fixed, whereas GORAG updates its index online over label rounds in Chapter 4.

5.6 Computational Properties

AGRAG's work is distributed across corpus indexing and query processing. Statistical extraction and passage-link construction operate over the corpus. Query processing includes triple mapping, filtering, graph propagation, subgraph selection, hybrid retrieval, and generation. Language-model calls contribute to relation extraction during indexing and to filtering and answer generation online.

The source's graph-operation analysis assigns $O(|\mathcal{V}||\mathcal{T}_c|)$ time to entity extraction and passage association, where $\mathcal{V}$ is the graph node set and $\mathcal{T}_c$ the chunk collection. Graph propagation is expressed in terms of node and edge counts, and semantic edge weighting visits the indexed edges. The Mehlhorn-based initial tree stage [93] is reported with $O(|\mathcal{E}| + |\mathcal{V}| \log |\mathcal{V}|)$ complexity, where $\mathcal{E}$ is the graph edge set. Expansion additionally depends on the selected subgraph and its neighborhood. These expressions characterize graph operations; the measured end-to-end results include model inference under the experimental configuration.

A fixed corpus index can be reused across queries. Mapping, filtering, graph selection, and generation recur for each question, so index preparation can be amortized while query latency still depends on graph processing and generated output length.

The index stores entity strings, relations, embeddings, passage associations, and graph adjacency. Its memory use depends on their representations. The efficiency experiments measure time and tokens.

5.7 Experimental Evaluation

5.7.1 Tasks, corpora, and model settings

We evaluate AGRAG on Novel and Medical from GraphRAG-bench [100], together with WOS [95] and IFS-Rel [10] for classification. Novel contains 5,140 chunks, 63,887 graph entities, and 163,169 graph relations. Medical contains 1,030 chunks, 12,426 entities, and 52,054 relations. The reported classification indexes contain 24 chunks for WOS and 11 for IFS-Rel, with 9,396 and 2,560 evaluated texts and 133 and 64 labels, respectively.

The Novel evaluation contains 67 creative-generation queries, 362 contextual-summarization queries, 610 complex-reasoning queries, and 971 fact-retrieval queries. The corresponding Medical counts are 166, 289, 509, and 1,098.

The main backbone is Qwen2.5-14B-Instruct [101], deployed through vLLM [102]. The machine has an Intel Xeon Gold 5220R CPU and one NVIDIA A100 GPU with 80 GB memory. Contriever [103] supplies dense representations for AGRAG and HippoRAG2; the other baselines use bge-large-en-v1.5 according to the reported configuration. Qwen3-Reranker-0.6B performs raw-triplet filtering. The retrieval top-k setting, where k is the number of retrieved items, is 5; AGRAG's raw mapped triple count and hybrid chunk count are also 5.

GraphRAG uses local search. LightRAG uses hybrid search for the QA tasks, with its classification-specific configuration retained where applicable. The comparison includes NaiveRAG, LongLLMLingua [96], RAPTOR [104], SiReRAG [105], G-Retriever [106], MIXRAG [107], GraphRAG [14], LightRAG [59], HippoRAG2 [97], and Fast-GraphRAG [108]. These methods represent passage retrieval, compression, hierarchical organization, connected subgraph selection, neighborhood retrieval, and graph propagation.

5.7.2 Metric interpretation

Classification accuracy is the proportion of predicted labels matching gold labels. Classification recall is macro recall. For the other tasks, the reported accuracy is a model-assessed score with values 0, 0.5, or 1 for inaccurate, partially accurate, or accurate answers. These quantities appear under the same ACC heading in the paper but evaluate different outputs.

Complex reasoning and fact retrieval additionally report the source's ROUGE-L measure [109]. Contextual summarization and creative generation report coverage, the fraction of required reference evidence represented in the answer. Creative generation also reports faithfulness, the fraction of answer claims supported by retrieved context. The evaluation uses

the RAGAS framework [110, 111] and the task-specific protocol. Coverage concerns inclusion of required information, whereas faithfulness concerns support for generated claims.

For generated answer sequence A_q and reference sequence A_q^{ref}, the source defines its reported lexical score as

$$\text{ROG}(A_q, A_q^{\text{ref}}) = \frac{\text{LCS}(A_q, A_q^{\text{ref}})}{|A_q|}. \tag{5.15}$$

Here A_q is the generated answer, A_q^{ref} the reference answer, and $\text{LCS}(A_q, A_q^{\text{ref}})$ the length of their longest common subsequence. The denominator is the generated answer length. This is the source's reported ROUGE-L quantity, rather than an F1 combination of longest-common-subsequence precision and recall.

Let $\mathscr{E}_{\text{ref}}$ denote the reference evidence items and let $M(e, A_q)$ indicate whether answer A_q covers item e. Coverage is

$$\text{COV}(A_q, \mathscr{E}_{\text{ref}}) = \frac{|\{e \in \mathscr{E}_{\text{ref}} : M(e, A_q)\}|}{|\mathscr{E}_{\text{ref}}|}. \tag{5.16}$$

Here A_q is the response, $\mathscr{E}_{\text{ref}}$ the reference evidence set, and $M(e, A_q)$ indicates whether the response covers evidence item e. For the answer's claim set $\mathscr{A}_q$ and retrieved context C_q, let $S(a, C_q)$ indicate that claim a is supported. Faithfulness is

$$\text{FS}(\mathscr{A}_q, C_q) = \frac{|\{a \in \mathscr{A}_q : S(a, C_q)\}|}{|\mathscr{A}_q|}. \tag{5.17}$$

Here $\mathscr{A}_q$ is the answer claim set, C_q the retrieved context, and $S(a, C_q)$ indicates whether claim a is supported by that context. Coverage measures required reference information, while faithfulness measures support for the claims actually generated. Omitting a required fact lowers coverage; adding an unsupported claim lowers faithfulness. These judgments use the model-based assessment procedure.

5.7.3 Answer quality across tasks

The main results are split by task in Tables 5.1–5.5. On Novel creative generation, AGRAG reports faithfulness 0.513, accuracy 0.339, and coverage 0.386. HippoRAG2 reports faithfulness 0.496, while RAPTOR reports accuracy 0.333 and NaiveRAG coverage 0.384.

On Medical creative generation, AGRAG reports faithfulness 0.356, accuracy 0.598, and coverage 0.589. NaiveRAG has accuracy 0.577 and coverage 0.554. HippoRAG2 has faithfulness 0.347.

Table 5.1: Creative generation with Qwen2.5-14B-Instruct. Results from the AGRAG main experiment [15].

Method	Novel FS	Novel ACC	Novel COV	Medical FS	Medical ACC	Medical COV
NaiveRAG	0.205	0.307	0.384	0.176	0.577	0.554
L.LLMLingua	0.213	0.316	0.377	0.225	0.575	0.496
RAPTOR	0.365	0.333	0.373	0.314	0.526	0.553
SiReRAG	0.324	0.328	0.365	0.295	0.544	0.456
G-retriever	0.294	0.167	0.200	0.169	0.371	0.468
MIXRAG	0.200	0.286	0.337	0.255	0.351	0.306
GraphRAG	0.427	0.310	0.339	0.345	0.475	0.325
LightRAG	0.201	0.314	0.329	0.189	0.567	0.549
HippoRAG2	0.496	0.295	0.325	0.347	0.563	0.451
F.GraphRAG	0.286	0.287	0.376	0.217	0.500	0.396
AGRAG	**0.513**	**0.339**	**0.386**	**0.356**	**0.598**	**0.589**

Table 5.2: Contextual summarization with Qwen2.5-14B-Instruct. Results from the AGRAG main experiment [15].

Method	Novel COV	Novel ACC	Medical COV	Medical ACC
NaiveRAG	0.745	0.621	0.677	0.576
L.LLMLingua	0.687	0.627	0.657	0.583
RAPTOR	0.760	0.626	0.633	0.583
SiReRAG	0.754	0.617	0.592	0.560
G-retriever	0.594	0.404	0.302	0.395
MIXRAG	0.695	0.538	0.409	0.464
GraphRAG	0.758	0.620	0.625	0.506
LightRAG	0.753	0.628	0.625	0.585
HippoRAG2	0.735	0.586	0.496	0.560
F.GraphRAG	0.706	0.528	0.507	0.587
AGRAG	**0.778**	**0.643**	**0.690**	**0.604**

Table 5.3: Complex reasoning with Qwen2.5-14B-Instruct. Results from the AGRAG main experiment [15].

Method	Novel ROG	Novel ACC	Medical ROG	Medical ACC
NaiveRAG	0.120	0.447	0.117	0.635
L.LLMLingua	0.092	0.434	0.080	0.633
RAPTOR	0.290	0.460	0.185	0.649
SiReRAG	0.307	0.459	0.269	0.646
G-retriever	0.146	0.441	0.102	0.396
MIXRAG	0.109	0.386	0.156	0.434
GraphRAG	0.246	0.449	0.250	0.606
LightRAG	0.223	0.448	0.255	0.637
HippoRAG2	0.314	0.453	0.332	0.622
F.GraphRAG	0.237	0.371	0.262	0.643
AGRAG	**0.319**	**0.467**	**0.343**	**0.665**

Table 5.4: Fact retrieval with Qwen2.5-14B-Instruct. Results from the AGRAG main experiment [15].

Method	Novel ROG	Novel ACC	Medical ROG	Medical ACC
NaiveRAG	0.117	0.431	0.182	0.604
L.LLMLingua	0.115	0.420	0.105	0.591
RAPTOR	0.133	0.465	0.132	0.609
SiReRAG	0.311	0.451	0.349	0.558
G-retriever	0.199	0.209	0.148	0.326
MIXRAG	0.102	0.366	0.160	0.463
GraphRAG	0.240	0.445	0.217	0.521
LightRAG	0.253	0.460	0.283	0.607
HippoRAG2	0.310	0.470	0.343	0.598
F.GraphRAG	0.318	0.439	0.352	0.610
AGRAG	**0.324**	**0.490**	**0.357**	**0.628**

Table 5.5: Text classification with Qwen2.5-14B-Instruct. Results from the AGRAG main experiment [15].

Method	WOS ACC	WOS REC	IFS-Rel ACC	IFS-Rel REC
NaiveRAG	0.237	0.257	0.351	0.197
L.LLMLingua	0.274	0.231	0.378	0.220
RAPTOR	0.409	0.290	0.293	0.212
SiReRAG	<u>0.430</u>	<u>0.294</u>	0.325	0.220
G-retriever	0.188	0.129	0.311	0.185
MIXRAG	0.244	0.235	0.362	0.203
GraphRAG	0.253	0.201	0.279	0.202
LightRAG	0.273	0.224	0.265	0.220
HippoRAG2	0.424	0.270	<u>0.389</u>	<u>0.246</u>
F.GraphRAG	0.283	0.245	0.307	0.221
AGRAG	**0.438**	**0.307**	**0.403**	**0.251**

For contextual summarization, AGRAG obtains coverage and accuracy of 0.778 and 0.643 on Novel, and 0.690 and 0.604 on Medical. Graph facts and passage context jointly supply information distributed across the corpus.

For complex reasoning, the reported ROUGE-L and accuracy values are 0.319 and 0.467 on Novel and 0.343 and 0.665 on Medical. For fact retrieval they are 0.324 and 0.490 on Novel and 0.357 and 0.628 on Medical. The tasks use different query collections and answer formats.

Classification accuracy and recall are 0.438 and 0.307 on WOS and 0.403 and 0.251 on IFS-Rel. SiReRAG reports WOS accuracy 0.430 and recall 0.294, while HippoRAG2 reports IFS-Rel accuracy 0.389 and recall 0.246. AGRAG's classifier receives graph-derived candidate labels under the source protocol.

5.7.4 Indexing and querying costs

Table 5.6 separates indexing time, querying time, input tokens, and output tokens. For Novel, AGRAG reports 65 minutes for indexing and 142 minutes for querying, compared with HippoRAG2's 95 and 134 minutes. For Medical the corresponding values are 16 and 138 minutes for AGRAG and 46 and 130 minutes for HippoRAG2. AGRAG reduces indexing time in both corpora but takes longer to process their queries.

The table also contains methods with lower costs in particular columns. NaiveRAG has lower

Table 5.6: Indexing / querying time in minutes. Qwen2.5-14B-Instruct; original paired measurements are preserved.

Method	Novel	Medical	WOS	IFS-Rel
NaiveRAG	13 / 371	7 / 245	2 / 292	2 / 225
L.LLMLingua	51 / 360	37 / 305	4 / 358	3 / 315
RAPTOR	464 / 250	117 / 319	30 / 162	24 / 121
SiReRAG	85 / 233	40 / 207	26 / 139	18 / 117
G-Retriever	19 / 119	12 / 145	8 / 158	3 / 164
MIXRAG	17 / 388	11 / 453	5 / 257	2 / 193
GraphRAG	347 / 205	194 / 158	85 / 240	64 / 217
LightRAG	166 / 221	116 / 164	42 / 211	31 / 197
HippoRAG2	95 / 134	46 / 130	19 / 140	15 / 127
F.GraphRAG	106 / 198	48 / 173	29 / 186	27 / 164
AGRAG	65 / 142	16 / 138	16 / 132	14 / 124

Table 5.7: Input / output tokens per query. Same source experiment as the preceding time table.

Method	Novel	Medical	WOS	IFS-Rel
NaiveRAG	9,437 / 1,708	7,473 / 379	7,292 / 29	4,521 / 16
L.LLMLingua	8,466 / 1,750	5,686 / 390	5,861 / 31	2,757 / 14
RAPTOR	7,243 / 777	4,817 / 695	6,007 / 41	3,807 / 33
SiReRAG	7,759 / 800	4,918 / 701	6,012 / 47	3,821 / 36
G-Retriever	6,344 / 116	6,290 / 157	6,762 / 12	4,002 / 21
MIXRAG	6,083 / 237	4,238 / 230	4,193 / 31	3,597 / 29
GraphRAG	12,686 / 2,109	8,385 / 1,017	11,809 / 351	9,009 / 287
LightRAG	13,145 / 1,233	9,535 / 426	11,510 / 227	7,667 / 199
HippoRAG2	8,138 / 1,073	6,264 / 595	5,066 / 93	2,440 / 102
F.GraphRAG	7,490 / 1,252	6,475 / 384	4,957 / 35	2,217 / 20
AGRAG	6,487 / 1,140	5,145 / 413	3,970 / 101	1,890 / 93

Table 5.8: Creative generation ablation with Qwen2.5-14B-Instruct. Values retain the original source ablation precision.

Method	Novel FS	Novel ACC	Novel COV	Medical FS	Medical ACC	Medical COV
AGRAG w. HR Only	0.491	0.322	0.378	0.303	0.526	0.549
AGRAG w. LLM ER	0.428	0.302	0.365	0.352	0.526	0.587
AGRAG w.o. HR	0.519	0.306	0.343	0.342	0.521	0.570
AGRAG w.o. MCMI	0.488	0.332	0.358	0.351	0.530	0.575
AGRAG w.o. Cle.	0.478	0.311	0.361	0.336	0.517	0.569
AGRAG	0.513	0.339	0.386	0.356	0.598	0.589

Table 5.9: Contextual summarization ablation with Qwen2.5-14B-Instruct. Values retain the original source ablation precision.

Method	Novel COV	Novel ACC	Medical COV	Medical ACC
AGRAG w. HR Only	0.739	0.522	0.535	0.583
AGRAG w. LLM ER	0.755	0.552	0.600	0.566
AGRAG w.o. HR	0.677	0.534	0.420	0.578
AGRAG w.o. MCMI	0.748	0.592	0.600	0.588
AGRAG w.o. Cle.	0.712	0.597	0.616	0.577
AGRAG	0.778	0.643	0.690	0.604

indexing times, while G-Retriever has a shorter Novel querying total.

On WOS and IFS-Rel, AGRAG reports querying times of 132 and 124 minutes and input token counts of 3,970 and 1,890 per query. The corresponding HippoRAG2 values are 140 and 127 minutes and 5,066 and 2,440 tokens. Output lengths differ between methods.

Table 5.10: Complex reasoning ablation with Qwen2.5-14B-Instruct. Values retain the original source ablation precision.

Method	Novel ROG	Novel ACC	Medical ROG	Medical ACC
AGRAG w. HR Only	0.303	0.453	0.334	0.636
AGRAG w. LLM ER	0.320	0.465	0.333	0.624
AGRAG w.o. HR	0.318	0.434	0.324	0.598
AGRAG w.o. MCMI	0.324	0.465	0.329	0.633
AGRAG w.o. Cle.	0.301	0.450	0.334	0.628
AGRAG	0.319	0.467	0.343	0.665

Table 5.11: Fact retrieval ablation with Qwen2.5-14B-Instruct. Values retain the original source ablation precision.

Method	Novel ROG	Novel ACC	Medical ROG	Medical ACC
AGRAG w. HR Only	0.323	0.487	0.355	0.619
AGRAG w. LLM ER	0.322	0.480	0.353	0.613
AGRAG w.o. HR	0.314	0.487	0.344	0.615
AGRAG w.o. MCMI	0.322	0.470	0.359	0.621
AGRAG w.o. Cle.	0.313	0.478	0.354	0.620
AGRAG	0.324	0.490	0.357	0.628

Table 5.12: Ablation time and token costs. Time is in minutes. Paired token columns retain the source I/G headings; its caption describes input/output token counts.

Method	Novel time I/Q	Novel tokens I/G	Medical time I/Q	Medical tokens I/G
AGRAG w. HR Only	11.7 / 137.9	5,117 / 193	11.4 / 117.9	5,215 / 199
AGRAG w. LLM ER	83.3 / 161.2	8,030 / 887	31.9 / 148.8	5,457 / 414
AGRAG w.o. HR	61.7 / 138.9	6,986 / 741	9.4 / 114.4	4,820 / 315
AGRAG w.o. MCMI	63.9 / 130.4	6,710 / 757	9.5 / 125.0	5,230 / 331
AGRAG w.o. Cle.	65.6 / 140.1	6,988 / 154	9.1 / 142.9	5,252 / 158
AGRAG	65.6 / 142.0	6,988 / 639	9.1 / 145.3	5,252 / 306

5.7.5 Component ablations

The ablation compares hybrid retrieval alone, language-model entity extraction, graph retrieval without hybrid context, an initial tree without MCMI expansion, and retrieval without the raw-triplet filtering step.

The hybrid-only variant has Novel creative-generation faithfulness 0.491, accuracy 0.322, and coverage 0.378. Full AGRAG reports 0.513, 0.339, and 0.386. For Novel contextual summarization, hybrid-only coverage and accuracy are 0.739 and 0.522, compared with 0.778 and 0.643 for the complete system. Adding graph retrieval improves creative-generation scores and summarization accuracy.

Removing hybrid retrieval produces a different pattern. Novel creative-generation faithfulness is 0.519, exceeding the complete system's 0.513, while accuracy and coverage decrease to 0.306 and 0.343. Medical contextual-summarization coverage decreases from 0.690 to 0.420. The graph-only variant can produce well-grounded claims while supplying less of the required information.

Replacing statistical entity extraction with model-based extraction gives Novel creative-generation faithfulness 0.428 and accuracy 0.302, with indexing time 83.3 minutes in the ab-

Table 5.13: Creative generation in the low-resource setting. Qwen3-4B-Instruct-2507 on two 2080 Ti GPUs; values and emphasis preserved.

Method	Novel FS	Novel ACC	Novel COV	Medical FS	Medical ACC	Medical COV
SiReRAG-L	0.324	0.412	0.299	0.421	0.758	0.475
HippoRAG2-L	0.433	0.481	0.335	0.425	0.755	0.433
AGRAG-L	**0.441**	**0.519**	**0.354**	**0.436**	**0.777**	**0.490**

Table 5.14: Contextual summarization in the low-resource setting. Qwen3-4B-Instruct-2507 on two 2080 Ti GPUs; values and emphasis preserved.

Method	Novel COV	Novel ACC	Medical COV	Medical ACC
SiReRAG-L	0.647	0.673	0.536	0.738
HippoRAG2-L	0.696	0.657	0.522	0.731
AGRAG-L	**0.745**	**0.706**	**0.545**	**0.751**

lation setting. The complete configuration reports 65.6 minutes in that table. This experiment changes entity extraction while retaining the rest of the method.

The variant without MCMI expansion uses a Steiner-tree representation. Its Novel contextual-summarization accuracy is 0.592, compared with 0.643 for full AGRAG, and its Medical creative-generation accuracy is 0.530, compared with 0.598. Some fact-retrieval metrics are close or favor the tree variant. Expansion improves these summarization and creative-generation scores, while the initial tree remains competitive on fact retrieval.

Removing raw-triplet filtering gives Novel contextual-summarization coverage 0.712 and accuracy 0.597. These are below 0.778 and 0.643 for full AGRAG. Filtering the mapped triples improves coverage and accuracy.

The main efficiency and ablation tables report different Medical indexing and querying totals. Component comparisons use the measurements in the ablation table.

5.7.6 Additional resource setting and parameter study

The low-resource experiment uses Qwen3-4B-Instruct-2507 on two NVIDIA 2080 Ti GPUs. Both the backbone and accelerators differ from the main experiment, so their individual effects are not isolated.

In that setting, AGRAG-L reports Novel creative-generation accuracy 0.519 and contextual-summarization accuracy 0.706. Its Novel index/query totals are 44.7 and 132.1 minutes. Medical totals are 14.4 and 167.3 minutes. HippoRAG2-L has shorter query totals in both corpora, while

Table 5.15: Complex reasoning in the low-resource setting. Qwen3-4B-Instruct-2507 on two 2080 Ti GPUs; values and emphasis preserved.

Method	Novel ROG	Novel ACC	Medical ROG	Medical ACC
SiReRAG-L	0.260	0.426	0.308	0.698
HippoRAG2-L	0.280	0.507	0.298	0.681
AGRAG-L	**0.293**	**0.552**	**0.312**	**0.739**

Table 5.16: Fact retrieval in the low-resource setting. Qwen3-4B-Instruct-2507 on two 2080 Ti GPUs; values and emphasis preserved.

Method	Novel ROG	Novel ACC	Medical ROG	Medical ACC
SiReRAG-L	0.271	0.416	0.343	0.661
HippoRAG2-L	0.288	0.513	0.338	0.654
AGRAG-L	**0.297**	**0.560**	**0.351**	**0.701**

Table 5.17: Novel low-resource efficiency. Qwen3-4B-Instruct-2507 on two 2080 Ti GPUs.

Method	Index min	Query min	Input tokens	Output tokens
SiReRAG-L	52.0	183.5	7,898	**1,003**
HippoRAG2-L	55.7	**131.2**	7,964	1,354
AGRAG-L	**44.7**	132.1	**6,625**	1,172

Table 5.18: Medical low-resource efficiency. Qwen3-4B-Instruct-2507 on two 2080 Ti GPUs.

Method	Index min	Query min	Input tokens	Output tokens
SiReRAG-L	18.1	182.3	**5,113**	827
HippoRAG2-L	21.8	**162.4**	5,577	584
AGRAG-L	**14.4**	167.3	5,282	**534**

AGRAG-L has shorter indexing totals.

The parameter study varies the maximum n-gram length over $\{1, 3, 5, 7\}$ and the extraction threshold over $\{0.1, 0.3, 0.5, 0.7\}$. The original heatmaps are reproduced in Figure 5.3. They show accuracy across the tested combinations for Novel and Medical. The main experiment uses maximum n-gram length $b = 3$ and extraction threshold $\tau = 0.5$.

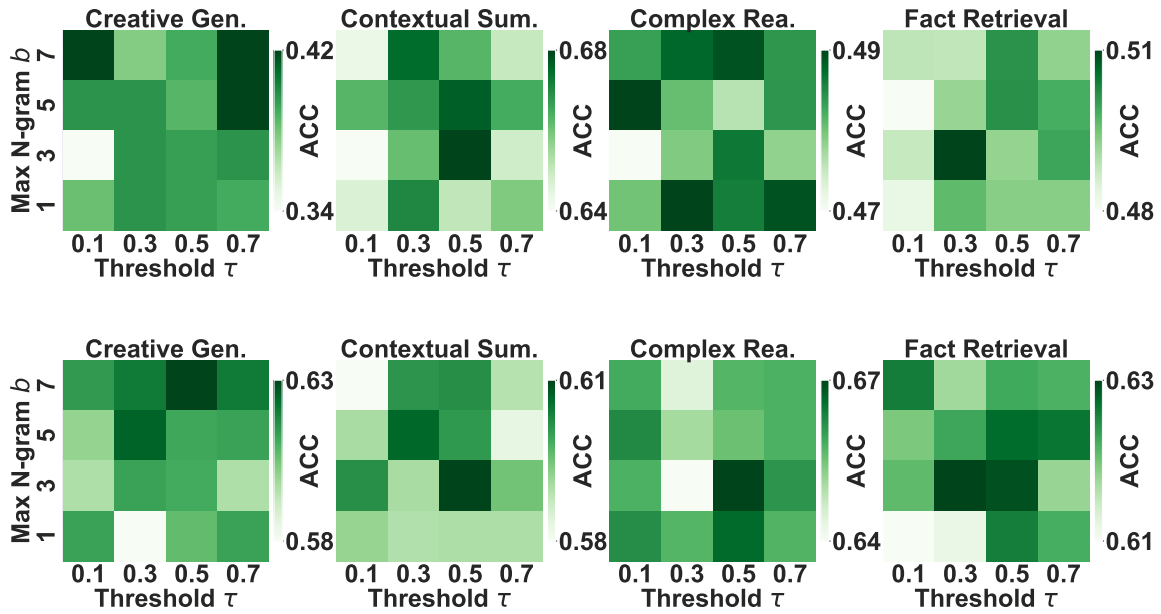


Figure 5.3: AGRAG parameter sensitivity on Novel (top) and Medical (bottom). The original panels vary the extraction threshold and maximum n-gram length and display task accuracy. Reproduced from Figure 3 of the source paper [15].

5.8 Discussion and Chapter Summary

We proposed AGRAG to retrieve connected facts while retaining their source context. Query-conditioned influence and semantic edge costs guide subgraph expansion, and hybrid passage retrieval supplements the selected graph.

Removing hybrid context can retain high faithfulness while reducing coverage. Graph facts can support the claims that are generated yet omit information needed for the full answer. Expansion beyond the initial tree also helps summarization and creative generation, while its effect on fact retrieval is smaller.

Relations are model-extracted, query anchors are model-filtered, and several QA metrics use model assessment. Errors in the extracted relations or filtered anchors can change the evidence supplied to generation.

AGRAG extends task-related graph retrieval to answers requiring connected facts and

broader information coverage. Its corpus graph supports repeated textual queries through query-conditioned evidence selection. Chapter 6 examines the complementary problem of multimodal evidence, where the query guides graph construction and a requested relation may correspond to a path with a different topology.

CHAPTER 6

SEMANTIC GRAPH MATCHING FOR MULTIMODAL REASONING WITH MGRAG

This chapter is based on *MGRAG: Semantic Subgraph Matching and Graph-Aware Caching for Multimodal Retrieval-Augmented Generation*, published in the *Proceedings of the VLDB Endowment*, volume 19, number 9, pages 2480–2493, 2026 [19]. We propose MGRAG to construct query-specific multimodal graphs, match query relations to semantic paths, and guide selective recomputation of cached model states.

6.1 Multimodal Evidence and Query Requirements

Constructing graphs from task data removes the need for a predefined external knowledge graph, as explored in GORAG. AGRAG further studies how connected textual facts support answer generation. For multimodal question answering, the facts to extract can depend on the question, and connectivity alone leaves their semantic correspondence to the query unresolved. A passage may name a person, an image may show an action, and another passage may link that person to an event. To meet these requirements, we construct query-related graphs and match their paths to the relationships requested by the question.

This chapter considers a corpus $\mathscr{D} = \mathscr{D}_{\text{txt}} \cup \mathscr{D}_{\text{img}}$, comprising text documents $\mathscr{D}_{\text{txt}}$ and images $\mathscr{D}_{\text{img}}$, and a query q that can contain text, an image, or both. A retrieval module selects evidence from the corpus, and a multimodal language model generates an answer using that evidence. The research question is how to construct and retrieve a query-relevant relational representation while controlling the cost of preparing and processing the context. The answer-generation task remains grounded in the original documents and images. The graph supplies an additional representation of their query-related content.

The first challenge is extracting the right facts. A corpus-wide graph is constructed before the question is known, and its extraction prompt may omit a visual detail needed later. We instead include the query in extraction and incrementally construct an evidence graph from ranked chunks.

The second challenge is matching the relations requested by the query to those expressed by

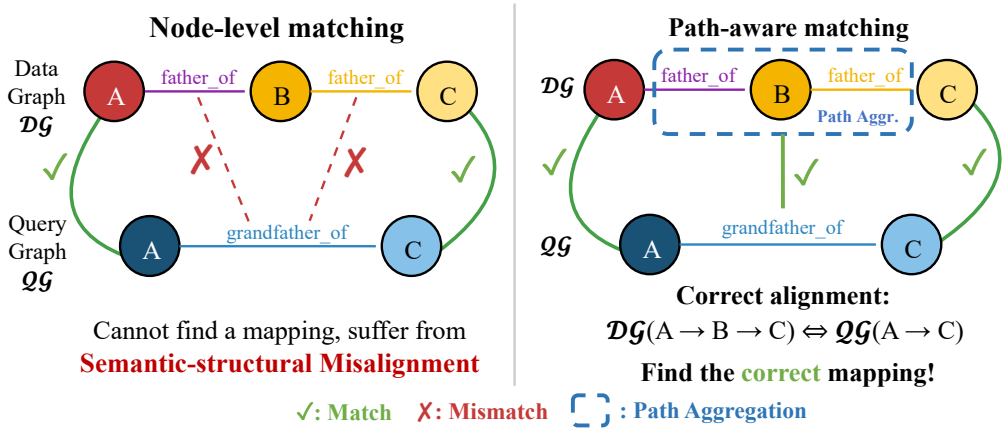


Figure 6.1: Path-level correspondence in the MGRAG motivating example. A `grandfather_of` relation corresponds to two `father_of` edges. Original Figure 1, p. 2481 [19].

the evidence. A query and its supporting documents need not use the same decomposition of a fact. The published motivating example describes a data graph containing two `father_of` edges and a query containing one `grandfather_of` edge. The requested relation is represented by a two-hop path in the data. Requiring the same edge topology would exclude this correspondence. Figure 6.1 reproduces the example. MGRAG evaluates semantic representations of paths so that a query relation can be supported by a sequence of evidence relations.

The third challenge is reducing repeated computation when processing retrieved evidence. Graph construction already sends evidence to a language model to extract entities and relations. Some of that content appears again when the matched graph and its source chunks are used for answering. Reusing the earlier key-value (KV) states can reduce repeated computation. Their suitability depends on the new query and context, so MGRAG selectively recomputes token states. It combines attention scores with topological features of the facts associated with those tokens. Structure thus enters both evidence selection and the subsequent use of the selected context.

6.2 Framework and Evidence Representations

Figure 6.2 shows three stages. Coarse-grained ranking orders text chunks and images according to their relevance to the query. Fine-grained retrieval consumes this ranked list, extracts query-related facts, and searches the growing graph for a semantic match. Graph-aware KV recomputation then determines which cached token states to update for answer generation. The stopping decision in the middle stage connects ranking, graph construction, and matching. Each additional chunk can change whether the graph contains a suitable correspondence.

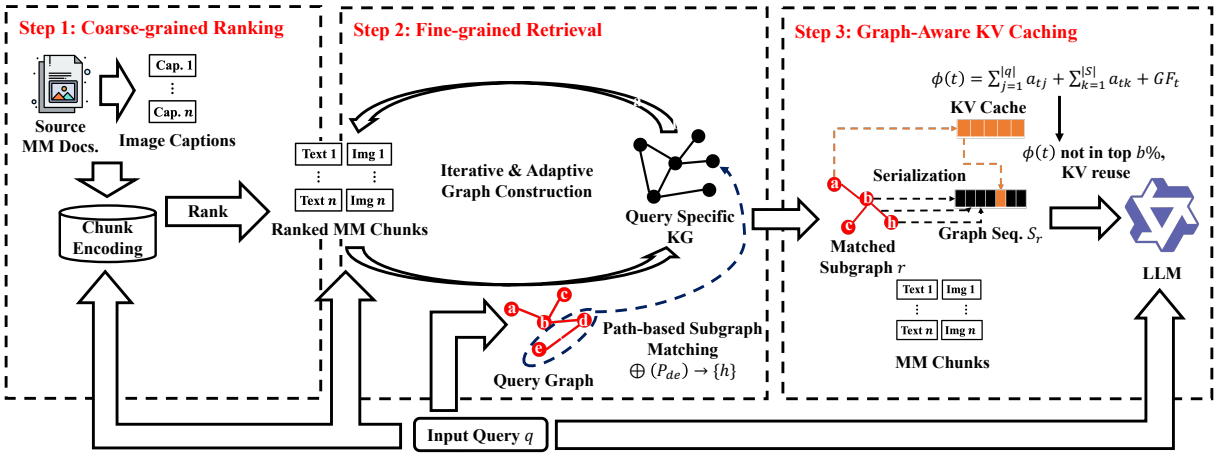


Figure 6.2: The MGRAG pipeline. Ranked chunks support incremental query-aware graph construction and path-based matching. The matched graph also supplies features for selective KV recomputation. Original Figure 2, p. 2484 [19].

Two graphs serve different purposes. The query graph $\mathcal{QG}_q = (\widehat{\mathcal{V}}_q, \widehat{\mathcal{E}}_q)$ describes the entities and relations requested by query q, including placeholders for unknown information. Its node and edge sets are $\widehat{\mathcal{V}}_q$ and $\widehat{\mathcal{E}}_q$. The evidence graph $\mathcal{KG}_q = (\mathcal{V}_q, \mathcal{E}_q)$ contains facts extracted from the chunks considered for that query, with node set $\mathcal{V}_q$ and edge set $\mathcal{E}_q$. A mapping connects query entities to evidence entities. Paths between the mapped endpoints supply candidate evidence for the query relations.

For generation, MGRAG combines the graph sequence S_r, the original chunks $\mathcal{C}_q$ consumed during construction, and the query q. The chunks retain source text and original images, including details omitted from the extracted triples.

6.3 Coarse-grained Ranking

MGRAG first divides textual documents into overlapping chunks. The published setting uses a chunk length $l_c = 256$ tokens and overlap $l_o = 32$ tokens. Images form the other members of the chunk collection, giving $\mathcal{C} = \mathcal{T} \cup \mathcal{D}_{\text{img}}$, where $\mathcal{T}$ denotes the text chunks. This collection is the input to the coarse retrieval stage. Ranking identifies the order in which content will be examined, rather than directly defining the final graph context.

The implementation uses the text encoder bge-base-en-v1.5 [112]. Text chunks are encoded directly. An image is represented for ranking by its caption, which can be supplied by the dataset or generated by a language model. Both types of evidence can therefore be compared in the text embedding space. Let $\mathcal{M}$ denote that encoder and $\mathbf{h}_c$ the representation of a candidate chunk. For an image chunk, the input to $\mathcal{M}$ is its caption. Ranking therefore depends on the caption,

while extraction and answering can use the original image.

For a query containing text q_{txt} and an image caption q_{cap}, MGRAG averages the two representations,

$$\mathbf{h}_q = \frac{\mathscr{M}(q_{\text{txt}}) + \mathscr{M}(q_{\text{cap}})}{2}. \tag{6.1}$$

Here $\mathbf{h}_q$ is the query embedding, $\mathscr{M}$ is the text encoder, and q_{txt} and q_{cap} are the query text and image caption. A query with one modality uses its corresponding textual representation. The distance to a candidate is

$$d(q, c) = \|\mathbf{h}_q - \mathbf{h}_c\|_2, \tag{6.2}$$

where $d(q, c)$ is the Euclidean distance between query embedding $\mathbf{h}_q$ and candidate chunk embedding $\mathbf{h}_c$; the chunks are ordered by increasing distance. Equations 6.1 and 6.2 define the query representation and ranking distance.

Fine-grained retrieval processes chunks in coarse-ranking order. Each chunk can add facts needed for a match, and construction stops when matching succeeds or the iteration budget is exhausted.

Caption quality affects which images receive an early position. A caption can omit a visual detail that becomes relevant during graph construction, or foreground content unrelated to the query. The published CLIP substitution experiment tests an alternative coarse representation within this pipeline. Section 6.11 reports the encoder comparison.

6.4 Query-aware Graph Construction

6.4.1 Representing the query

We extract entities and relations from the query. Known entities retain their descriptions, while requested values use semantic placeholders such as *Person* or *Year*. The matcher can compare these placeholders with evidence entities before the answer values are known.

Figure 6.3 shows the query and evidence graph representations. The placeholder boxes describe missing knowledge in the query. Passage nodes connect extracted entities back to their chunks. These connections let the graph carry both relational content and its association with the original multimodal evidence. In a multimodal corpus, a passage node can represent an image as well as a text chunk.

The original example asks which year an Irandhir Santos film was released when its poster shows people doing martial arts. The query graph connects the actor to an unknown film, the film

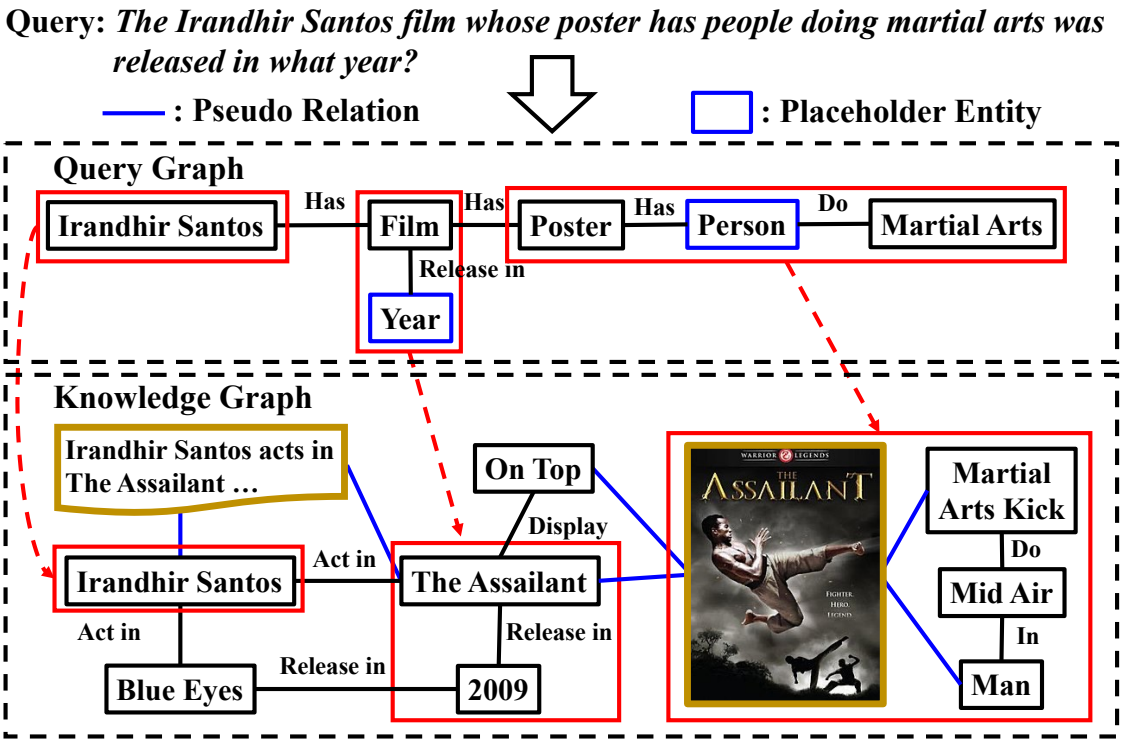


Figure 6.3: Query and evidence graphs in path-based semantic matching. Placeholder entities express unknown query information; pseudo-relations connect passage nodes to extracted entities. Original Figure 3, p. 2485 [19].

to a poster, and a person on the poster to an action. It also includes a release-year requirement. The evidence graph contains named film candidates and a visual description of a man performing a martial-arts kick in midair. The depicted film, *The Assailant*, is connected to the year 2009.

The actor name leads to several film candidates. The poster action distinguishes the requested film, whose release relation then supplies the year. Matching must therefore connect the actor, film, visual action, and release date.

The film placeholder aligns with a candidate film entity, and the year placeholder marks the requested release value. Path matching checks the relationships connecting those entities.

Extraction errors in query placeholders or relations can change the evidence accepted by the matcher.

6.4.2 Extracting facts from ranked chunks

For a chunk c, the extraction model receives the chunk, query, and extraction prompt p_{ext}. Its output contains query-related entities and relations,

$$(\mathcal{V}_q^c, \mathcal{E}_q^c) = \text{LLM}(c, q, p_{\text{ext}}). \tag{6.3}$$

Here $\mathcal{V}_q^c$ and $\mathcal{E}_q^c$ are the entities and relations extracted from chunk c for query q, and p_{ext} is the extraction prompt. Including q makes the extraction selective. The model is asked to identify events and relations relevant to the current question, rather than to enumerate every describable fact in the chunk. For an image, this permits attention to query-related visual events; for text, it permits selection of the entities and relationships needed to describe the requested evidence.

The chunk is also added as a passage node. MGRAG connects this node to the entities extracted from it using the pseudo-relation `consists_in`, retaining the published relation name and orientation from chunk to entity. If $\mathcal{C}_q^{(i)}$ is the prefix of ranked chunks consumed after iteration i, the cumulative graph contains

$$\mathcal{V}_q^{(i)} = \bigcup_{c \in \mathcal{C}_q^{(i)}} (\mathcal{V}_q^c \cup \{c\}), \tag{6.4}$$

$$\mathcal{E}_q^{(i)} = \bigcup_{c \in \mathcal{C}_q^{(i)}} \left(\mathcal{E}_q^c \cup \{(c, \texttt{consists_in}, v) : v \in \mathcal{V}_q^c\} \right). \tag{6.5}$$

Here $\mathcal{V}_q^{(i)}$ and $\mathcal{E}_q^{(i)}$ are the cumulative nodes and edges after iteration i, $\mathcal{C}_q^{(i)}$ is the processed chunk prefix, and $\mathcal{V}_q^c$, $\mathcal{E}_q^c$ are the entities and relations extracted from chunk c. The added edges connect each chunk c to an extracted entity v.

Passage nodes connect extracted facts to their source chunks and can link facts from the same source item.

6.4.3 Matching as a stopping decision

After each graph update, MGRAG attempts path-based semantic subgraph matching against the query graph. A valid match ends the construction loop. If the matcher returns an empty result, the next ranked chunk is used to enrich the graph. The loop is bounded by the maximum number of construction iterations μ. Thus, the amount of graph construction depends on both the ranking order and the matching outcome.

If a required endpoint or path is missing, another chunk can supply it. Matching thus determines whether construction continues, using the relational requirements of the query graph.

The iteration cap limits how far construction proceeds through the ranked list. A valid match can stop it earlier, while relevant evidence below the cap remains unexamined.

6.5 Path-based Semantic Subgraph Matching

6.5.1 From edges to paths

A relation in the query graph can be expressed by several relations in the evidence graph. MGRAG assigns text embeddings to entity and relation descriptions. For a path $P(u, v)$ containing relations $r_1, \ldots, r_k$, it defines a semantic representation by aggregating the relation embeddings,

$$\mathbf{h}_{P(u,v)} = \bigoplus_{j=1}^{k} \mathbf{h}_{r_j}. \tag{6.6}$$

Here $\mathbf{h}_{r_j}$ is the embedding of relation r_j, $\mathbf{h}_{P(u,v)}$ represents the path from u to v, and $\bigoplus$ is an aggregation function, such as mean pooling, over its k relation embeddings. Greedy extension additionally compares the current path, candidate edge, and candidate endpoint with the query relation.

Let $I : \widehat{\mathscr{V}_q} \to \mathscr{V}_q$ map query entities to evidence entities. For a query edge $(\hat{u}, \hat{r}, \hat{v})$, let $\mathscr{P}(I(\hat{u}), I(\hat{v}))$ be the evidence paths between its mapped endpoints. The matching objective is

$$\max_{I} \sum_{(\hat{u},\hat{r},\hat{v}) \in \widehat{\mathscr{E}_q}} \max_{P \in \mathscr{P}(I(\hat{u}), I(\hat{v}))} \cos\big(\mathbf{h}_{\hat{r}}, \mathbf{h}_P\big). \tag{6.7}$$

Here $\widehat{\mathscr{E}_q}$ is the query edge set, $\mathbf{h}_{\hat{r}}$ and $\mathbf{h}_P$ represent the query relation and evidence path, and cos measures their cosine similarity. Candidate mappings must satisfy the semantic threshold θ for the mapped entities and for every query relation's supporting path,

$$\cos\big(\mathbf{h}_{\hat{u}}, \mathbf{h}_{I(\hat{u})}\big) > \theta \qquad \text{for all } \hat{u} \in \widehat{\mathscr{V}_q}, \tag{6.8}$$

$$\max_{P \in \mathscr{P}(I(\hat{u}), I(\hat{v}))} \cos\big(\mathbf{h}_{\hat{r}}, \mathbf{h}_P\big) > \theta \qquad \text{for all } (\hat{u}, \hat{r}, \hat{v}) \in \widehat{\mathscr{E}_q}. \tag{6.9}$$

Here $\hat{u}, \hat{v}$ are query entities, $I(\hat{u}), I(\hat{v})$ are their evidence counterparts, and θ is the cosine-similarity threshold. The constraints validate each query edge against an associated evidence path. The selected evidence is the subgraph formed from the mapped entities and accepted paths, which can include intermediate entities absent from the query graph.

For a fixed entity mapping, the inner maximization in Equation 6.7 selects a path for each query relation. The outer search compares mappings using these path scores. Similar endpoints can therefore produce different matching scores when their connecting paths express different relations.

The returned subgraph is the union of the accepted paths, which can share nodes and edges. Intermediate entities allow a relation stated compactly in the question to match a longer description in the evidence.

In the family-relation example, two father relations support one grandfather relation. The path similarity must pass validation, and its endpoints must also align with the query entities.

6.5.2 Anchor selection and candidate mappings

MGRAG first computes pairwise cosine similarities between query entities and evidence entities. The highest-similarity pair supplies an anchor entity in the query graph and its candidate counterpart in the evidence graph. If no valid match is obtained from that anchor, the method tries other anchors. For the selected query anchor γ, Algorithm 1 considers each evidence counterpart v_a exceeding the similarity threshold and enumerates mappings consistent with that correspondence.

Candidate mappings are enumerated using a similarity-based modification of VF2 [113]. VF2 supplies a graph-matching search procedure; the MGRAG adaptation uses the cosine-similarity threshold θ in semantic feasibility checks. Each candidate is then validated relation by relation through greedy path search. A candidate entity correspondence can consequently be rejected because one of its required relations lacks an acceptable evidence path.

For each accepted mapping, the algorithm takes the union of the paths that support its query edges. It accumulates their semantic similarities in s and divides by the number of query edges to obtain the average score. The highest-scoring accepted subgraph is returned. Dividing by the number of query edges does not change the ranking of candidate mappings for the same query. It provides an average semantic score with a common range across queries whose graph sizes differ.

6.5.3 Greedy semantic path growth

The path search begins at the mapped source entity and maintains its current endpoint x. For each neighboring entity y and connecting edge (x, r, y), MGRAG evaluates the candidate extension using

$$\cos\left(\mathbf{h}_P \oplus \mathbf{h}_{e(x,r,y)} \oplus \mathbf{h}_y, \mathbf{h}_{\hat{e}(\hat{u},\hat{r},\hat{v})}\right). \tag{6.10}$$

Here $\mathbf{h}_P$ represents the current path, $\mathbf{h}_{e(x,r,y)}$ the candidate edge, $\mathbf{h}_y$ its endpoint, and $\mathbf{h}_{\hat{e}(\hat{u},\hat{r},\hat{v})}$ the query edge. The operator $\oplus$ aggregates these representations and cos compares the resulting extension with the query edge. Among extensions exceeding the similarity threshold θ, it selects

Algorithm 1 Path-based semantic matching in MGRAG

Require: Query graph $\mathcal{QG}_q$, query anchor γ, evidence graph $\mathcal{KG}_q$, threshold θ

Initialize valid results $\mathcal{R} \leftarrow \varnothing$
for each $v_a \in \mathcal{V}_q$ with $\cos\left(\mathbf{h}_{v_a}, \mathbf{h}_{\gamma}\right) > \theta$ **do**
  Enumerate candidate mappings consistent with $I(\gamma) = v_a$
  for each candidate mapping I **do**
    Initialize the subgraph with mapped query entities; set $s \leftarrow 0$
    for each query edge $(\hat{u}, \hat{r}, \hat{v})$ **do**
      Search greedily for a semantically valid path from $I(\hat{u})$ to $I(\hat{v})$
      if the path search fails **then**
        Reject I and stop validating this mapping
      else
        Add the path's nodes and edges to the candidate subgraph
        Add its cosine similarity to s
      end if
    end for
    if every query edge has a valid path **then**
      Add the subgraph and score $s/|\widehat{\mathcal{E}}_q|$ to $\mathcal{R}$
    end if
  end for
end for
Return the highest-scoring result, or $\varnothing$ if $\mathcal{R}$ is empty

the highest-scoring one, appends that edge to the path, and updates the endpoint to y. This repeats until the path reaches the mapped target. The search fails when no qualifying extension is available. The experiments also impose a maximum path length l_p to bound expansion.

A path must reach the mapped target and pass the semantic checks. Greedy selection can follow a locally attractive extension that blocks a valid route, so search failure does not imply that no valid path exists. The algorithm returns the best accepted candidate it finds.

A mapping is accepted only when every query edge has a valid path. Accepted paths may have different lengths. The construction loop then stops; otherwise, it adds the next ranked chunk and repeats matching.

6.5.4 Published algorithmic properties

The publication states that path-based semantic subgraph matching is NP-hard. Its proof sketch reduces subgraph isomorphism to a restricted instance with uniform embeddings and unit path lengths. The result motivates the use of semantic path growth and constrained mapping enumeration. The complexity of enumerating mappings remains relevant even when each individual extension is selected greedily.

The published Theorem 2 gives a threshold-based quality condition for a valid returned match. Every matched query relation has a supporting path with cosine similarity at least the threshold θ. Under the strict acceptance test in Equation 6.9, the average score of an accepted mapping exceeds θ as well. This condition describes the semantic validation of returned evidence. It is conditional on finding a valid match and does not require the algorithm to recover one whenever such a subgraph exists.

The technical report discusses the threshold in terms of a bottleneck transport cost [114]. With a per-edge cost of one minus cosine similarity, the threshold limits the largest accepted local cost. Section 6.9 evaluates graph-matching substitutions and answer quality.

6.5.5 Serializing selected evidence

After matching, MGRAG serializes the selected triples into S_r, separating facts with `[SEP]`. This sequence is combined with the query q and the original chunks consumed during construction. The serialization supplies an explicit list of entities and relationships to the answering model. Original chunks remain available because extraction is a selective description of the evidence, and some linguistic or visual details may lie outside the triples.

Construction and answering process overlapping content. We reuse cached states from construction and recompute selected tokens for the answering context.

6.6 Graph-aware KV Recomputation

6.6.1 Reuse and selective updates

MGRAG reuses key and value states from graph construction when their content appears in the answering context. Because the assembled context differs from the earlier extraction input, it recomputes selected token states and reuses the rest.

Let $\phi(t)$ be the recomputation score of token t. At each transformer layer, MGRAG recomputes the tokens in the highest $b\%$ of scores and reuses the cached states for the remaining tokens. The published description uses a fixed-size cache with least-recently-used eviction. Its proposed policy concerns the ranking of token states for selective recomputation. The budget is represented by b; the numerical experiment tables do not specify a fixed value for that budget.

Figure 6.4 shows how the graph contributes. Query-related events are represented by entities and relations. The score combines attention to the query, attention to the graph sequence, and a

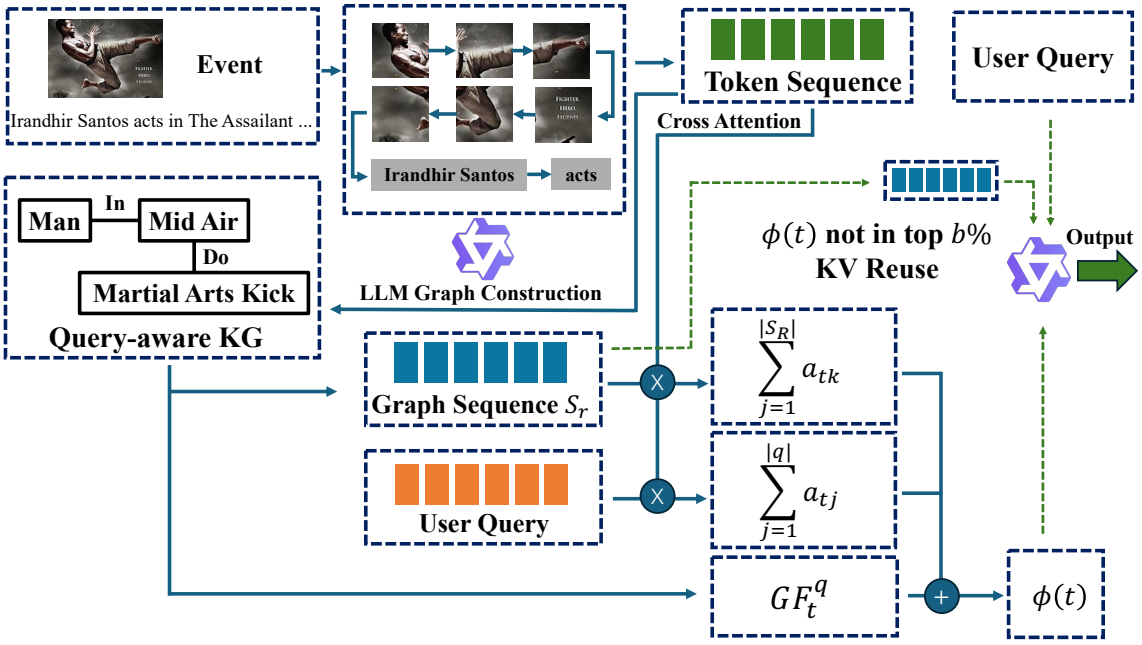


Figure 6.4: Graph features supplement attention in token recomputation scoring. The diagram retains the original graphical content. Original Figure 4, p. 2487 [19].

feature derived from the topology of the associated fact. The goal of the added term is to give tokens access to an event-level structural signal alongside token-level attention.

6.6.2 Topological features of facts

MGRAG uses degree, betweenness, eigenvector, and closeness centrality to characterize a node. Degree centrality measures its normalized number of adjacent connections. Betweenness characterizes its participation in paths connecting other nodes. Eigenvector centrality weights connections by the centrality of their neighbors. Closeness uses distances to other graph nodes. The published method averages these four quantities,

$$f(u) = \frac{dc(u) + bc(u) + ec(u) + cc(u)}{4}. \tag{6.11}$$

Here $f(u)$ is the feature of node u, and dc, bc, ec, and cc denote degree, betweenness, eigenvector, and closeness centrality, respectively. For a token associated with a triple (u, r, v), its graph feature is the mean of the endpoint scores,

$$GF_t^q = \frac{f(u) + f(v)}{2}. \tag{6.12}$$

Here GF_t^q is the graph feature of token t for query q, and $f(u)$ and $f(v)$ are the features of the triple endpoints. The token-to-fact association follows the entity and relation token spans in the serialized representation. Tokens referring to the same fact therefore receive a shared structural component.

The structural feature measures an entity's position in the query-specific graph. Nodes connecting several extracted facts can receive different scores from isolated endpoints.

The graphs contain fewer than twenty entities on average in the published evaluation. We compute the four centrality measures on these query-specific graphs.

6.6.3 Combined recomputation score

Let a_{tj} denote attention between token t and token j, and let $|q|$ and $|S_r|$ denote the query and serialized-graph lengths. The score is

$$\phi(t) = \sum_{j=1}^{|q|} a_{tj} + \sum_{k=1}^{|S_r|} a_{tk} + GF_t^q. \tag{6.13}$$

Here $\phi(t)$ is the score for token t, a_{tj} and a_{tk} are its attention scores to query token j and graph-sequence token k, and GF_t^q is its graph feature for query q. The score adds attention to the question, attention to the serialized graph, and the token's graph feature. Tokens with higher combined scores receive priority for recomputation.

The publication compares the policy with vLLM prefix caching [102] and CacheBlend [72]. Their retrieval pipelines and cache variants are identified separately in the evaluation. The inverse ablations also evaluate graph-aware recomputation within an existing retrieval pipeline. These comparisons measure how combining retrieval and caching changes answer quality and processing time.

6.7 Computational Cost

The published offline encoding cost is $O(|\mathscr{C}|d_{\text{emb}}l_c^2)$, where $\mathscr{C}$ is the chunk collection, d_{emb} is the embedding dimension, and l_c is the text chunk length. Using k processed chunks, query-graph vertex and edge sets $\widehat{\mathscr{V}}$ and $\widehat{\mathscr{E}}$, evidence-graph sets $\mathscr{V}$ and $\mathscr{E}$, maximum path length l_p,

and language-model hidden dimension d_{LLM}, its reported online bound is

$$O\big(d_{\mathrm{emb}} k l_c^2 + l_p |\widehat{\mathscr{E}}|\, |\mathscr{V}|^{|\widehat{\mathscr{V}}|+1} + |\mathscr{V}||\mathscr{E}| + d_{\mathrm{LLM}}(|S_r|^2 + |q|^2)\big). \tag{6.14}$$

Here k is the number of processed chunks, l_c their text token length, l_p the path-length bound, d_{emb} and d_{LLM} are embedding and model hidden dimensions, and $|S_r|$ and $|q|$ are graph-sequence and query lengths. The sets $\widehat{\mathscr{V}}$, $\widehat{\mathscr{E}}$ and $\mathscr{V}$, $\mathscr{E}$ contain query-graph and evidence-graph nodes and edges, respectively. The mapping-enumeration term is exponential in query-graph size. Query-specific construction and a bounded path length restrict the graphs processed in the evaluated setting. Module-level runtime measurements show where the computation is spent. In particular, graph construction calls the language model repeatedly, whereas matching operates on the extracted graph. Section 6.10 reports these costs individually.

6.8 Experimental Protocol

We evaluate MGRAG on MMQA [4] and the Arxiv, Manual, Web, Wiki, Wit, and Recipe subsets of MRAMG-Bench [71]. Table 6.1 preserves the published dataset statistics. MMQA includes textual and image queries. The six MRAMG subsets contain textual queries over multimodal corpora. Their query-hop ranges and corpus sizes differ, so results are reported by dataset.

Table 6.1: Evaluation datasets, reproduced from published Table 3, p. 2488 [19]. T and I denote textual and image query modalities. The columns comprise MMQA and six MRAMG subsets.

Statistic	MMQA	Wiki	Wit	Recipe	Arxiv	Manual	Web
Query Modality	T / I	T	T	T	T	T	T
Queries	951	500	600	2,360	200	390	750
Texts	218,285	538	639	1,528	101	40	1,500
Images	57,058	538	639	8,569	337	2,607	1,500
Corpus Size	22.5M	680k	478k	1.1M	136k	420k	215k
Query Hops	1–2	2–5	2–4	1–6	3–4	2–6	2–3
Avg Query Hops	1.5	3.0	2.8	3.3	3.1	3.7	2.4

The shared generation backbone is Qwen2.5-VL-7B-Instruct [115]. The experiments run on an Intel Xeon Gold 5220R CPU at 2.20 GHz and one NVIDIA A100-SXM4 GPU with 40 GB memory. The retrieval amount is set to five, text chunks use 256 tokens, and overlap is 32 tokens. HippoRAG2 uses Contriever for dense passage retrieval [103]; the other reported embedding configurations use bge-base-en-v1.5. Similarity-based retrieval represents images through their captions.

VanillaRAG is implemented with LangChain and FAISS [116]. The GraphRAG implementation is nano-graphrag [117], and the remaining baseline systems use their released implementations. The protocol selects hybrid search for LightRAG and local search for GraphRAG. vLLM accelerates the tested baseline systems, while MGRAG uses its graph-aware cache procedure. Other baseline hyperparameters follow the implementations' defaults.

Four metrics characterize answers. Semantic similarity (SIM.) is the cosine similarity between bge-base-en-v1.5 representations of generated and reference answers. Accuracy (ACC.) is the reported proportion of correct answers. ROUGE-L (ROG.) evaluates longest-common-subsequence overlap [109], and BLEU-1 evaluates modified unigram precision [118]. The paper uses the RAGAS evaluation framework [111]. The tables preserve its metric names and precision, including the label BLEU used for BLEU-1.

The publication reports point estimates for these experiments.

The complete-system baselines use different retrieval units. VanillaRAG returns dense-retrieved chunks, and R1-Searcher incorporates reinforcement learning into search and question answering. GraphRAG organizes its graph into communities and prepares summaries; LightRAG retrieves graph elements and neighborhoods through its selected search mode. HippoRAG2 adds passage nodes and uses Personalized PageRank for retrieval.

Component substitutions evaluate graph construction, semantic matching, and cache policy within MGRAG. Inverse ablations add matching and recomputation to other retrieval systems.

6.9 Answer Quality and Structural Retrieval

6.9.1 Complete-system comparisons

Table 6.2 reproduces the complete-system results. MGRAG has the highest value for all four metrics on each dataset. On MMQA, its accuracy is 0.7442, compared with 0.7053 for VanillaRAG and 0.6335 for HippoRAG2. Its semantic similarity is 0.8533, ROUGE-L is 0.6763, and BLEU-1 is 0.7215. GraphRAG and LightRAG have missing MMQA entries because those experiments exceeded three days.

Table 6.2: Answer quality with Qwen2.5-VL-7B-Instruct, reproduced from published Table 4, p. 2489 [19]. Vanilla., R1., Graph., Light., and Hippo. denote VanillaRAG, R1-Searcher, GraphRAG, LightRAG, and HippoRAG2. A dash means that the experiment exceeded three days; it is not a zero score. Bold and underlining retain the publication's highlighting.

Dataset	Metric	Vanilla.	R1.	Graph.	Light.	Hippo.	MGRAG
MMQA	SIM.	0.8362	0.5478	-	-	0.8381	**0.8533**
MMQA	ACC.	0.7053	0.3567	-	-	0.6335	**0.7442**
MMQA	ROG.	0.5942	0.1041	-	-	0.6659	**0.6763**
MMQA	BLEU	0.6645	0.0817	-	-	0.5917	**0.7215**
Arxiv	SIM.	0.8664	0.7051	0.7943	0.7404	0.8293	**0.8760**
Arxiv	ACC.	0.6638	0.3775	0.2350	0.3813	0.4688	**0.6800**
Arxiv	ROG.	0.2713	0.1189	0.1553	0.1580	0.2331	**0.2726**
Arxiv	BLEU	0.2452	0.1092	0.1690	0.1370	0.1281	**0.2498**
Manual	SIM.	0.8166	0.6046	0.5453	0.7255	0.6852	**0.8221**
Manual	ACC.	0.5881	0.2071	0.1436	0.4096	0.2641	**0.6173**
Manual	ROG.	0.3509	0.1078	0.0974	0.1595	0.2047	**0.3839**
Manual	BLEU	0.2767	0.0740	0.0718	0.1385	0.1314	**0.3093**
Web	SIM.	0.6937	0.6620	0.5470	0.5513	0.5642	**0.7009**
Web	ACC.	0.4703	0.4200	0.1053	0.2000	0.3607	**0.5123**
Web	ROG.	0.1508	0.1142	0.1133	0.0905	0.0616	**0.1664**
Web	BLEU	0.0326	0.0785	0.1031	0.0206	0.0133	**0.1456**
Wiki	SIM.	0.8824	0.5478	0.6966	0.7517	0.8204	**0.8838**
Wiki	ACC.	0.7800	0.3567	0.2000	0.3660	0.6500	**0.7845**
Wiki	ROG.	0.3559	0.1418	0.1361	0.2008	0.3042	**0.3579**
Wiki	BLEU	0.3053	0.1146	0.1007	0.1863	0.1851	**0.3174**
Wit	SIM.	0.7851	0.6110	0.6211	0.6973	0.7230	**0.8033**
Wit	ACC.	0.6675	0.6017	0.1500	0.2483	0.5138	**0.6967**
Wit	ROG.	0.3255	0.1307	0.1319	0.2416	0.2712	**0.3589**
Wit	BLEU	0.2095	0.0934	0.1142	0.1928	0.1200	**0.2382**
Recipe	SIM.	0.8599	0.6155	0.8166	0.8118	0.7792	**0.8673**
Recipe	ACC.	0.6525	0.2180	0.2771	0.3334	0.4917	**0.6814**
Recipe	ROG.	0.3714	0.1065	0.1745	0.2128	0.3303	**0.4220**
Recipe	BLEU	0.2599	0.0773	0.1527	0.1648	0.1941	**0.3102**

The advantage over the strongest baseline varies by dataset and metric. On Wiki, accuracy is 0.7845 for MGRAG and 0.7800 for VanillaRAG, while their semantic similarities are 0.8838

and 0.8824. On Web, the corresponding accuracies are 0.5123 and 0.4703. On Recipe, MGRAG reports ROUGE-L of 0.4220 compared with 0.3714 for VanillaRAG.

Several graph-based baselines score below VanillaRAG in this multimodal protocol. For example, on Manual, GraphRAG, LightRAG, and HippoRAG2 report accuracies of 0.1436, 0.4096, and 0.2641, compared with VanillaRAG's 0.5881.

6.9.2 Construction and matching substitutions

Table 6.3 reorganizes every value from the published graph-construction and matching comparison. Stanford OpenIE [119] and the statistical extraction alternative STAT replace the construction mechanism. VF2 and ISMAGS [113, 120] provide exact-matching alternatives. SM, VJ, RRWM, and IPFP represent approximate graph matching [121–124]. BGE, SLOTAlign, and FGWAlign supply semantic alternatives [112, 125, 126]. The table preserves the publication's three-decimal precision for this experiment.

Table 6.3: Graph-construction and graph-matching substitutions, transposed from published Table 7, p. 2490 [19]. All twelve method columns and all four metric values per dataset are retained. O.IE and STAT replace graph construction; the other alternatives replace matching. The shared backbone is Qwen2.5-VL-7B-Instruct.

Dataset	Method	SIM.	ACC.	ROG.	BLEU
MMQA	O.IE	0.642	0.571	0.510	0.483
MMQA	STAT	0.655	0.602	0.519	0.500
MMQA	VF2	0.851	0.735	0.673	0.620
MMQA	ISMAGS	0.833	0.706	0.638	0.590
MMQA	SM	0.832	0.702	0.632	0.582
MMQA	VJ	0.812	0.715	0.646	0.606
MMQA	RRWM	0.838	0.698	0.647	0.599
MMQA	IPFP	0.773	0.633	0.601	0.585
MMQA	BGE	0.832	0.703	0.635	0.586
MMQA	SLOT	0.838	0.700	0.650	0.602
MMQA	FGW	0.837	0.697	0.647	0.598
MMQA	MGRAG	0.853	0.744	0.676	0.722
Arxiv	O.IE	0.618	0.526	0.177	0.138
Arxiv	STAT	0.640	0.576	0.184	0.145
Arxiv	VF2	0.871	0.650	0.270	0.239
Arxiv	ISMAGS	0.868	0.643	0.266	0.231

Table 6.3 (continued)

Dataset	Method	SIM.	ACC.	ROG.	BLEU
Arxiv	SM	0.866	0.670	0.263	0.244
Arxiv	VJ	0.872	0.667	0.269	0.245
Arxiv	RRWM	0.874	0.675	0.257	0.243
Arxiv	IPFP	0.814	0.581	0.212	0.203
Arxiv	BGE	0.874	0.668	0.270	0.244
Arxiv	SLOT	0.872	0.653	0.269	0.233
Arxiv	FGW	0.868	0.651	0.264	0.230
Arxiv	MGRAG	0.876	0.680	0.273	0.250
Manual	O.IE	0.520	0.459	0.245	0.209
Manual	STAT	0.509	0.443	0.234	0.182
Manual	VF2	0.817	0.612	0.377	0.306
Manual	ISMAGS	0.817	0.612	0.382	0.307
Manual	SM	0.813	0.614	0.378	0.303
Manual	VJ	0.808	0.608	0.372	0.298
Manual	RRWM	0.811	0.614	0.377	0.303
Manual	IPFP	0.800	0.594	0.369	0.289
Manual	BGE	0.815	0.615	0.375	0.303
Manual	SLOT	0.813	0.611	0.375	0.301
Manual	FGW	0.820	0.600	0.374	0.292
Manual	MGRAG	0.822	0.617	0.384	0.309
Web	O.IE	0.466	0.290	0.052	0.020
Web	STAT	0.485	0.298	0.101	0.095
Web	VF2	0.677	0.507	0.156	0.088
Web	ISMAGS	0.675	0.509	0.155	0.079
Web	SM	0.676	0.507	0.156	0.056
Web	VJ	0.679	0.508	0.160	0.056
Web	RRWM	0.675	0.503	0.157	0.066
Web	IPFP	0.666	0.493	0.149	0.055
Web	BGE	0.687	0.510	0.156	0.095
Web	SLOT	0.676	0.498	0.154	0.116
Web	FGW	0.676	0.480	0.147	0.096
Web	MGRAG	0.701	0.512	0.166	0.146

Table 6.3 (continued)

Dataset	Method	SIM.	ACC.	ROG.	BLEU
Wiki	O.IE	0.504	0.498	0.230	0.200
Wiki	STAT	0.525	0.510	0.253	0.218
Wiki	VF2	0.877	0.752	0.352	0.300
Wiki	ISMAGS	0.879	0.754	0.355	0.303
Wiki	SM	0.878	0.750	0.353	0.298
Wiki	VJ	0.873	0.751	0.349	0.297
Wiki	RRWM	0.878	0.743	0.355	0.301
Wiki	IPFP	0.869	0.735	0.348	0.296
Wiki	BGE	0.878	0.746	0.355	0.301
Wiki	SLOT	0.879	0.749	0.356	0.301
Wiki	FGW	0.878	0.747	0.352	0.300
Wiki	MGRAG	0.884	0.785	0.358	0.317
Wit	O.IE	0.492	0.465	0.233	0.210
Wit	STAT	0.489	0.453	0.205	0.198
Wit	VF2	0.799	0.693	0.354	0.234
Wit	ISMAGS	0.797	0.694	0.355	0.231
Wit	SM	0.784	0.686	0.353	0.229
Wit	VJ	0.786	0.691	0.356	0.227
Wit	RRWM	0.795	0.679	0.347	0.230
Wit	IPFP	0.792	0.678	0.346	0.231
Wit	BGE	0.789	0.689	0.356	0.230
Wit	SLOT	0.793	0.693	0.351	0.227
Wit	FGW	0.794	0.691	0.349	0.223
Wit	MGRAG	0.803	0.697	0.359	0.238
Recipe	O.IE	0.551	0.425	0.211	0.202
Recipe	STAT	0.545	0.401	0.198	0.172
Recipe	VF2	0.859	0.679	0.413	0.304
Recipe	ISMAGS	0.861	0.668	0.407	0.302
Recipe	SM	0.862	0.678	0.411	0.305
Recipe	VJ	0.857	0.672	0.413	0.295
Recipe	RRWM	0.852	0.670	0.406	0.298
Recipe	IPFP	0.847	0.661	0.395	0.221

Table 6.3 (continued)

Dataset	Method	SIM.	ACC.	ROG.	BLEU
Recipe	BGE	0.864	0.674	0.415	0.308
Recipe	SLOT	0.861	0.675	0.412	0.306
Recipe	FGW	0.860	0.677	0.409	0.305
Recipe	MGRAG	0.867	0.681	0.422	0.310

On MMQA, the full method reports accuracy 0.744, compared with 0.571 for OpenIE and 0.602 for STAT. Query-aware construction outperforms these extraction alternatives on MMQA.

Among MMQA matching alternatives, VF2 reaches accuracy 0.735, while MGRAG reaches 0.744. VF2 also has ROUGE-L of 0.673 compared with 0.676. On Web, BGE reaches accuracy 0.510 and MGRAG 0.512. On Wiki, ISMAGS reaches 0.754 and MGRAG 0.785. MGRAG leads the displayed metric rows, with smaller MMQA and Web accuracy margins than the Wiki accuracy margin.

6.9.3 Queries requiring multiple hops

Table 6.4 reports results grouped by MRAMG query-hop count. For one-hop questions, VanillaRAG reports accuracy 0.613 and MGRAG 0.624. For six-hop questions, the reported values are 0.661 and 0.738. MGRAG also reports higher ROUGE-L and BLEU-1 for each hop group. VanillaRAG is faster in every hop group. The one-hop times are 0.13 and 0.35 minutes, and the six-hop times are 0.18 and 0.43 minutes.

Table 6.4: Results by query hop count, reorganized from published Table 8, p. 2490 [19]. Time is in minutes, as reported. Each original row is shown as two method rows, with the query count repeated.

Hops	Queries	Method	Time	ACC.	ROG.	BLEU
1-hop	201	VanillaRAG	**0.13**	0.613	0.379	0.240
1-hop	201	MGRAG	0.35	**0.624**	**0.425**	**0.286**
2-hop	1,237	VanillaRAG	**0.10**	0.588	0.304	0.193
2-hop	1,237	MGRAG	0.37	**0.601**	**0.335**	**0.230**
3-hop	2,357	VanillaRAG	**0.16**	0.633	0.329	0.238
3-hop	2,357	MGRAG	0.41	**0.659**	**0.355**	**0.260**
4-hop	499	VanillaRAG	**0.19**	0.697	0.338	0.259
4-hop	499	MGRAG	0.44	**0.712**	**0.373**	**0.293**

Table 6.4 (continued)

Hops	Queries	Method	Time	ACC.	ROG.	BLEU
5-hop	286	VanillaRAG	**0.18**	0.678	0.319	0.205
5-hop	286	MGRAG	0.45	**0.733**	**0.392**	**0.282**
6-hop	220	VanillaRAG	**0.18**	0.661	0.305	0.176
6-hop	220	MGRAG	0.43	**0.738**	**0.377**	**0.249**

The six-hop group contains 220 queries, whereas the three-hop group contains 2,357. The comparison shows a larger accuracy gap on the longest-hop group than on the shortest-hop group, alongside additional processing time.

6.10 Cache Quality and End-to-end Costs

Table 6.5 preserves the MMQA cache-system comparison. MGRAG reports accuracy 0.7442 and total time 4.3 hours. The vLLM and CacheBlend variants paired with VanillaRAG report accuracies 0.6593 and 0.6601, with total time 4.5 hours for both. Paired with HippoRAG2, they report accuracies 0.6335 and 0.6267, with total times of 8.8 and 8.3 hours.

Table 6.5: Cache-system comparison on MMQA, reproduced from published Table 6, p. 2489 [19]. V. and H. identify VanillaRAG and HippoRAG2 retrieval pipelines. TTFT is in seconds and total time is in hours.

System	SIM.	ACC.	ROG.	BLEU	TTFT	Time
vLLM (V.)	0.8315	0.6593	0.6277	0.5867	3.2 s	4.5 h
CacheBlend (V.)	0.8315	0.6601	0.6417	0.5964	**3.1 s**	4.5 h
vLLM (H.)	0.8381	0.6335	0.6659	0.5917	4.6 s	8.8 h
CacheBlend (H.)	0.8415	0.6267	0.6636	0.5917	4.4 s	8.3 h
MGRAG	**0.8533**	**0.7442**	**0.6763**	**0.7215**	3.2 s	**4.3 h**

Time to first token (TTFT) gives a different view from total time. CacheBlend with VanillaRAG reports 3.1 seconds, while MGRAG and vLLM with VanillaRAG report 3.2 seconds. MGRAG combines the highest answer-quality values with a total time of 4.3 hours; CacheBlend with VanillaRAG has the shortest TTFT.

The inverse-ablation table also measures quality when graph-aware recomputation is added after path matching. Within the HippoRAG2 composition, accuracy changes from 0.7372 with path matching to 0.7442 with both components, while the total time changes from 4.1 to 4.3

hours. Selective recomputation improves answer quality in this composition with an increase in total time.

Section 6.11 compares these component additions within the HippoRAG2 and VanillaRAG pipelines.

Table 6.6 reports time and token costs by module. On MMQA, graph construction accounts for 193.0 minutes, matching for 0.14 minutes, and answer generation for 8.9 minutes; the reported end-to-end time is 248.1 minutes. The construction and generation token entries are 3,085 and 1,431, with end-to-end token entry 4,516. The listed module times do not sum to the reported end-to-end total; the publication does not reconcile the difference in this breakdown.

Table 6.6: Module costs reproduced from published Table 5, p. 2489 [19]. Time is in minutes. KGC, QA, and E2E cells retain the published time/token pairs for graph construction, answering, and end-to-end processing. Match denotes matching time. Token counts and E2E totals are preserved as reported.

Dataset	Queries	KGC	Match	QA	E2E
MMQA	951	193.0 / 3,085	0.14	8.9 / 1,431	248.1 / 4,516
Arxiv	200	61.9 / 3,946	0.23	16.2 / 2,944	83.9 / 6,890
Manual	390	76.1 / 4,285	0.04	23.9 / 2,556	109.7 / 6,841
Web	750	143.8 / 4,623	0.27	42.9 / 3,101	196.5 / 7,723
Wiki	500	164.2 / 6,629	0.30	52.1 / 6,267	241.5 / 12,896
Wit	600	140.6 / 7,660	0.21	42.6 / 6,539	196.5 / 14,199
Recipe	2,360	525.7 / 4,351	1.74	191.3 / 3,173	808.7 / 7,525

Graph construction is the largest listed time component on all seven datasets. Recipe reports 525.7 minutes for construction, 1.74 for matching, and 191.3 for answering, with end-to-end time 808.7 minutes. On the smaller Arxiv query set, the corresponding values are 61.9, 0.23, and 16.2 minutes, with end-to-end time 83.9. Repeated extraction accounts for most of the listed computation time despite the matcher's exponential worst-case bound.

Although mapping enumeration has an exponential bound, matching processes small query-specific graphs in these experiments. The measured bottleneck is repeated multimodal extraction. Reducing the number of processed chunks or the cost per extraction is therefore a target for runtime improvement.

6.11 Ablations and Parameter Sensitivity

6.11.1 Component ablations

The MMQA ablations in Table 6.7 change matching, coarse encoding, and pipeline composition. Removing graph matching constructs the query-specific graph from a fixed number of chunks and reports accuracy 0.7260, compared with 0.7442 for MGRAG. Replacing caption-based coarse ranking with direct CLIP encoding [127] reports accuracy 0.7376. The Vanilla MGRAG variant applies path matching and graph-aware caching within the VanillaRAG pipeline with retrieval amount $k = 5$; it reports accuracy 0.6887.

Table 6.7: MMQA component ablations, reproduced from published Table 9, p. 2490 [19]. For Vanilla MGRAG, k is the retrieval amount. Time is in hours.

Model	SIM.	ACC.	ROG.	BLEU	Time
Without graph matching	0.8362	0.7260	0.6657	0.5774	4.4 h
With CLIP encoder	0.8513	0.7376	0.6744	0.7008	4.5 h
Vanilla MGRAG, $k = 5$	0.8260	0.6887	0.6266	0.5773	4.7 h
MGRAG	**0.8533**	**0.7442**	**0.6763**	**0.7215**	**4.3 h**

The full system also has the largest SIM., ROUGE-L, and BLEU-1 entries in this table. Its reported time is 4.3 hours, compared with 4.4 hours without graph matching, 4.5 hours with CLIP, and 4.7 hours for Vanilla MGRAG. Caption-based ranking performs better than the CLIP substitution on MMQA with this pipeline and backbone.

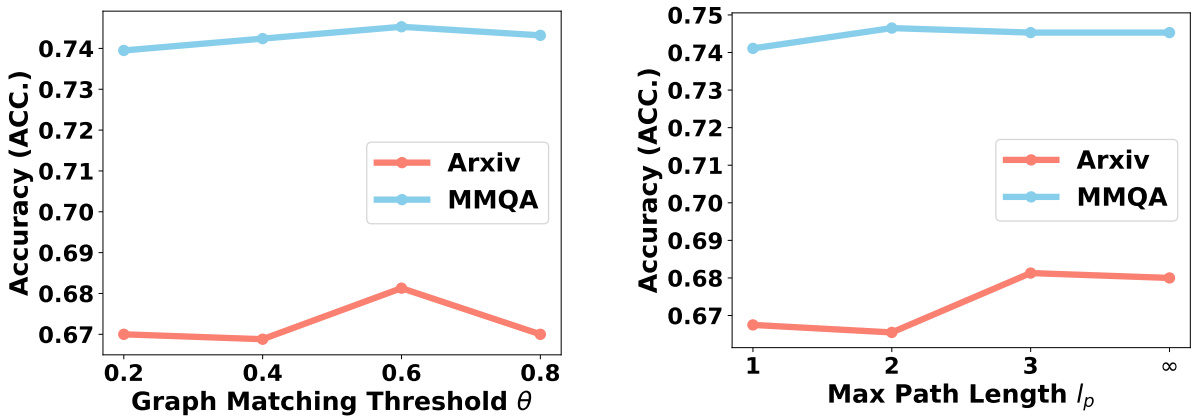
6.11.2 Adding components to other pipelines

Table 6.8 retains the inverse ablations. Adding graph-aware KV recomputation to HippoRAG2 changes accuracy from 0.6335 to 0.6855 and time from 7.5 to 7.6 hours. Adding path-based semantic matching gives accuracy 0.7372 and time 4.1 hours. Combining both gives accuracy 0.7442 and time 4.3 hours. The latter configuration shares MGRAG's reported values because the publication identifies its retrieval and generation setting with the full method once both components are incorporated.

For VanillaRAG, the base accuracy is 0.7053 at 2.6 hours. Adding path matching gives 0.7175 at 4.7 hours, and adding graph-aware recomputation as well gives 0.7232 at 4.8 hours. Thus, the component additions increase quality in these measured compositions, while their time

Table 6.8: MMQA inverse ablations, reproduced from published Table 10, p. 2491 [19]. PSGM and GKV denote path-based semantic graph matching and graph-aware KV recomputation. Time is in hours.

Model	SIM.	ACC.	ROG.	BLEU	Time
Hippo.	0.8381	0.6335	0.6659	0.5917	7.5 h
Hippo.+GKV	0.8392	0.6855	0.6651	0.7013	7.6 h
Hippo.+PSGM	0.8416	0.7372	0.6720	0.7128	4.1 h
Hippo.+PSGM+GKV	**0.8533**	**0.7442**	**0.6763**	**0.7215**	4.3 h
Vanilla.	0.8362	0.7053	0.5942	0.6645	**2.6 h**
Vanilla.+PSGM	0.8460	0.7175	0.6466	0.6973	4.7 h
Vanilla.+PSGM+GKV	0.8485	0.7232	0.6563	0.7001	4.8 h
MGRAG	**0.8533**	**0.7442**	**0.6763**	**0.7215**	4.3 h



(a) Similarity threshold θ.

(b) Maximum path length l_p.

Figure 6.5: MGRAG matching-parameter sensitivity. Original subfigures of Figure 5, p. 2491 [19].

changes depend on the host pipeline.

6.11.3 Matching threshold and path length

Figure 6.5 reproduces the published parameter curves. The experiments vary the semantic threshold θ and maximum path length l_p, including lengths 1, 2, 3, and unrestricted length. The reported discussion identifies similarity threshold $\theta = 0.6$ and maximum path length $l_p = 3$ as effective choices in these tests.

The threshold changes which entity correspondences and path extensions can pass validation. A higher value demands closer embedding agreement, while a lower value permits more candidates. The path-length cap determines how many relations may be used to express a query relation. Its effect therefore concerns both representational flexibility and search.

6.12 Discussion and Chapter Summary

We proposed MGRAG to construct and match multimodal evidence for individual queries. Query-aware extraction selects relevant facts, semantic path matching connects them to the query, and graph-aware recomputation reuses processing from construction during answer generation.

MGRAG obtains the highest values across the seven datasets' four answer metrics. Construction and matching substitutions show the benefits of query-dependent extraction and path semantics. The hop-group results show larger quality gains for questions requiring more relational steps, with added runtime. Cache and inverse-ablation results show how recomputation changes quality and cost within different retrieval pipelines.

Caption omissions and extraction errors can exclude relevant facts before matching. An accepted match ends construction, so an incorrect correspondence can prevent examination of later chunks. The construction and path-length caps further restrict the search.

MGRAG makes the query part of graph construction and matches its requirements to semantic paths across multimodal evidence. The same graph guides which cached token states are recomputed for answering. The next chapter discusses the progression from existing knowledge graphs and supervised representation learning to task-related graph retrieval, connected evidence selection, and query-aware multimodal reasoning.

CHAPTER 7

DISCUSSION

Our research progressed from knowledge-enriched representations to graph construction from task data and prediction without model fine-tuning. Extending retrieval beyond classification introduced requirements for evidence coverage, multimodal correspondence, and context-processing efficiency. This chapter discusses how these requirements shape the construction and use of knowledge structures.

7.1 From Knowledge Enrichment to Task-driven Retrieval

KGLink uses the semantic information already represented in an external knowledge graph. Graph processing filters linked entities and selects candidate types and feature descriptions before the encoder reads the table. Supervised training learns how these descriptions correspond to annotation labels. The method therefore depends on both the coverage of the external graph and the labeled data available for learning the annotation vocabulary.

GORAG changes the source of the structure and the way the system adapts. Support examples supply keyword–label associations, and later observations expand the index. A large language model uses retrieved labels and their descriptions for classification without task-specific parameter updates. New knowledge enters the graph, allowing the system to incorporate evolving labels and vocabulary without relying on a predefined external knowledge graph.

The required retrieval output becomes richer when moving from classification to generation. Candidate labels constrain a classification decision, while an answer can require several facts and their explanations. AGRAG selects connected facts from a reusable corpus graph and adds source passages to improve information coverage. This makes both the relationships among facts and their original context available to the generator.

For multimodal retrieval, MGRAG addresses the additional need to select and match evidence for each question. The question determines which details to extract from text and images, and semantic matching checks whether evidence paths express the requested relations. Matching controls whether another chunk is processed, while graph features guide the recomputation of cached token states for answering.

Table 7.1: The roles of structure across the four studies. The entries summarize the methods in their respective research chapters.

Study	Structure source	Selected or represented evidence	Model role
KGLink	Existing knowledge graph and table links	Candidate types and graph-derived column features	Fine-tuned table encoding and type prediction
GORAG	Labeled examples and prescribed online updates	Connected keyword–label evidence and candidate labels	Label-description generation and classification
AGRAG	Corpus terms, extracted relations, and passage links	Query-conditioned connected subgraph and hybrid passages	Relation extraction, filtering, and generation
MGRAG	Query and ranked multimodal chunks	Semantic path matches, original chunks, and graph features	Query-aware extraction, QA, and selective KV recomputation

Table 7.1 summarizes how the representation changes with the output requirements. Graph descriptions enrich column representations, keyword–label connections constrain candidate classes, and fact paths support generated answers. In MGRAG, the structure also allocates computation when the retrieved context is processed.

7.2 Static, Online, and Query-dependent Structures

The transition away from a predefined knowledge graph introduces a choice about when to construct and update the replacement structure. KGLink reuses an external graph and adapts evidence selection to each table. Its supervised encoder learns how to use the selected descriptions.

GORAG updates a task index as labels and text arrive. Associations from earlier predictions become available to later queries, so classification performance depends on the sequence of index states.

AGRAG prepares the corpus graph once, then changes query anchors, node influence, edge costs, and subgraph selection for each question. This amortizes graph preparation across repeated queries.

MGRAG extracts facts for each query as it consumes ranked chunks. Matching can stop construction before the full list is processed. Query-aware extraction targets the relationships needed by the current question and adds construction and matching costs to query processing.

GORAG's online updates respond to changes in the task's data, while AGRAG's query-conditioned selection responds to different questions over a shared corpus. MGRAG moves

Table 7.2: Main experimental conditions across the four studies. Hardware denotes the reported local setup; GORAG's IFS-Rel generation uses the GPT-4o service. Full task-specific controls appear in the corresponding research chapters.

Study	Main model and hardware	Supervision and evidence state	Reported outcome
KGLink	BERT; Tesla V100	7:1:2 train/validation/test; external KG; three-run averages	Annotation accuracy and weighted F1 on a percentage scale
GORAG	Qwen2.5-7B on CAD/WOS; GPT-4o on IFS-Rel; A100 40 GB	1/5/10 support texts per label; cumulative labels; prediction-dependent index updates	Exact-label accuracy by round and support size
AGRAG	Qwen2.5-14B-Instruct; A100 80 GB	Fixed corpus graph; query-conditioned scores and subgraph; one training passage per label for classification	Task-specific QA assessment/overlap, coverage, faithfulness, and classification metrics
MGRAG	Qwen2.5-VL-7B-Instruct; A100 40 GB	Query-specific multimodal graph; semantic matching; selective KV recomputation	Answer quality, TTFT, and complete-system/module costs

query dependence into extraction itself. The choice therefore depends on how often the source data change, how many queries share the corpus, and whether relevant facts can be extracted before the question is known.

7.3 Evidence Quality and Computational Cost

The relevant quality criterion follows the task. A column representation must support the expected type label. Classification retrieval must retain the correct candidate label. Generation requires both sufficient answer coverage and support for the claims produced. Cache reuse must preserve answer quality while reducing processing cost.

Table 7.2 compares the main experimental conditions. KGLink evaluates a trained encoder; the other studies evaluate generative pipelines with different backbones. Shared datasets also have different supervision and update protocols. GORAG's WOS and IFS-Rel experiments accumulate labels and prediction-derived keyword–label associations over rounds, whereas AGRAG indexes one training passage per label for classification.

The assessment rule follows the output being evaluated. GORAG's accuracy requires the predicted class to match the gold label. AGRAG's QA accuracy uses model judgments scored 0, 0.5, or 1, while its classification accuracy uses label correctness. Its source-defined ROUGE-L quantity divides longest-common-subsequence length by generated-answer length. MGRAG reports answer accuracy alongside semantic similarity and lexical overlap.

The AGRAG ablations show why coverage and faithfulness must be considered together. Without hybrid passages, the generator can produce supported claims while omitting required information. Graph connectivity helps organize facts, while passages preserve context that extraction or subgraph selection can omit.

MGRAG separates evidence retrieval from reuse of its computation. Cache comparisons measure combined retrieval and serving choices. Inverse ablations test additions within one retrieval pipeline, where improved answer quality can require extra runtime.

Graph preparation and query processing contribute to total cost in different ways. KGLink reports preprocessing and model-processing costs for annotation. GORAG measures the resource demands of its evolving index and retrieval. AGRAG separates corpus indexing from querying. MGRAG reports construction, matching, answer-generation, and end-to-end costs.

7.4 Information Preservation and Semantic Granularity

The hyper-relational study in Chapter 2 supplies another perspective on structural representations. A main triple and its qualifiers express different aspects of a fact. Decomposing that representation can retain some relationships while losing the association that gives them their meaning. The study examines this issue through decomposition methods, qualifier capture, graph reception, and link-prediction results.

A related design question appears in each core chapter. KGLink must translate graph-derived type descriptions into the annotation task's vocabulary. GORAG associates words with labels whose relevance depends on the classification task. AGRAG selects a graph serialization while retaining a passage channel for information absent from that serialization. MGRAG matches relations to paths that can express a corresponding meaning through a different topology. Each method specifies an evidence unit and a way to relate that unit to the task.

A column vector, candidate label, graph fact, path, and source passage preserve different information. Paths expose intermediate relations, while passages and images retain details that may disappear during extraction. Moving between these representations changes what the next stage must infer.

A useful structure must preserve the relationships needed for the task. Connectivity identifies available links, semantic matching checks their correspondence to a query, and supervision determines the required annotation category. The hyper-relational study examines how graph transformation can alter these relationships before prediction.

7.5 Error Propagation Across Structural Decisions

Errors propagate differently depending on where structure enters the pipeline. An incorrect KGLink entity link changes the descriptions offered to the encoder for a table. In GORAG, an incorrect predicted label can become an association available to later queries. AGRAG reuses a corpus graph, so an extracted relation can influence several query-specific selections. MGRAG constructs a new graph for each query, but a mistaken accepted match can end construction before later chunks are examined.

Feedback can target these error sources directly. Entity disambiguation can revise cell links, online correction can revise inserted associations, source verification can check extracted relations, and matching diagnosis can inspect entity alignment and path choices. Chapter 8 develops these directions.

7.6 Limitations and Open Questions

KGLink's knowledge source is limited by external graph coverage, while its supervised predictor learns the label vocabulary represented in training. Its no-linkage analysis examines how the encoder performs when graph evidence is unavailable.

Constructing a graph from task data shifts the coverage problem to the examples and sources supplied to the system. GORAG starts from sparse support data and can retain errors through prediction-derived updates. Longer streams and different patterns of label evolution remain open evaluation settings.

AGRAG and MGRAG rely on relation extraction, and MGRAG also uses captions to rank images. A useful fact must first survive these preparation steps before structural retrieval can select it. Improving extraction coverage and correcting stored relationships are therefore important for extending the methods to new corpora. Model-assessed QA results additionally depend on the evaluator and its assessment rules.

AGRAG's 2-approximation statement concerns the cost of its initial terminal-connecting tree, and MGRAG's threshold condition concerns accepted path similarities.

CHAPTER 8

CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

In this thesis, we investigated how explicit knowledge structures support language-model prediction and reasoning. Our research progressed from enrichment with an existing knowledge graph to graph construction from task data and retrieval without model fine-tuning. We further studied answers requiring connected textual facts or relationships distributed across text and images.

We first proposed KGLink to enrich column representations with external graph descriptions. Cross-column filtering selects relevant descriptions, and contextual type-representation training learns their correspondence to annotation labels. The trained encoder also predicts from table values when KG evidence is unavailable.

To remove the dependence on existing knowledge graphs and model fine-tuning, we proposed GORAG for dynamic few-shot classification. GORAG constructs a keyword–label graph from task data and retrieves candidate classes for a large language model. Online insertion incorporates newly observed vocabulary without updating model parameters. Experiments show the benefit of connected retrieval and online updates as labels accumulate.

We then extended task-driven retrieval to the connected facts needed for question answering and generation. AGRAG constructs a graph from a text corpus and selects connected facts using query-dependent node influence and semantic edge costs. Expanding the initial tree supplies additional answer components, while hybrid passages retain context beyond the selected graph. The ablations show how these evidence channels contribute to information coverage and answer faithfulness.

To support answers that combine text and image evidence, we proposed MGRAG to construct query-specific graphs and match requested relations to semantic paths. Graph-aware recomputation extends the use of structure to context processing. The experiments relate answer quality to query-aware extraction and matching, and measure cache reuse quality and cost.

Our hyper-relational knowledge-graph study examined how graph transformations preserve fact information. Qualifier capture and graph reception explain how the representation and the model jointly determine which relationships remain available for prediction.

8.2 Future Directions

8.2.1 Feedback for structure maintenance

Future work can use feedback to maintain evidence structures. A retrieved relation may prove irrelevant, its source passage may be insufficient, or a prediction inserted into an index may later be corrected. These observations could guide which associations are retained, revised, or removed.

Feedback timing determines which updates are possible. An online correction becomes available after the original prediction, and an answer assessment can guide only subsequent retrieval. Recording this timing and the feedback source would allow correction and retention policies to be evaluated under explicit supervision conditions.

8.2.2 Multimodal structure and provenance

Multimodal graph construction could retain a link from each extracted fact to its supporting text span or image region. Caption-based ranking and query-aware extraction both select visual information, and source links would make their omissions easier to identify.

Source links would help distinguish missing visual evidence from extraction, relation-representation, and path-search errors. They would also support checking whether an accepted path is grounded in the original source at the granularity required by the answer.

8.2.3 Joint retrieval and context computation

Retrieval and context computation can be optimized jointly. Retrieved evidence determines input length, modality, and relationships; cache availability and recomputation policy determine processing cost. MGRAG uses graph features for one such decision.

Further work could compare evidence structures under a fixed serving budget, measuring answer quality alongside construction, matching, cache reuse, and generation costs. Keeping the query workload and hardware fixed would show which structures provide useful information at an acceptable processing cost.

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